```
;Disassembly of the running p-system

; 80-column screen memory
; 2000-251E used for other purposes during boot time

2000 3430 40 STCR *R0+,0
2002 3630 60 STCR *R0+,8
2004 2034  4 COC  *R4+,R0
2006 3037 07 LDCR *R7+,0
2008 3020 0  LDCR @4070,0
200A 4070 @p
200C 2020    COC  @2020,R0
200E 2020
2010 2020    COC  @2020,R0
2012 2020
2014 2020    COC  @2020,R0
2016 2020
2018 2020    COC  @2020,R0
201A 2020
201C 2020    COC  @2020,R0
201E 2020
2020 2020    COC  @2020,R0
2022 2020
2024 2020    COC  @2020,R0
2026 2020
2028 2020    COC  @2020,R0
202A 2020
202C 2020    COC  @2020,R0
202E 2020
2030 2020    COC  @2020,R0
2032 2020
2034 2020    COC  @2020,R0
2036 2020
2038 2020    COC  @2020,R0
203A 2020
203C 2020    COC  @2020,R0
203E 2020
2040 2020    COC  @2020,R0
2042 2020
2044 2020    COC  @2020,R0
2046 2020
2048 2020    COC  @2020,R0
204A 2020
204C 2020    COC  @2020,R0
204E 2020
2050 3430 40 STCR *R0+,0
2052 3632 62 STCR *R2+,8
2054 2030  0 COC  *R0+,R0
2056 3641 6A STCR R1,9
2058 3020 0  LDCR @2020,0
205A 2020
205C 2042  B COC  R2,R1
205E 4C20 L  SZC  @2020,*R0+
2060 2020
2062 4032 @2 SZC  *R2+,R0
2064 3030 00 LDCR *R0+,0
2066 3420 4  STCR @2020,0
2068 2020
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206A 2020    COC  @2020,R0
206C 2020
206E 2020    COC  @2020,R0
2070 2020
2072 2020    COC  @2020,R0
2074 2020
2076 2020    COC  @2020,R0
2078 2020
207A 2020    COC  @2020,R0
207C 2020
207E 2020    COC  @2020,R0
2080 2020
2082 2020    COC  @2020,R0
2084 2020
2086 2020    COC  @2020,R0
2088 2020
208A 2020    COC  @2020,R0
208C 2020
208E 2020    COC  @2020,R0
2090 2020
2092 2020    COC  @2020,R0
2094 2020
2096 2020    COC  @2020,R0
2098 2020
209A 2020    COC  @2020,R0
209C 2020
209E 2020    COC  @3430,R0
20A0 3430 40
20A2 3634 64 STCR *R4+,8
20A4 2032  2 COC  *R2+,R0
20A6 3030 00 LDCR *R0+,0
20A8 3420 4  STCR @2020,0
20AA 2020
20AC 2020    COC  @2020,R0
20AE 2020
20B0 2020    COC  @2020,R0
20B2 2020
20B4 2020    COC  @2020,R0
20B6 2020
20B8 2020    COC  @2020,R0
20BA 2020
20BC 2020    COC  @2020,R0
20BE 2020
20C0 2020    COC  @2020,R0
20C2 2020
20C4 2020    COC  @2020,R0
20C6 2020
20C8 2020    COC  @2020,R0
20CA 2020
20CC 2020    COC  @2020,R0
20CE 2020
20D0 2020    COC  @2020,R0
20D2 2020
20D4 2020    COC  @2020,R0
20D6 2020
20D8 2020    COC  @2020,R0
20DA 2020
20DC 2020    COC  @2020,R0
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20DE 2020
20E0 2020    COC  @2020,R0
20E2 2020
20E4 2020    COC  @2020,R0
20E6 2020
20E8 2020    COC  @2020,R0
20EA 2020
20EC 2020    COC  @2020,R0
20EE 2020
20F0 3430 40 STCR *R0+,0
20F2 3636 66 STCR *R6+,8
20F4 2030  0 COC  *R0+,R0
20F6 3230 20 LDCR *R0+,8
20F8 4320 C  SZC  @2020,R12
20FA 2020
20FC 204C  L COC  R12,R1
20FE 4920 I  SZC  @2020,@5231(R4)
2100 2020
2102 5231 R1
2104 322C 2, LDCR @3833(R12),8
2106 3833 83
2108 3030 00 LDCR *R0+,0
210A 2020    COC  @2020,R0
210C 2020
210E 2020    COC  @2020,R0
2110 2020
2112 2020    COC  @2020,R0
2114 2020
2116 2020    COC  @2020,R0
2118 2020
211A 2020    COC  @2020,R0
211C 2020
211E 2020    COC  @2020,R0
2120 2020
2122 2020    COC  @2020,R0
2124 2020
2126 2020    COC  @2020,R0
2128 2020
212A 2020    COC  @2020,R0
212C 2020
212E 2020    COC  @2020,R0
2130 2020
2132 2020    COC  @2020,R0
2134 2020
2136 2020    COC  @2020,R0
2138 2020
213A 2020    COC  @2020,R0
213C 2020
213E 2020    COC  @3430,R0
2140 3430 40
2142 3638 68 STCR *R8+,8
2144 2038  8 COC  *R8+,R0
2146 3330 30 LDCR *R0+,12
2148 3020 0  LDCR @2020,0
214A 2020
214C 2020    COC  @2020,R0
214E 2020
2150 2020    COC  @2020,R0
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2152 2020
2154 2020    COC  @2020,R0
2156 2020
2158 2020    COC  @2020,R0
215A 2020
215C 2020    COC  @2020,R0
215E 2020
2160 2020    COC  @2020,R0
2162 2020
2164 2020    COC  @2020,R0
2166 2020
2168 2020    COC  @2020,R0
216A 2020
216C 2020    COC  @2020,R0
216E 2020
2170 2020    COC  @2020,R0
2172 2020
2174 2020    COC  @2020,R0
2176 2020
2178 2020    COC  @2020,R0
217A 2020
217C 2020    COC  @2020,R0
217E 2020
2180 2020    COC  @2020,R0
2182 2020
2184 2020    COC  @2020,R0
2186 2020
2188 2020    COC  @2020,R0
218A 2020
218C 2020    COC  @2020,R0
218E 2020
2190 3430 40 STCR *R0+,0
2192 3641 6A STCR R1,9
2194 2030  0 COC  *R0+,R0
2196 3435 45 STCR *R5+,0
2198 4320 C  SZC  @205C,R12
219A 205C  \
219C 2042  B COC  R2,R1
219E 2020    COC  @2020,R0
21A0 2020
21A2 2A52 *R XOR  *R2,R9
21A4 3132 12 LDCR *R2+,4
21A6 2020    COC  @2020,R0
21A8 2020
21AA 2020    COC  @2020,R0
21AC 2020
21AE 2020    COC  @2020,R0
21B0 2020
21B2 2020    COC  @2020,R0
21B4 2020
21B6 2020    COC  @2020,R0
21B8 2020
21BA 2020    COC  @2020,R0
21BC 2020
21BE 2020    COC  @2020,R0
21C0 2020
21C2 2020    COC  @2020,R0
21C4 2020
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21C6 2020    COC  @2020,R0
21C8 2020
21CA 2020    COC  @2020,R0
21CC 2020
21CE 2020    COC  @2020,R0
21D0 2020
21D2 2020    COC  @2020,R0
21D4 2020
21D6 2020    COC  @2020,R0
21D8 2020
21DA 2020    COC  @2020,R0
21DC 2020
21DE 2020    COC  @3430,R0
21E0 3430 40
21E2 3643 6C STCR R3,9
21E4 2034  4 COC  *R4+,R0
21E6 4534 E4 SZC  *R4+,*R4
21E8 4620 F  SZC  @4E4F,*R8
21EA 4E4F NO
21EC 2053  S COC  *R3,R1
21EE 5A43 ZC SZCB R3,@2020(R9)
21F0 2020
21F2 5231 R1 SZCB *R1+,R8
21F4 352C 5, STCR @2A52(R12),4
21F6 2A52 *R
21F8 392B 9+ MPY  @2020(R11),R4
21FA 2020
21FC 2020    COC  @2020,R0
21FE 2020
2200 2020    COC  @2020,R0
2202 2020
2204 2020    COC  @2020,R0
2206 2020
2208 2020    COC  @2020,R0
220A 2020
220C 2020    COC  @2020,R0
220E 2020
2210 2020    COC  @2020,R0
2212 2020
2214 2020    COC  @2020,R0
2216 2020
2218 2020    COC  @2020,R0
221A 2020
221C 2020    COC  @2020,R0
221E 2020
2220 2020    COC  @2020,R0
2222 2020
2224 2020    COC  @2020,R0
2226 2020
2228 2020    COC  @2020,R0
222A 2020
222C 2020    COC  @2020,R0
222E 2020
2230 3430 40 STCR *R0+,0
2232 3645 6E STCR R5,9
2234 2034  4 COC  *R4+,R0
2236 3734 74 STCR *R4+,12
2238 4620 F  SZC  @474F,*R8
```

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223A 474F GO
223C 2053  S COC  *R3,R1
223E 5A43 ZC SZCB R3,@2020(R9)
2240 2020
2242 5231 R1 SZCB *R1+,R8
2244 352C 5, STCR @2A52(R12),4
2246 2A52 *R
2248 3133 13 LDCR *R3+,4
224A 2020    COC  @2020,R0
224C 2020
224E 2020    COC  @2020,R0
2250 2020
2252 2020    COC  @2020,R0
2254 2020
2256 2020    COC  @2020,R0
2258 2020
225A 2020    COC  @2020,R0
225C 2020
225E 2020    COC  @2020,R0
2260 2020
2262 2020    COC  @2020,R0
2264 2020
2266 2020    COC  @2020,R0
2268 2020
226A 2020    COC  @2020,R0
226C 2020
226E 2020    COC  @2020,R0
2270 2020
2272 2020    COC  @2020,R0
2274 2020
2276 2020    COC  @2020,R0
2278 2020
227A 2020    COC  @2020,R0
227C 2020
227E 2020    COC  @3430,R0
2280 3430 40
2282 3730 70 STCR *R0+,12
2284 2030  0 COC  *R0+,R0
2286 3230 20 LDCR *R0+,8
2288 3320 3  LDCR @2020,12
228A 2020
228C 204C  L COC  R12,R1
228E 4920 I  SZC  @2020,@5233(R4)
2290 2020
2292 5233 R3
2294 2C30 ,0 XOP  *R0+,0
2296 3030 00 LDCR *R0+,0
2298 3420 4  STCR @2020,0
229A 2020
229C 2020    COC  @2020,R0
229E 2020
22A0 2020    COC  @2020,R0
22A2 2020
22A4 2020    COC  @2020,R0
22A6 2020
22A8 2020    COC  @2020,R0
22AA 2020
22AC 2020    COC  @2020,R0
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22AE 2020
22B0 2020    COC  @2020,R0
22B2 2020
22B4 2020    COC  @2020,R0
22B6 2020
22B8 2020    COC  @2020,R0
22BA 2020
22BC 2020    COC  @2020,R0
22BE 2020
22C0 2020    COC  @2020,R0
22C2 2020
22C4 2020    COC  @2020,R0
22C6 2020
22C8 2020    COC  @2020,R0
22CA 2020
22CC 2020    COC  @2020,R0
22CE 2020
22D0 3430 40 STCR *R0+,0
22D2 3732 72 STCR *R2+,12
22D4 2030  0 COC  *R0+,R0
22D6 3030 00 LDCR *R0+,0
22D8 3420 4  STCR @2020,0
22DA 2020
22DC 2020    COC  @2020,R0
22DE 2020
22E0 2020    COC  @2020,R0
22E2 2020
22E4 2020    COC  @2020,R0
22E6 2020
22E8 2020    COC  @2020,R0
22EA 2020
22EC 2020    COC  @2020,R0
22EE 2020
22F0 2020    COC  @2020,R0
22F2 2020
22F4 2020    COC  @2020,R0
22F6 2020
22F8 2020    COC  @2020,R0
22FA 2020
22FC 2020    COC  @2020,R0
22FE 2020
2300 2020    COC  @2020,R0
2302 2020
2304 2020    COC  @2020,R0
2306 2020
2308 2020    COC  @2020,R0
230A 2020
230C 2020    COC  @2020,R0
230E 2020
2310 2020    COC  @2020,R0
2312 2020
2314 2020    COC  @2020,R0
2316 2020
2318 2020    COC  @2020,R0
231A 2020
231C 2020    COC  @2020,R0
231E 2020
2320 3430 40 STCR *R0+,0
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2322 3734 74 STCR *R4+,12
2324 2030  0 COC  *R0+,R0
2326 3230 20 LDCR *R0+,8
2328 3120 1  LDCR @2020,4
232A 2020
232C 204C  L COC  R12,R1
232E 4920 I  SZC  @2020,@5231(R4)
2330 2020
2332 5231 R1
2334 2C38 ,8 XOP  *R8+,0
2336 3338 38 LDCR *R8+,12
2338 4120 A  SZC  @2020,R4
233A 2020
233C 2020    COC  @2020,R0
233E 2020
2340 2020    COC  @2020,R0
2342 2020
2344 2020    COC  @2020,R0
2346 2020
2348 2020    COC  @2020,R0
234A 2020
234C 2020    COC  @2020,R0
234E 2020
2350 2020    COC  @2020,R0
2352 2020
2354 2020    COC  @2020,R0
2356 2020
2358 2020    COC  @2020,R0
235A 2020
235C 2020    COC  @2020,R0
235E 2020
2360 2020    COC  @2020,R0
2362 2020
2364 2020    COC  @2020,R0
2366 2020
2368 2020    COC  @2020,R0
236A 2020
236C 2020    COC  @2020,R0
236E 2020
2370 3430 40 STCR *R0+,0
2372 3736 76 STCR *R6+,12
2374 2038  8 COC  *R8+,R0
2376 3338 38 LDCR *R8+,12
2378 4120 A  SZC  @2020,R4
237A 2020
237C 2020    COC  @2020,R0
237E 2020
2380 2020    COC  @2020,R0
2382 2020
2384 2020    COC  @2020,R0
2386 2020
2388 2020    COC  @2020,R0
238A 2020
238C 2020    COC  @2020,R0
238E 2020
2390 2020    COC  @2020,R0
2392 2020
2394 2020    COC  @2020,R0
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2396 2020
2398 2020    COC  @2020,R0
239A 2020
239C 2020    COC  @2020,R0
239E 2020
23A0 2020    COC  @2020,R0
23A2 2020
23A4 2020    COC  @2020,R0
23A6 2020
23A8 2020    COC  @2020,R0
23AA 2020
23AC 2020    COC  @2020,R0
23AE 2020
23B0 2020    COC  @2020,R0
23B2 2020
23B4 2020    COC  @2020,R0
23B6 2020
23B8 2020    COC  @2020,R0
23BA 2020
23BC 2020    COC  @2020,R0
23BE 2020
23C0 3430 40 STCR *R0+,0
23C2 3738 78 STCR *R8+,12
23C4 2044  D COC  R4,R1
23C6 4336 C6 SZC  *R6+,R12
23C8 3420 4  STCR @2064,0
23CA 2064  d
23CC 204D  M COC  R13,R1
23CE 4F56 OV SZC  *R6,*R13+
23D0 4220 B  SZC  @4046,R8
23D2 4046 @F
23D4 4246 BF SZC  R6,R9
23D6 4528 E( SZC  @5234(R8),*R4
23D8 5234 R4
23DA 292C ), XOR  @2A52(R12),R4
23DC 2A52 *R
23DE 312B 1+ LDCR @2020(R11),4
23E0 2020
23E2 2020    COC  @2020,R0
23E4 2020
23E6 2020    COC  @2020,R0
23E8 2020
23EA 2020    COC  @2020,R0
23EC 2020
23EE 2020    COC  @2020,R0
23F0 2020
23F2 2020    COC  @2020,R0
23F4 2020
23F6 2020    COC  @2020,R0
23F8 2020
23FA 2020    COC  @2020,R0
23FC 2020
23FE 2020    COC  @2020,R0
2400 2020
2402 2020    COC  @2020,R0
2404 2020
2406 2020    COC  @2020,R0
2408 2020
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240A 2020    COC  @2020,R0
240C 2020
240E 2020    COC  @3430,R0
2410 3430 40
2412 3741 7A STCR R1,13
2414 2046  F COC  R6,R1
2416 4246 BF SZC  R6,R9
2418 4520 E  SZC  @2020,*R4
241A 2020
241C 2020    COC  @2020,R0
241E 2020
2420 2020    COC  @2020,R0
2422 2020
2424 2020    COC  @2020,R0
2426 2020
2428 2020    COC  @2020,R0
242A 2020
242C 2020    COC  @2020,R0
242E 2020
2430 2020    COC  @2020,R0
2432 2020
2434 2020    COC  @2020,R0
2436 2020
2438 2020    COC  @2020,R0
243A 2020
243C 2020    COC  @2020,R0
243E 2020
2440 2020    COC  @2020,R0
2442 2020
2444 2020    COC  @2020,R0
2446 2020
2448 2020    COC  @2020,R0
244A 2020
244C 2020    COC  @2020,R0
244E 2020
2450 2020    COC  @2020,R0
2452 2020
2454 2020    COC  @2020,R0
2456 2020
2458 2020    COC  @2020,R0
245A 2020
245C 2020    COC  @2020,R0
245E 2020
2460 3430 40 STCR *R0+,0
2462 3743 7C STCR R3,13
2464 2030  0 COC  *R0+,R0
2466 3630 60 STCR *R0+,8
2468 3320 3  LDCR @2020,12
246A 2020
246C 2044  D COC  R4,R1
246E 4543 EC SZC  R3,*R5
2470 2020    COC  @5233,R0
2472 5233 R3
2474 2020    COC  @2020,R0
2476 2020
2478 2020    COC  @2020,R0
247A 2020
247C 2020    COC  @2020,R0
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247E 2020
2480 2020    COC  @2020,R0
2482 2020
2484 2020    COC  @2020,R0
2486 2020
2488 2020    COC  @2020,R0
248A 2020
248C 2020    COC  @2020,R0
248E 2020
2490 2020    COC  @2020,R0
2492 2020
2494 2020    COC  @2020,R0
2496 2020
2498 2020    COC  @2020,R0
249A 2020
249C 2020    COC  @2020,R0
249E 2020
24A0 2020    COC  @2020,R0
24A2 2020
24A4 2020    COC  @2020,R0
24A6 2020
24A8 2020    COC  @2020,R0
24AA 2020
24AC 2020    COC  @2020,R0
24AE 2020
24B0 3430 40 STCR *R0+,0
24B2 3745 7E STCR R5,13
24B4 2031  1 COC  *R1+,R0
24B6 3646 6F STCR R6,9
24B8 4320 C  SZC  @2020,R12
24BA 2020
24BC 204A  J COC  R10,R1
24BE 4E45 NE SZC  R5,*R9+
24C0 2020    COC  @3430,R0
24C2 3430 40
24C4 3738 78 STCR *R8+,12
24C6 2020    COC  @2020,R0
24C8 2020
24CA 2020    COC  @2020,R0
24CC 2020
24CE 2020    COC  @2020,R0
24D0 2020
24D2 2020    COC  @2020,R0
24D4 2020
24D6 2020    COC  @2020,R0
24D8 2020
24DA 2020    COC  @2020,R0
24DC 2020
24DE 2020    COC  @2020,R0
24E0 2020
24E2 2020    COC  @2020,R0
24E4 2020
24E6 2020    COC  @2020,R0
24E8 2020
24EA 2020    COC  @2020,R0
24EC 2020
24EE 2020    COC  @2020,R0
24F0 2020
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24F2 2020    COC  @2020,R0
24F4 2020
24F6 2020    COC  @2020,R0
24F8 2020
24FA 2020    COC  @2020,R0
24FC 2020
24FE 2020    COC  @4428,R0
2500 4428 D(
2502 6973 is S    *R3+,@6173(R5)
2504 6173 as
2506 732C s, SB   @5128(R12),R12
2508 5128 Q(
250A 7569 ui SB   @7420(R9),*R5
250C 7420 t
250E 2020    COC  @2020,R0
2510 2020
2512 2020    COC  @2020,R0
2514 2020
2516 2020    COC  @2020,R0
2518 2020
251A 2020    COC  @2020,R0
251C 2020
251E 2020    COC  @2020,R0
2520 2020
2522 2020    COC  @2020,R0
2524 2020
2526 2020    COC  @2020,R0
2528 2020
252A 2020    COC  @2020,R0
252C 2020
252E 2020    COC  @2020,R0
2530 2020
2532 2020    COC  @2020,R0
2534 2020
2536 2020    COC  @2020,R0
2538 2020
253A 2020    COC  @2020,R0
253C 2020
253E 2020    COC  @2020,R0
2540 2020
2542 2020    COC  @2020,R0
2544 2020
2546 2020    COC  @2020,R0
2548 2020
254A 2020    COC  @2020,R0
254C 2020
254E 2020    COC  @2020,R0
2550 2020
2552 2020    COC  @2020,R0
2554 2020
2556 2020    COC  @2020,R0
2558 2020
255A 2020    COC  @2020,R0
255C 2020
255E 2020    COC  @2020,R0
2560 2020
2562 2020    COC  @2020,R0
2564 2020
```

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2566 2020    COC  @2020,R0
2568 2020
256A 2020    COC  @2020,R0
256C 2020
256E 2020    COC  @2020,R0
2570 2020
2572 2020    COC  @2020,R0
2574 2020
2576 2020    COC  @2020,R0
2578 2020
257A 2020    COC  @2020,R0
257C 2020
257E 2020    COC  @2020,R0
2580 2020
2582 2020    COC  @2020,R0
2584 2020
2586 2020    COC  @2020,R0
2588 2020
258A 2020    COC  @2020,R0
258C 2020
258E 2020    COC  @2020,R0
2590 2020
2592 2020    COC  @2020,R0
2594 2020
2596 2020    COC  @2020,R0
2598 2020
259A 2020    COC  @2020,R0
259C 2020
259E 2020    COC  @4F75,R0
25A0 4F75 Ou
25A2 7470 tp SB   *R0+,*R1
25A4 7574 ut SB   *R4+,*R5
25A6 2066  f COC  @696C(R6),R1
25A8 696C il
25AA 6520 e  S    @5B43,*R4
25AC 5B43 [C
25AE 4F4E ON SZC  R14,*R13+
25B0 534F SO SZCB R15,R13
25B2 4C45 LE SZC  R5,*R1+
25B4 3A5D :] MPY  *R13,R9
25B6 3F20 ?  DIV  @7072,R12
25B8 7072 pr
25BA 696E in S    @7465(R14),@723A(R5)
25BC 7465 te
25BE 723A r:
25C0 2020    COC  @2020,R0
25C2 2020
25C4 2020    COC  @2020,R0
25C6 2020
25C8 2020    COC  @2020,R0
25CA 2020
25CC 2020    COC  @2020,R0
25CE 2020
25D0 2020    COC  @2020,R0
25D2 2020
25D4 2020    COC  @2020,R0
25D6 2020
25D8 2020    COC  @2020,R0
```

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25DA 2020
25DC 2020    COC  @2020,R0
25DE 2020
25E0 2020    COC  @2020,R0
25E2 2020
25E4 2020    COC  @2020,R0
25E6 2020
25E8 2020    COC  @2020,R0
25EA 2020
25EC 2020    COC  @2020,R0
25EE 2020
25F0 2020    COC  @2020,R0
25F2 2020
25F4 2020    COC  @2020,R0
25F6 2020
25F8 2020    COC  @2020,R0
25FA 2020
25FC 2020    COC  @2020,R0
25FE 2020
2600 2020    COC  @2020,R0
2602 2020
2604 2020    COC  @2020,R0
2606 2020
2608 2020    COC  @2020,R0
260A 2020
260C 2020    COC  @2020,R0
260E 2020
2610 2020    COC  @2020,R0
2612 2020
2614 2020    COC  @2020,R0
2616 2020
2618 2020    COC  @2020,R0
261A 2020
261C 2020    COC  @2020,R0
261E 2020
2620 2020    COC  @2020,R0
2622 2020
2624 2020    COC  @2020,R0
2626 2020
2628 2020    COC  @2020,R0
262A 2020
262C 2020    COC  @2020,R0
262E 2020
2630 2020    COC  @2020,R0
2632 2020
2634 2020    COC  @2020,R0
2636 2020
2638 2020    COC  @2020,R0
263A 2020
263C 2020    COC  @2020,R0
263E 2020
2640 5374 St SZCB *R4+,R13
2642 6172 ar S    *R2+,R5
2644 7461 ta SB   @6464(R1),*R1
2646 6464 dd
2648 7265 re SB   @7373(R5),R9
264A 7373 ss
264C 2028  ( COC  @3420(R8),R0
```

```
264E 3420 4
2650 6865 he S    @7820(R5),@6469(R1)
2652 7820 x
2654 6469 di
2656 6769 gi S    @7473(R9),*R13
2658 7473 ts
265A 293F )? XOR  *R15+,R4
265C 2032  2 COC  *R2+,R0
265E 3030 00 LDCR *R0+,0
2660 3020 0  LDCR @2020,0
2662 2020
2664 2020    COC  @2020,R0
2666 2020
2668 2020    COC  @2020,R0
266A 2020
266C 2020    COC  @2020,R0
266E 2020
2670 2020    COC  @2020,R0
2672 2020
2674 2020    COC  @2020,R0
2676 2020
2678 2020    COC  @2020,R0
267A 2020
267C 2020    COC  @2020,R0
267E 2020
2680 2020    COC  @2020,R0
2682 2020
2684 2020    COC  @2020,R0
2686 2020
2688 2020    COC  @2020,R0
268A 2020
268C 2020    COC  @2020,R0
268E 2020
2690 2020    COC  @2020,R0
2692 2020
2694 2020    COC  @2020,R0
2696 2020
2698 2020    COC  @2020,R0
269A 2020
269C 2020    COC  @2020,R0
269E 2020
26A0 2020    COC  @2020,R0
26A2 2020
26A4 2020    COC  @2020,R0
26A6 2020
26A8 2020    COC  @2020,R0
26AA 2020
26AC 2020    COC  @2020,R0
26AE 2020
26B0 2020    COC  @2020,R0
26B2 2020
26B4 2020    COC  @2020,R0
26B6 2020
26B8 2020    COC  @2020,R0
26BA 2020
26BC 2020    COC  @2020,R0
26BE 2020
26C0 2020    COC  @2020,R0
```

```
26C2 2020
26C4 2020    COC  @2020,R0
26C6 2020
26C8 2020    COC  @2020,R0
26CA 2020
26CC 2020    COC  @2020,R0
26CE 2020
26D0 2020    COC  @2020,R0
26D2 2020
26D4 2020    COC  @2020,R0
26D6 2020
26D8 2020    COC  @2020,R0
26DA 2020
26DC 2020    COC  @2020,R0
26DE 2020
26E0 5374 St SZCB *R4+,R13
26E2 6F70 op S    *R0+,*R13+
26E4 6164 ad S    @6472(R4),R5
26E6 6472 dr
26E8 6573 es S    *R3+,*R5
26EA 7320 s  SB   @2028,R12
26EC 2028  (
26EE 3420 4  STCR @6865,0
26F0 6865 he
26F2 7820 x  SB   @6469,@6769
26F4 6469 di
26F6 6769 gi
26F8 7473 ts SB   *R3+,*R1
26FA 293F )? XOR  *R15+,R4
26FC 2034  4 COC  *R4+,R0
26FE 3030 00 LDCR *R0+,0
2700 3020 0  LDCR @2020,0
2702 2020
2704 2020    COC  @2020,R0
2706 2020
2708 2020    COC  @2020,R0
270A 2020
270C 2020    COC  @2020,R0
270E 2020
2710 2020    COC  @2020,R0
2712 2020
2714 2020    COC  @2020,R0
2716 2020
2718 2020    COC  @2020,R0
271A 2020
271C 2020    COC  @2020,R0
271E 2020
2720 2020    COC  @2020,R0
2722 2020
2724 2020    COC  @2020,R0
2726 2020
2728 2020    COC  @2020,R0
272A 2020
272C 2020    COC  @2020,R0
272E 2020
2730 2020    COC  @2020,R0
2732 2020
2734 2020    COC  @2020,R0
```

```
2736 2020
2738 2020    COC  @2020,R0
273A 2020
273C 2020    COC  @2020,R0
273E 2020
2740 2020    COC  @2020,R0
2742 2020
2744 2020    COC  @2020,R0
2746 2020
2748 2020    COC  @2020,R0
274A 2020
274C 2020    COC  @2020,R0
274E 2020
2750 2020    COC  @2020,R0
2752 2020
2754 2020    COC  @2020,R0
2756 2020
2758 2020    COC  @2020,R0
275A 2020
275C 2020    COC  @2020,R0
275E 2020
2760 2020    COC  @2020,R0
2762 2020
2764 2020    COC  @2020,R0
2766 2020
2768 2020    COC  @2020,R0
276A 2020
276C 2020    COC  @2020,R0
276E 2020
2770 2020    COC  @2020,R0
2772 2020
2774 2020    COC  @2020,R0
2776 2020
2778 2020    COC  @2020,R0
277A 2020
277C 2020    COC  @2020,R0
277E 2020
; End of 80-column screen

INTMEM    2780 0DF8    .WORD   ;Start of primary code pool. Init to
0DF8H.
TOPMEM    2782 3E3C >< .WORD   ;End of primary code pool. Init to 3EDFH.
2784 0001    UNDEFINED                    ;Sound data
2786 0001    UNDEFINED
2788 0001    UNDEFINED
278A 0001    UNDEFINED
278C 5975 Yu SZCB *R5+,@596F(R5)
278E 595F Y_
2790 5958 YX SZCB *R8,@5953(R5)
2792 5943 YC
2794 0000    UNDEFINED
2796 0000    UNDEFINED
2798 0000    UNDEFINED
279A 0000    UNDEFINED
279C 0000    UNDEFINED
279E 0000    UNDEFINED
27A0 0000    UNDEFINED
27A2 0000    UNDEFINED
```

```
27A4 0000    UNDEFINED
27A6 0000    UNDEFINED
27A8 0000    UNDEFINED
27AA 0000    UNDEFINED
27AC 0000    UNDEFINED
27AE 0000    UNDEFINED
27B0 0000    UNDEFINED
27B2 0000    UNDEFINED
27B4 0000    UNDEFINED
27B6 0000    UNDEFINED                    ; End of sound data

27B8 0000    UNDEFINED
27BA 0000    UNDEFINED
27BC 0000    UNDEFINED
27BE 0000    UNDEFINED
27C0 0000    UNDEFINED
27C2 0000    UNDEFINED
27C4 0000    UNDEFINED
27C6 0000    UNDEFINED
27C8 0000    UNDEFINED
27CA 0000    UNDEFINED
27CC 0000    UNDEFINED
27CE 0000    UNDEFINED
27D0 0000    UNDEFINED
27D2 0000    UNDEFINED
27D4 0000    UNDEFINED
27D6 0000    UNDEFINED

27D8 0000    UNDEFINED                    ;Sound data
27DA 0000    UNDEFINED
27DC 9FBF    CB   *R15+,*R14+
27DE DFFF    MOVB *R15+,*R15+
27E0 0000    UNDEFINED
27E2 0000    UNDEFINED
27E4 0000    UNDEFINED
27E6 0000    UNDEFINED                    ;End of sound data

27E8 0000    UNDEFINED
27EA 0000    UNDEFINED
27EC 0000    UNDEFINED
27EE 0000    UNDEFINED
27F0 0000    UNDEFINED
27F2 0000    UNDEFINED
27F4 0000    UNDEFINED
27F6 0000    UNDEFINED
27F8 0000    UNDEFINED
27FA 0000    UNDEFINED
27FC 0000    UNDEFINED
27FE 0000    UNDEFINED
2800 0000    UNDEFINED
2802 0000    UNDEFINED
2804 0000    UNDEFINED
2806 0000    UNDEFINED

2808 7FFF    .WORD   ;Last sprite coincidence counter
280A 0000    .WORD            ;Bit map for moving sprites
280C 0000    .WORD            ;Bit map for moving sprites
280E 3348 3H .WORD
```

```
VDPCOPY   2810 81F0    .WORD   ;Copy of VDP registers 1-7 standard mode
          2812 8202    .WORD             ;Used when booting p-code card
          2814 832F  / .WORD
          2816 8400    .WORD
          2818 8518    .WORD
          281A 8600    .WORD
          281C 001F    .WORD

281E CB00    .WORD   ;VCP status copy

2820 FFFF
2822 0000    UNDEFINED
2824 0000    UNDEFINED
2826 0000    UNDEFINED
2828 0000    UNDEFINED
282A 0000    UNDEFINED
282C 0000    UNDEFINED
282E 0000    UNDEFINED
2830 0000    UNDEFINED
2832 0000    UNDEFINED
2834 0000    UNDEFINED
2836 0000    UNDEFINED
2838 0000    UNDEFINED
283A 0000    UNDEFINED
283C 0000    UNDEFINED
283E 0000    UNDEFINED
2840 0000    UNDEFINED
2842 0000    UNDEFINED
2844 0000    UNDEFINED
2846 0000    UNDEFINED
2848 0000    UNDEFINED
284A 0000    UNDEFINED
284C 0000    UNDEFINED
284E 0000    UNDEFINED
2850 0000    UNDEFINED
2852 0000    UNDEFINED
2854 0000    UNDEFINED
2856 0000    UNDEFINED
2858 0000    UNDEFINED
285A 0000    UNDEFINED
285C 0000    UNDEFINED
285E 0000    UNDEFINED
2860 0000    UNDEFINED
2862 0000    UNDEFINED
2864 0000    UNDEFINED
2866 0000    UNDEFINED
2868 0000    UNDEFINED
286A 0000    UNDEFINED
286C 0000    UNDEFINED
286E 0000    UNDEFINED
2870 0000    UNDEFINED
2872 0000    UNDEFINED
2874 0000    UNDEFINED
2876 0000    UNDEFINED
2878 0000    UNDEFINED
287A 0000    UNDEFINED
287C 0000    UNDEFINED
```

```
287E 0000    UNDEFINED
2880 0000    UNDEFINED
2882 0000    UNDEFINED
2884 0000    UNDEFINED
2886 0000    UNDEFINED                    ;Cursor addressing flag
2888 0000    UNDEFINED

288A 0BE0    .WORD   ;Keyboard buffer pointer
288C 0D80    .WORD   ;Keyboard layout area pointer
288E 6BA4 k  .WORD   ;Screen blanking timer
2890 0000    .WORD   ;Stop flag. False=0, true=FFFF
2892 0000    .WORD   ;Alpha lock flag
2894 0000    .WORD   ;Flush flag

;Subsidiary workspace. Used at boot too
2896 0000    .WORD
2898 0000    .WORD
289A 0000    .WORD
289C 0BB9    .WORD
289E 29A8 )  .WORD
28A0 5B02 [  .WORD
28A2 FF40  @ .WORD   ;R6 key buffer mode flag FF40=empty, FF00=something,
FF80=full
28A4 000A    .WORD   ;R7 key buffer store pointer
28A6 000A    .WORD   ;R8 key buffer fetch pointer
28A8 0000    .WORD
28AA 8C02    .WORD   ;R10 VDPWA
28AC 41BE A  .WORD
28AE 2AA6 *  .WORD
28B0 28B6 (  .WORD
28B2 3368 3h .WORD
28B4 D400    .WORD

; Simulated interrupt workspace
28B6 0000    UNDEFINED
28B8 0004    UNDEFINED
28BA 0006    UNDEFINED
28BC 0000    UNDEFINED
28BE 0000    UNDEFINED
28C0 0000    UNDEFINED
28C2 0000    UNDEFINED
28C4 0000    UNDEFINED
28C6 0000    UNDEFINED
28C8 0000    UNDEFINED
28CA 280C (  XOR  R12,R0
28CC 3518 5  STCR *R8,4
28CE 0000    UNDEFINED
28D0 8380    C    R0,R14       ;R13 has been called from Pascal WS
28D2 51F6 Q  SZCB *R6+,R7
28D4 C000    MOV  R0,R0

28D6 0000    UNDEFINED
28D8 0000    UNDEFINED
28DA 0000    UNDEFINED
28DC 0000    UNDEFINED
28DE 0000    UNDEFINED
28E0 0000    UNDEFINED
28E2 0000    UNDEFINED
```

```
28E4 0000    UNDEFINED
28E6 0000    UNDEFINED
28E8 0000    UNDEFINED
28EA 0000    UNDEFINED
28EC 0000    UNDEFINED
28EE 0000    UNDEFINED
28F0 0000    UNDEFINED
28F2 0000    UNDEFINED
28F4 0000    UNDEFINED
28F6 0000    UNDEFINED
28F8 0000    UNDEFINED
28FA 0000    UNDEFINED
28FC 0000    UNDEFINED
28FE 0000    UNDEFINED
2900 0000    UNDEFINED
2902 0000    UNDEFINED
2904 0000    UNDEFINED
2906 0000    UNDEFINED
2908 0000    UNDEFINED
290A 0000    UNDEFINED
290C 0000    UNDEFINED
290E 0000    UNDEFINED
2910 0000    UNDEFINED
2912 0000    UNDEFINED
2914 0000    UNDEFINED
2916 0000    UNDEFINED
2918 0000    UNDEFINED
291A 0000    UNDEFINED
291C 0000    UNDEFINED
291E 0000    UNDEFINED
2920 0000    UNDEFINED
2922 0000    UNDEFINED
2924 0000    UNDEFINED
2926 0000    UNDEFINED
2928 0000    UNDEFINED
292A 0000    UNDEFINED
292C 0000    UNDEFINED
292E 0000    UNDEFINED
2930 0000    UNDEFINED
2932 0000    UNDEFINED
2934 0000    UNDEFINED
2936 0000    UNDEFINED
2938 0000    UNDEFINED
293A 0000    UNDEFINED
293C 0000    UNDEFINED
293E 0000    UNDEFINED
2940 0000    UNDEFINED
2942 0000    UNDEFINED
2944 0000    UNDEFINED
2946 0000    UNDEFINED
2948 0000    UNDEFINED
294A 0000    UNDEFINED
294C 0000    UNDEFINED
294E 0000    UNDEFINED
2950 0000    UNDEFINED
2952 0000    UNDEFINED
2954 0000    UNDEFINED
2956 0000    UNDEFINED
```

```
2958 0000    UNDEFINED
295A 0000    UNDEFINED
295C 0000    UNDEFINED
295E 0000    UNDEFINED

2960 3E90 >  DIV  *R0,R10      ;PCB pointer
2962 0300    LIMI 0            ;Opcode to be inserted into PAB

; Workspace for IO
2964 0000           ;R0
2966 507A Pz        ;R1
2968 C000 ;R2
296A 0000 ;R3
296C 0008 ;R4
296E 83E0           ;R5
2970 3E98 >         ;R6
2972 4174 At        ;R7
2974 0700           ;R8
2976 1300           ;R9
2978 0108           ;R10
297A 0000           ;R11
297C 1F00           ;R12
297E 83C0           ;R13
2980 5B08 [         ;R14
2982 C000           ;R15

2984 0000    UNDEFINED                  ;Interrupt table index

; CRU addresses of cards with interrupt routines.
2986 1400    JHE  2988
2988 1300    JEQ  298A
298A 0000    UNDEFINED                  ;End of list
298C 0000    UNDEFINED
298E 0000    UNDEFINED
2990 0000    UNDEFINED
2992 0000    UNDEFINED
2994 0000    UNDEFINED
2996 0000    UNDEFINED
2998 0000    UNDEFINED
299A 0000    UNDEFINED
299C 0000    UNDEFINED
299E 0000    UNDEFINED
29A0 0000    UNDEFINED
29A2 0000    UNDEFINED

29A4 8300    C    R0,R12
IOASM     29A6 0000    UNDEFINED                  ;True when unit IO
called as assembly procedure
; IO control block
29A8 0002    .WORD            ;Opcode. Used as index in table at 2C68.
29AA 0A00    .WORD            ;Block number
29AC 0001    .WORD            ;Byte count
29AE ACE6    .WORD            ;Buffer address
29B0 0006    .WORD            ;Drive number
29B2 0000    .WORD            ;Control parameter
29B4 0000    .WORD            ;IO result code
29B6 0000    .WORD            ;Physical drive number
; One byte buffers for serial devices
```

```
29B8 000D    .WORD
29BA 2000    .WORD
29BC 0000    .WORD
29BE 0D00    .WORD
29C0 0000    .WORD
29C2 0000    .WORD
29C4 00FF    .WORD
29C6 00FF    .WORD
29C8 FFFF    .WORD
29CA FFFF    .WORD
29CC FFFF    .WORD
29CE FFFF    .WORD
29D0 FFFF    .WORD
29D2 FFFF    .WORD
29D4 FFFF    .WORD
29D6 FF00    .WORD
29D8 0000    .WORD
29DA 0000    .WORD

; PCB pointer table for units
29DC 0000    .WORD   ;Unit #0
29DE 0000    .WORD   ;Unit #1 CONSOLE:
29E0 0000    .WORD   ;Unit #2 KEYBOARD:
29E2 0000    .WORD   ;Unit #3
29E4 3ECD >  .WORD   ;Unit #4 Drive 1
29E6 3ECD >  .WORD   ;Unit #5 Drive 2
29E8 3E90 >  .WORD   ;Unit #6 PRINTER:
29EA 3E53 >S .WORD   ;Unit #7 REMIN:
29EC 3E53 >S .WORD   ;Unit #8 REMOUT:
29EE 3ECD >  .WORD   ;Unit #9 Drive 3
29F0 3ECD >  .WORD   ;Unit #10 Drive 4
29F2 3ECD >  .WORD   ;Unit #11 Drive 5
29F4 3ECD >  .WORD   ;Unit #12 Drive 6
29F6 3ECD >  .WORD   ;Unit #13
29F8 3ECD >  .WORD   ;Unit #14 OS:
29FA 3ECD >  .WORD   ;Unit #15
29FC 3ECD >  .WORD   ;Unit #16
29FE 3ECD >  .WORD   ;Unit #17
2A00 3ECD >  .WORD   ;Unit #18
2A02 3ECD >  .WORD   ;Unit #19
2A04 3ECD >  .WORD   ;Unit #20
2A06 3ECD >  .WORD   ;Unit #21
2A08 3ECD >  .WORD   ;Unit #22
2A0A 3ECD >  .WORD   ;Unit #23
2A0C 3ECD >  .WORD   ;Unit #24
2A0E 3ECD >  .WORD   ;Unit #25
2A10 3ECD >  .WORD   ;Unit #26
2A12 3ECD >  .WORD   ;Unit #27
2A14 3ECD >  .WORD   ;Unit #28
2A16 3ECD >  .WORD   ;Unit #29
2A18 3ECD >  .WORD   ;Unit #30
2A1A 0000    .WORD   ;Unit #31
2A1C 3E3D >= .WORD   ;Unit #32

2A1E 0000    UNDEFINED
2A20 0000    UNDEFINED
2A22 0000    UNDEFINED
; Register save area R0-R7.
```

```
2A24 5036 P6 .WORD
2A26 AD98    .WORD
2A28 4D6E Mn .WORD
2A2A 0004    .WORD
2A2C 008F    .WORD
2A2E AD6A  j .WORD
2A30 AD96    .WORD
2A32 FB52  R .WORD
2A34 4F48 OH .WORD   ;Save area R11

2A36 0000    UNDEFINED
2A38 0000    UNDEFINED
2A3A 0000    UNDEFINED
2A3C 0000    UNDEFINED
2A3E 8300    C    R0,R12
2A40 0001    UNDEFINED
2A42 0000    UNDEFINED
2A44 0000    UNDEFINED
2A46 0019    .WORD   ;I exec error, 25 at start
2A48 FFB4    SOCB *R4+,*R14+
2A4A FDD0    SOCB *R0,*R7+
2A4C FF04    SOCB R4,*R12+
2A4E A45E  ^ A    *R14,*R1
2A50 A3DA    A    *R10,R15
2A52 A406    A    R6,*R0
2A54 FF88    SOCB R8,*R14+
2A56 FF5C  \ SOCB *R12,*R13+
2A58 FF30  0 SOCB *R0+,*R12+
2A5A A432  2 A    *R2+,*R0
2A5C A356  V A    *R6,R13
2A5E FE54  T SOCB *R4,*R9+
2A60 FED8    SOCB *R8,*R11+
2A62 FEAC    SOCB @FE28(R12),*R10+
2A64 FE28  (
2A66 FD78  x SOCB *R8+,*R5+
2A68 FCF4    SOCB *R4+,*R3+
2A6A FCC8    SOCB R8,*R3+
2A6C FC9C    SOCB *R12,*R2+
2A6E FC70  p SOCB *R0+,*R1+
2A70 FC44  D SOCB R4,*R1+
2A72 FC18    SOCB *R8,*R0+
2A74 FBEC    SOCB @FBC0(R12),@FB94(R15)
2A76 FBC0
2A78 FB94
2A7A 0011    UNDEFINED
2A7C 0000    UNDEFINED
2A7E 0000    UNDEFINED
2A80 0000    UNDEFINED
2A82 0000    UNDEFINED
2A84 0000    UNDEFINED
2A86 0000    UNDEFINED
2A88 0000    UNDEFINED
2A8A 0000    UNDEFINED
2A8C 0000    UNDEFINED
2A8E 0000    UNDEFINED
2A90 0000    UNDEFINED
2A92 0000    UNDEFINED
2A94 0000    UNDEFINED
```

```
2A96 0000    UNDEFINED
2A98 0000    UNDEFINED
2A9A 0000    UNDEFINED
2A9C 0000    UNDEFINED
2A9E 0000    UNDEFINED
; Keyscan data for 83C6-83CA
2AA0 0100    UNDEFINED
2AA2 FFFF    SOCB *R15+,*R15+
2AA4 FF83    SOCB R3,*R14+

2AA6 2AA6 *  XOR  @0AD8(R6),R10
2AA8 0AD8

VDPINT    2AAA 28B6 (  WORKSPACE         ;Vector for VDP "interrupt"
          2AAC 32DC 2  ADDRESS

KEYSCAN   2AAE 2896 (  WORKSPACE         ;Vector for keyboard scan and
buffer storage
          2AB0 41BA A  ADDRESS

; Various PME data for task switching and similar
CURSIB    2AB2 AD2A  * .WORD   ;Current SIB pointer
          CURSIB
ERECP     2AB4 AD20    .WORD   ;Current environment record pointer
          E_Rec_P
READYQ    2AB6 A000    .WORD   ;Poiner to TIB at head of ready queue
          READYQ
EVECP     2AB8 A8FE    .WORD   ;Current environment vector pointer
          E_Vec_P
CURTSK    2ABA 2BD0 +  .WORD   ;Current TIB pointer
          CURTSK
          2ABC 0000    .WORD   ;Base for Load processor register
SEGCNAB   2ABE B95C  \ .WORD   ;Current segment constant pool pointer
(absolute).
SEGCNRL   2AC0 0BC6    .WORD   ;Current segment constant pool pointer
(relative).
2AC2 00      ..BYTE
CURPRNO   2AC3 01      ..BYTE  ;Current procedure number (byte)
2AC4 FFFF    .WORD   ;For bit manipulation SRS
2AC6 0002    .WORD
2AC8 0000    .WORD
2ACA 0009    .WORD

IPCSAVE   2ACC B8C2    .WORD   ;P-code IPC save area. Used by PME during
fault recovery

2ACE 0000
2AD0 FFFF    .WORD   ;Bit manipulation SRS (Subrange set)
2AD2 0000    .WORD   ;Bit manipulation SRS
2AD4 0006    .WORD   ;Bit manipulation status SRS
2AD6 0000    .WORD
2AD8 0000    UNDEFINED

RETSAV1   2ADA 5C22 \" .WORD   ;Return address (R11) save area when unit
I/O is called as subroutines.
RETSAV2   2ADC 4D02 M  .WORD   ;Return address (R11) save area

2ADE 0000    UNDEFINED
```

```
CURERE C   2AE0 AD20    .WORD   ;Current environment record
SEGBASE    2AE2 AD96    .WORD   ;Segment base. IPC is segment relative
DICTPNT    2AE4 BDD2    .WORD   ;Routine dictionary pointer
           2AE6 0000    UNDEFINED

PMEFLIP    2AE8 0001    .WORD   ;PME flip indicator
FLTCNT     2AEA 0000    .WORD   ;Fault counter
GLOBPNT    2AEC A91C    .WORD   ;Global data pointer when error
           2AEE 0000
STKMISS    2AF0 000C    .WORD   ;Stack fault data missing

SAVPACK    2AF2 BCE2    .WORD   ;Save area (packing)
2AF4 0007     .WORD   ;SRS
2AF6 0D0D    UNDEFINED
2AF8 5365 Se TEXT 'Segment ________ not found:'
2AFA 676D gm
2AFC 656E en
2AFE 7420 t
2B00 5F5F __
2B02 5F5F __
2B04 5F5F __
2B06 5F5F __
2B08 206E  n
2B0A 6F74 ot
2B0C 2066  f
2B0E 6F75 ou
2B10 6E64 nd
2B12 3A0D :
2B14 5075 Pu TEXT 'Put volume _______ in unit #--'
2B16 7420 t
2B18 766F vo
2B1A 6C75 lu
2B1C 6D65 me
2B1E 205F  _
2B20 5F5F __
2B22 5F5F __
2B24 5F5F __
2B26 2069  i
2B28 6E20 n
2B2A 756E un
2B2C 6974 it
2B2E 2023  # COC
2B30 2D2D --
2B32 0D50  P TEXT 'Press spacebar to continue...'
2B34 7265 re
2B36 7373 ss
2B38 2073  s
2B3A 7061 pa
2B3C 6365 ce
2B3E 6261 ba
2B40 7220 r
2B42 746F to
2B44 2063  c
2B46 6F6E on
2B48 7469 ti
2B4A 6E75 nu
2B4C 652E e.
```

```
2B4E 2E2E ..
2B50 0001    UNDEFINED
2B52 0000    UNDEFINED
2B54 0000    UNDEFINED
2B56 005A  Z UNDEFINED
2B58 0000    UNDEFINED
2B5A 2AF6 *  XOR  *R6+,R11
2B5C 0002    UNDEFINED
2B5E 0000    UNDEFINED
2B60 0000    UNDEFINED
2B62 0001    UNDEFINED
2B64 0000    UNDEFINED
2B66 2B00 +  XOR  R0,R12
2B68 0001    UNDEFINED
2B6A 001A    UNDEFINED
2B6C 000A    UNDEFINED
2B6E 0000    .WORD   ;Ioresult. Start of SYSCOM
2B70 0000    UNDEFINED
2B72 0004    UNDEFINED
2B74 0000    UNDEFINED
2B76 0000    UNDEFINED
2B78 0001    UNDEFINED
2B7A 0000    UNDEFINED
2B7C 0000    UNDEFINED         ;Real_Sem faulthandler start
2B7E A10E    A    R14,R4

FATIB     2B80 2BD0 +  XOR  *R0,R15      ;Fault_TIB
FAEREC    2B82 A432  2 A    *R2+,*R0     ;Fault_EREC
FAWORD    2B84 0000    UNDEFINED                  ;Fault_Words
FATYPE    2B86 0080    UNDEFINED                  ;Fault_type 80H
segment fault, 81H stack fault

2B88 AD74  t .WORD   ;SIB pointer
2B8A 0001    UNDEFINED
2B8C 0000    UNDEFINED
2B8E 0000    UNDEFINED
2B90 0000    UNDEFINED
2B92 0000    UNDEFINED
2B94 0000    UNDEFINED
2B96 0000    UNDEFINED
2B98 0000    UNDEFINED
2B9A 0000    UNDEFINED
2B9C 0000    UNDEFINED
2B9E 8483    C    R3,*R2
2BA0 F2F1    SOCB *R1+,R11
2BA2 8493    C    *R3,*R2
TIMELOW   2BA4 C085    .WORD   ;System clock (low)
TIMEHI    2BA6 0000    .WORD   ;System clock (high)
2BA8 020D    LI   R13,0000
2BAA 0000
2BAC 8200    C    R0,R8
2BAE 8D8E    C    R14,*R6+
2BB0 8B89    C    R9,@0008(R14)
2BB2 0008
2BB4 0000    UNDEFINED
2BB6 0000    UNDEFINED
2BB8 0018    .WORD   ;Screen height (24)
2BBA 0028  ( .WORD   ;Screen width (40)
```

```
2BBC 8A    ..BYTE              ;Code for arrow down
2BBD 8B    ..BYTE              ;Code for arrow up
2BBE 89    ..BYTE              ;Code for arrow right
2BBF 88    ..BYTE              ;Code for arrow left
2BC0 87    ..BYTE              ;Code for arrow FLUSH
2BC1 03    ..BYTE              ;Code for arrow etx
2BC2 8E    .BYTE               ;Code for arrow STOP
2BC3 11    .BYTE               ;Code for arrow BREAK
2BC4 3F    .BYTE               ;Code for arrow editor unknown character?
2BC5 08 ?  .BYTE               ;Code for arrow backspace
2BC6 1B    .BYTE               ;Code for arrow escape
2BC7 8F    .BYTE               ;Code for arrow line delete
2BC8 03    .BYTE               ;Code for arrow etx
2BC9 00    .BYTE               ;?
2BCA 8C    .BYTE               ;Code for alpha lock
2BCB 00    .BYTE               ;Character mask 7 or 8 bit
2BCC 0000    UNDEFINED
2BCE 0050  P UNDEFINED
2BD0 0000    .WORD   ;TIB for main task
2BD2 0080    .WORD
2BD4 D8D4    .WORD
2BD6 0000    .WORD
2BD8 F840  @ .WORD
2BDA F8F6    .WORD
2BDC 0000    .WORD
2BDE 021D    .WORD
2BE0 AD20    .WORD
2BE2 0008    .WORD
2BE4 0000    .WORD
2BE6 0000    .WORD
2BE8 0001    .WORD
2BEA FB80    .WORD

2BEC 0000    .WORD   ;Table with indexes into table at 2C68 device driver
entries
2BEE 0002    UNDEFINED
2BF0 0004    UNDEFINED
2BF2 0006    UNDEFINED

;User defined IO devices
2BF4 007F  ⌂ .WORD   ;Highest device number
2BF6 0000    .WORD   ;Pointer to unit pointer table
2BF8 0000    .WORD   ;Pointer to PCB pointer table
2BFA 0000    .WORD   ;Pointer to buffer address table
; Table with corresponding drive number for blocked units.
2BFC 00      .BYTE
2BFD 00      .BYTE
2BFE 00      .BYTE
2BFF 00      .BYTE
2C00 01      .BYTE
2C01 02        .BYTE
2C02 00      .BYTE
2C03 00      .BYTE
2C04 00      .BYTE
2C05 03      .BYTE
2C06 04      .BYTE
2C07 05      .BYTE
2C08 06      .BYTE
```

```
2C09 10      .BYTE
2C0A 11      .BYTE
2C0B 12      .BYTE
2C0C 13      .BYTE
2C0D 14      .BYTE
2C0E 15      .BYTE
2C0F 16      .BYTE
2C10 17      .BYTE
2C11 18      .BYTE
2C12 19      .BYTE
2C13 1A      .BYTE
2C14 1B      .BYTE
2C15 1C      .BYTE
2C16 1D      .BYTE
2C17 1E      .BYTE
2C18 1F      .BYTE
2C19 20      .BYTE
2C1A 21   !  .BYTE
2C1B 00      .BYTE
2C1C 00      .BYTE
2C1D 00      .BYTE

2C1E 0000    UNDEFINED

; Unit pointer table
2C20 0000    .WORD #0 (reserved)
2C22 2C68 ,h .WORD #1 CONSOLE
2C24 2C68 ,h .WORD #2 SYSTERM
2C26 0000    .WORD #3 (Graphic) Reserved for Terak microcomputers.
2C28 2C70 ,p .WORD #4 Diskette name
2C2A 2C70 ,p .WORD #5 Diskette name
2C2C 2C80 ,  .WORD #6 (reserved)
2C2E 2C80 ,  .WORD #7 (reserved)
2C30 2C80 ,  .WORD #8 (reserved)
2C32 2C70 ,p .WORD #9 Diskette name
2C34 2C70 ,p .WORD #10 Diskette name
2C36 0000    .WORD #11 (Diskette name)
2C38 0000    .WORD #12 (Diskette name)
2C3A 2C70 ,p .WORD #13 Assigned to unit loaded from tape
2C3C 2C70 ,p .WORD #14 OS
2C3E 2C70 ,p .WORD #15
2C40 2C70 ,p .WORD #16
2C42 2C70 ,p .WORD #17
2C44 2C70 ,p .WORD #18
2C46 2C70 ,p .WORD #19
2C48 2C70 ,p .WORD #20
2C4A 2C70 ,p .WORD #21
2C4C 2C70 ,p .WORD #22
2C4E 2C70 ,p .WORD #23
2C50 2C70 ,p .WORD #24
2C52 2C70 ,p .WORD #25
2C54 2C70 ,p .WORD #26
2C56 2C70 ,p .WORD #27
2C58 2C70 ,p .WORD #28
2C5A 2C70 ,p .WORD #29
2C5C 2C70 ,p .WORD #30
2C5E 2C88 ,  .WORD #31 TAPE
2C60 2C78 ,x .WORD #32 TP
```

```
2C62 0000    UNDEFINED
2C64 0000    UNDEFINED
2C66 0000    UNDEFINED

; Addresses in secondary ROM page for different unit IO operations
; Console
2C68 538E S  .WORD Read
2C6A 547E T~ .WORD Write
2C6C 5270 Rp .WORD Clear
2C6E 5360 S` .WORD Status
; Blocked devices
2C70 5796 W  .WORD Read
2C72 5740 W@ .WORD Write
2C74 568C V  .WORD Clear
2C76 5652 VR .WORD Status
;Thermal printer
2C78 0000    .WORD Read
2C7A 59FA Y  .WORD Write
2C7C 59CA Y  .WORD Clear
2C7E 0000    .WORD Status
; RS232 card
2C80 5ABE Z  .WORD Read
2C82 5AD8 Z  .WORD Write
2C84 5AB4 Z  .WORD Clear
2C86 0000    .WORD Status
; Tape
2C88 5B68 [h .WORD Read
2C8A 5B22 [" .WORD Write
2C8C 5B10 [  .WORD Clear
2C8E 0000    .WORD Status

2C90 80A0    C    @C0E0,R2
2C92 C0E0
2C94 90B0    CB   *R0+,R2
2C96 D0F0    MOVB *R0+,R3
2C98 9FBF    CB   *R15+,*R14+
2C9A DFFF    MOVB *R15+,*R15+
2C9C 01C3    UNDEFINED
2C9E 0006    UNDEFINED
2CA0 0064  d UNDEFINED
2CA2 C000    MOV  R0,R0
2CA4 8081    C    R1,R2
2CA6 C200    MOV  R0,R8
2CA8 C100    MOV  R0,R4
2CAA A000    A    R0,R0
2CAC AA00    .WORD DSR validation code
2CAE 0080
2CB0 81F0    .WORD VDP copy R1
2CB2 8202    .WORD VDP copy R2
2CB4 832F  / .WORD VDP copy R3
2CB6 8400    .WORD VDP copy R4
2CB8 8518    .WORD VDP copy R5
2CBA 8600    .WORD VDP copy R6
2CBC 001F    .WORD VDP copy R7
2CBE 2000    .WORD Pointer 80 column screen
2CC0 0017    .WORD Current line
2CC2 0000    .WORD Current column
2CC4 2730 '0 .WORD Current 80 column address
```

```
2CC6 0028  ( .WORD Screen width
2CC8 0018    .WORD Screen height
2CCA 0000    .WORD Current screen window
2CCC 0730  0 .WORD Number of bytes in 23 lines (for scroll)
2CCE 0040  @ UNDEFINED
2CD0 0200    .WORD Autorepeat counter
2CD2 0006    .WORD Keyscan limit (key buffering isn't done on every
interrupt)
2CD4 0001    .WORD Kyescan limit counter
2CD6 0002    UNDEFINED
2CD8 0004    UNDEFINED
2CDA 0009    .WORD IO error code "no such volume"
2CDC 0008    UNDEFINED
2CDE 17FE    JNC  2CDC
2CE0 0802    SRA  R2,0
2CE2 0000    UNDEFINED
2CE4 FA00    SOCB R0,@0100(R8)
2CE6 0100    .WORD Physical disk sector size
2CE8 0200    LI   R0,3F00
2CEA 3F00 ?  .WORD Address to disk IO buffer in VDP RAM
2CEC 1100    JLT  2CEE
; Disk sector access error code conversion table
2CEE 06      .BYTE
2CEF 09      .BYTE
2CF0 07      .BYTE
2CF1 11      .BYTE
2CF2 34      .BYTE
2CF3 10      .BYTE

2CF4 28D8 (  XOR  *R8,R3
2CF6 0300    LIMI 0
2CF8 0200
; Table...
2CFA 0000    UNDEFINED
2CFC 8800    C    R0,@5BFC
2CFE 5BFC [
2D00 9800    CB   R0,@9804
2D02 9804
2D04 9808    CB   R8,@980C
2D06 980C
2D08 9810    CB   *R0,@9814
2D0A 9814
2D0C 9818    CB   *R8,@981C
2D0E 981C
2D10 9820    CB   @9824,@9828
2D12 9824  $
2D14 9828  (
2D16 982C  , CB   @9830(R12),@9834
2D18 9830  0
2D1A 9834  4
2D1C 9838  8 CB   *R8+,@9810
2D1E 9810
; Table...
2D20 0000    UNDEFINED
2D22 FFFF    SOCB *R15+,*R15+
2D24 0001    UNDEFINED
2D26 0001    UNDEFINED
2D28 0001    UNDEFINED
```

```
2D2A 0001    UNDEFINED
2D2C 0001    UNDEFINED
2D2E 0001    UNDEFINED
2D30 0001    UNDEFINED
2D32 0001    UNDEFINED
2D34 0001    UNDEFINED
2D36 0001    UNDEFINED
2D38 0001    UNDEFINED
2D3A 0001    UNDEFINED
2D3C 0001    UNDEFINED
2D3E 0001    UNDEFINED
2D40 0001    UNDEFINED
2D42 0001    UNDEFINED
2D44 0001    UNDEFINED
2D46 3298 2  LDCR *R8,10
2D48 3250 2P LDCR *R0,9
2D4A 3250 2P LDCR *R0,9
2D4C 3250 2P LDCR *R0,9
2D4E 3250 2P LDCR *R0,9
2D50 3250 2P LDCR *R0,9
2D52 3250 2P LDCR *R0,9
2D54 3250 2P LDCR *R0,9
2D56 3250 2P LDCR *R0,9
2D58 3250 2P LDCR *R0,9
2D5A 3250 2P LDCR *R0,9
2D5C 3250 2P LDCR *R0,9
2D5E 3250 2P LDCR *R0,9
2D60 3250 2P LDCR *R0,9
2D62 3250 2P LDCR *R0,9
2D64 3250 2P LDCR *R0,9
2D66 3250 2P LDCR *R0,9
2D68 3250 2P LDCR *R0,9
2D6A 3250 2P LDCR *R0,9
; Some table
2D6C 0000    UNDEFINED
2D6E 0000    UNDEFINED
2D70 0000    UNDEFINED
2D72 6000 `  S    R0,R0
2D74 6000 `  S    R0,R0
2D76 6000 `  S    R0,R0
2D78 6000 `  S    R0,R0
2D7A 6000 `  S    R0,R0
2D7C 6000 `  S    R0,R0
2D7E 6000 `  S    R0,R0
2D80 6000 `  S    R0,R0
2D82 6000 `  S    R0,R0
2D84 6000 `  S    R0,R0
2D86 6000 `  S    R0,R0
2D88 6000 `  S    R0,R0
2D8A 6000 `  S    R0,R0
2D8C 6000 `  S    R0,R0
2D8E 6000 `  S    R0,R0
2D90 6000 `  S    R0,R0
; Bit masks for general use
2D92 0001    UNDEFINED
2D94 0002    UNDEFINED
2D96 0004    UNDEFINED
2D98 0008    UNDEFINED
```

```
2D9A 0010    UNDEFINED
2D9C 0020    UNDEFINED
2D9E 0040  @ UNDEFINED
2DA0 0080    UNDEFINED
2DA2 0100    UNDEFINED
2DA4 0200    LI   R0,0400
2DA6 0400
2DA8 0800    SRA  R0,0
2DAA 1000    JMP  2DAC
2DAC 2000    COC  R0,R0
2DAE 4000 @  SZC  R0,R0
2DB0 8000    C    R0,R0

; Address table for p-code interpreter. Opcode is a word index.
OPCODE     2DB2 4968 .WORD SLDC ;SLDC0 Short load constant
           2DB4 4968 .WORD SLDC
           2DB6 4968 .WORD SLDC
           2DB8 4968 .WORD SLDC
           2DBA 4968 .WORD SLDC
           2DBC 4968 .WORD SLDC
           2DBE 4968 .WORD SLDC
           2DC0 4968 .WORD SLDC
           2DC2 4968 .WORD SLDC
           2DC4 4968 .WORD SLDC
           2DC6 4968 .WORD SLDC
           2DC8 4968 .WORD SLDC
           2DCA 4968 .WORD SLDC
           2DCC 4968 .WORD SLDC
           2DCE 4968 .WORD SLDC
           2DD0 4968 .WORD SLDC
           2DD2 4968 .WORD SLDC
           2DD4 4968 .WORD SLDC
           2DD6 4968 .WORD SLDC
           2DD8 4968 .WORD SLDC
           2DDA 4968 .WORD SLDC
           2DDC 4968 .WORD SLDC
           2DDE 4968 .WORD SLDC
           2DE0 4968 .WORD SLDC
           2DE2 4968 .WORD SLDC
           2DE4 4968 .WORD SLDC
           2DE6 4968 .WORD SLDC
           2DE8 4968 .WORD SLDC
           2DEA 4968 .WORD SLDC
           2DEC 4968 .WORD SLDC
           2DEE 4968 .WORD SLDC
           2DF0 4968 .WORD SLDC ;SLDC31
           2DF2 4972 .WORD SLDL ;SLDL1 Short load local
           2DF4 4972 .WORD SLDL
           2DF6 4972 .WORD SLDL
           2DF8 4972 .WORD SLDL
           2DFA 4972 .WORD SLDL
           2DFC 4972 .WORD SLDL
           2DFE 4972 .WORD SLDL
           2E00 4972 .WORD SLDL
           2E02 4972 .WORD SLDL
           2E04 4972 .WORD SLDL
           2E06 4972 .WORD SLDL
           2E08 4972 .WORD SLDL
```

```
2E0A 4972 .WORD SLDL
2E0C 4972 .WORD SLDL
2E0E 4972 .WORD SLDL
2E10 4972 .WORD SLDL ;SLDL16
2E12 497E .WORD SLDO ;SLDO1 Short load global
2E14 497E .WORD SLDO
2E16 497E .WORD SLDO
2E18 497E .WORD SLDO
2E1A 497E .WORD SLDO
2E1C 497E .WORD SLDO
2E1E 497E .WORD SLDO
2E20 497E .WORD SLDO
2E22 497E .WORD SLDO
2E24 497E .WORD SLDO
2E26 497E .WORD SLDO
2E28 497E .WORD SLDO
2E2A 497E .WORD SLDO
2E2C 497E .WORD SLDO
2E2E 497E .WORD SLDO
2E30 497E .WORD SLDO ;SLDO1
2E32 48CC .WORD UNUSED
2E34 48CC .WORD UNUSED
2E36 48CC .WORD UNUSED
2E38 48CC .WORD UNUSED
2E3A 48CC .WORD UNUSED
2E3C 48CC .WORD UNUSED
2E3E 48CC .WORD UNUSED
2E40 48CC .WORD UNUSED
2E42 48CC .WORD UNUSED
2E44 48CC .WORD UNUSED
2E46 48CC .WORD UNUSED
2E48 48CC .WORD UNUSED
2E4A 48CC .WORD UNUSED
2E4C 48CC .WORD UNUSED
2E4E 48CC .WORD UNUSED
2E50 48CC .WORD UNUSED
2E52 48CC .WORD UNUSED
2E54 48CC .WORD UNUSED
2E56 48CC .WORD UNUSED
2E58 48CC .WORD UNUSED
2E5A 48CC .WORD UNUSED
2E5C 48CC .WORD UNUSED
2E5E 48CC .WORD UNUSED
2E60 48CC .WORD UNUSED
2E62 48CC .WORD UNUSED
2E64 48CC .WORD UNUSED
2E66 48CC .WORD UNUSED
2E68 48CC .WORD UNUSED
2E6A 48CC .WORD UNUSED
2E6C 48CC .WORD UNUSED
2E6E 48CC .WORD UNUSED
2E70 48CC .WORD UNUSED
2E72 4A08 .WORD SLLA ;SLLA1 Short load local address
2E74 4A08 .WORD SLLA
2E76 4A08 .WORD SLLA
2E78 4A08 .WORD SLLA
2E7A 4A08 .WORD SLLA
2E7C 4A08 .WORD SLLA
```

```
2E7E 4A08 .WORD SLLA
2E80 4A08 .WORD SLLA ;SLLA8
2E82 4A22 .WORD SSTL ;SSTL1 Short store local
2E84 4A22 .WORD SSTL
2E86 4A22 .WORD SSTL
2E88 4A22 .WORD SSTL
2E8A 4A22 .WORD SSTL
2E8C 4A22 .WORD SSTL
2E8E 4A22 .WORD SSTL
2E90 4A22 .WORD SSTL ;SSTL8
2E92 4D2A .WORD SCXG1          ;SCXG1 Short call external
global
2E94 4D36 .WORD SCXG
2E96 4D36 .WORD SCXG
2E98 4D36 .WORD SCXG
2E9A 4D36 .WORD SCXG
2E9C 4D36 .WORD SCXG
2E9E 4D36 .WORD SCXG
2EA0 4D36 .WORD SCXG ;SCXG8
2EA2 498A .WORD SIND ;SIND0 Short index
2EA4 498A .WORD SIND
2EA6 498A .WORD SIND
2EA8 498A .WORD SIND
2EAA 498A .WORD SIND
2EAC 498A .WORD SIND
2EAE 498A .WORD SIND
2EB0 498A .WORD SIND ;SIND7
2EB2 4B52 .WORD LDCB ;Load constant byte
2EB4 49E2 .WORD LDCI ;Load constant integer
2EB6 4C30 .WORD LCO  ;Load constant offset
2EB8 4BEC .WORD LDC  ;Load constant
2EBA 49FC .WORD LLA  ;Load local address
2EBC 4A30 .WORD LDO  ;Load global
2EBE 4A3E .WORD LAO  ;Load global address
2EC0 49EE .WORD LDL  ;Load local
2EC2 4A80 .WORD LDA  ;Load intermediate address
2EC4 4A6A .WORD LOD  ;Load intermediate
2EC6 48FE .WORD UJP  ;Unconditional jump
2EC8 4914 .WORD JPL  ;Unconditional jump long
2ECA 524C .WORD MPI  ;Multiply integer
2ECC 521E .WORD DVI  ;Divide integer
2ECE 4B28 .WORD STM  ;Store multiple
2ED0 520A .WORD MODI ;Modulo integer
2ED2 4CDC .WORD CLP  ;Call local procedure
2ED4 4CE4 .WORD CGP  ;Call global procedure
2ED6 4CC8 .WORD CIP  ;Call intermediate procedure
2ED8 4D22 .WORD CXL  ;Call external local
2EDA 4D4E .WORD CXG  ;Call external global
2EDC 4D06 .WORD CXI  ;Call external intermediate
2EDE 500E .WORD RPU  ;Return from procedure
2EE0 4D80 .WORD CFP  ;Call formal procedure
2EE2 49DA .WORD LDCN ;Load constant nil
2EE4 4B3E .WORD LSL  ;Load static link
2EE6 4ACA .WORD LDE  ;Load extended
2EE8 4AD8 .WORD LAE  ;Load extended address
2EEA 4996 .WORD NOP  ;No operation
2EEC 4C6A .WORD LPR  ;Load processor register
2EEE 48D4 .WORD BPT  ;Break point
```

```
            2EF0 51D8 .WORD BNOT ;Boolean not
            2EF2 51CC .WORD LOR  ;Logical or
            2EF4 51C4 .WORD LAND ;Logical and
            2EF6 51F8 .WORD ADI  ;Add integer
            2EF8 5204 .WORD SBI  ;Subtract integer
            2EFA 4A1A .WORD STL  ;Store local
            2EFC 4A4E .WORD SRO  ;Store global
            2EFE 4A98 .WORD STR  ;Store intermediate
            2F00 4998 .WORD LDB            ;Load byte
            2F02 48B6 .WORD NAT  ;Enter native code
            2F04 48C4 .WORD NAT_INFO       ;Native code information
            2F06 48CC .WORD UNUSED         ;Reserved
            2F08 50CE .WORD CAP  ;Copy array parameter
            2F0A 510A .WORD CSP  ;Copy string parameter
            2F0C 4A5A .WORD SLOD1          ;Short load intermediate
            2F0E 4A62 .WORD SLOD2          ;Short load intermediate
            2F10 48CC .WORD UNUSED         ;Unused
            2F12 52E6 .WORD EQUI           ;Equal integer
            2F14 52E6 .WORD EQUI ;Not equal integer
            2F16 52E6 .WORD EQUI           ;Less than or equal integer
            2F18 52E6 .WORD EQUI ;Greater than or equal integer
            2F1A 52E6 .WORD EQUI           ;Less than or equal unsigned
            2F1C 52E6 .WORD EQUI ;Greater than or equal unsigned
            2F1E 5638 .WORD EQPWR          ;Equal set
            2F20 5638 .WORD EQPWR          ;Less than or equal set
            2F22 5638 .WORD EQPWR          ;Greater than or equal set
            2F24 52F4 .WORD EQBYT          ;Equal byte array
            2F26 52F4 .WORD EQBYT          ;Less than or equal byte array
            2F28 52F4 .WORD EQBYT          ;Greater than or equal byte
array
            2F2A 5470 .WORD SRS  ;Subrange set
            2F2C 4C3E .WORD SWAP ;Swap
            2F2E 5F34 .WORD TNC  ;Truncate real
            2F30 5F04 .WORD RND  ;Round real
            2F32 5EB0 .WORD ADR  ;Add real
            2F34 5EB8 .WORD SBR  ;Subtract real
            2F36 5EC8 .WORD MPR  ;Multiply real
            2F38 5EC0 .WORD DVR  ;Divide real
            2F3A 4AAC .WORD STO  ;Store
            2F3C 4B5A .WORD MOV  ;Move
            2F3E 4C56 .WORD DUPR           ;Duplicate real
            2F40 540C .WORD ADJ  ;Adjust set
            2F42 49A6 .WORD STB  ;Store byte
            2F44 5150 .WORD LDP  ;Load packed
            2F46 5170 .WORD STP  ;Store packed
            2F48 5256 .WORD CHK  ;Check subrange bounds
            2F4A 5F24 .WORD FLT  ;Float
            2F4C 5EDC .WORD EQREAL         ;Equal real
            2F4E 5EDC .WORD EQREAL         ;Less than or equal real
            2F50 5EDC .WORD EQREAL         ;Greater than or equal real
            2F52 4AF4 .WORD LDM  ;Load multiple
            2F54 4C88 .WORD SPR            ;Store processor reqister
            2F56 4904 .WORD EFJ  ;Equal false jump
            2F58 490C .WORD NFJ  ;Not equal false jump
            2F5A 48E6 .WORD FJP  ;False jump
            2F5C 4932 .WORD FJPL           ;False long jump
            2F5E 493C .WORD XJP  ;Case jump
            2F60 49CC .WORD IXA  ;Index array
```

```
           2F62 51AA .WORD IXP  ;Index packed array
           2F64 4AE8 .WORD STE  ;Store extended
           2F66 554E .WORD INN  ;Set membership
           2F68 55EE .WORD UNI  ;Set union
           2F6A 55FE .WORD INT  ;Set intersection
           2F6C 5622 .WORD DIF  ;Set difference
           2F6E 56B4 .WORD SIGNAL         ;Signal
           2F70 5720 .WORD WAIT ;Wait
           2F72 51F2 .WORD ABI  ;Absolute value integer
           2F74 51FE .WORD NGI  ;Negate integer
           2F76 4C4C .WORD DUPI ;Duplicate one word
           2F78 5ED0 .WORD ABR  ;Absolute value real
           2F7A 5ED6 .WORD NGR  ;Negate real
           2F7C 51D2 .WORD LNOT ;Logical not
           2F7E 49BA .WORD IND  ;Index
           2F80 49C4 .WORD INC            ;Increment
           2F82 533C .WORD EQSTR          ;Equal string
           2F84 533C .WORD EQSTR          ;Less than or equal string
           2F86 533C .WORD EQSTR          ;Greater than or equal string
           2F88 53A0 .WORD ASTR ;Assign string
           2F8A 53EC .WORD CSTR           ;Check string index
           2F8C 51E6 .WORD INCI ;Increment integer
           2F8E 51EC .WORD DECI ;Decrement integer
           2F90 4CAA .WORD SCPI1          ;Short call intermediate
procedure
           2F92 4CB2 .WORD SCPI2          ;Short call intermediate
procedure
           2F94 48DC .WORD TJP  ;True jump
           2F96 4BCC .WORD LDCRL          ;Load constant real
           2F98 5E8A .WORD LDRL ;Load real
           2F9A 5E9E .WORD STRL ;Store real
           2F9C 5E88 .WORD UNUSED1        ;Unused, does nothing
           2F9E 5E88 .WORD UNUSED1        ;Unused, does nothing
           2FA0 48CC .WORD UNUSED         ;Unused, fails
           2FA2 48CC .WORD UNUSED         ;Unused
           2FA4 48CC .WORD UNUSED         ;Unused
           2FA6 48CC .WORD UNUSED         ;Reserved
           2FA8 48CC .WORD UNUSED         ;Reserved
           2FAA 48CC .WORD UNUSED         ;Reserved
           2FAC 48CC .WORD UNUSED         ;Reserved
           2FAE 48CC .WORD UNUSED         ;Reserved
           2FB0 4996 .WORD RESERVE6       ;Reserved, does nothing

; Address table for Kernel procedures
KERNPROC   2FB2

           2FB8 354C RELOCSEG

           2FCE 5856 MOVELEFT
           2FD0 5884 MOVERIGHT
           2FD4 34B6 UNITREAD
           2FD6 34CE UNITWRITE
           2FD8 58D6 TIME
...
           2FFC                ; End of Kernel procedure table

; Kernel procedure FLIPSEG.BYTES (?) Does a software reset (?)
2FFE 020C    LI   R12,1F00     ; P-code card
```

```
3000 1F00
3002 1E00    SBZ  0
3004 04CC    CLR  R12                    ; Reset sequence
3006 1D00    SBO  0
3008 1E0F    SBZ  15
300A 1E00    SBZ  0
300C 04E0    CLR  @83C2
300E 83C2
3010 04E0    CLR  @83C4        ; User interrupt link
3012 83C4
3014 0420    BLWP @0000
3016 0000

; Kernel procedure QUIET
3018 06A0    BL   @352C
301A 352C 5,
301C 06A0    BL   @52B2
301E 52B2 R
3020 06A0    BL   @353C
3022 353C 5<
3024 045C  \ B    *R12

; Table with pointers to some of the keywords in the next table. Each
pointer is followed by a value.
3026 3098 0  LDCR *R8,2                  ; AND 2
3028 0002    UNDEFINED
302A 30AC 0  LDCR @0001(R12),2 ; BEGIN 1
302C 0001
302E 30B6 0  LDCR *R6+,2       ; CASE 2
3030 0002    UNDEFINED
3032 30CA 0  LDCR R10,3                  ; DEFINITI 4
3034 0004    UNDEFINED
3036 30F2 0  LDCR *R2+,3       ; ELSE 3
3038 0003    UNDEFINED
303A 3110 1  LDCR *R0,4                  ; FOR 4
303C 0004    UNDEFINED
303E 3138 18 LDCR *R8+,4       ; GOTO 1
3040 0001    UNDEFINED
3042 308E 0  LDCR R14,2                  ; @@@@@@@@ 1
3044 0001    UNDEFINED
3046 3142 1B LDCR R2,5                   ; IF 4
3048 0004    UNDEFINED
304A 308E 0  LDCR R14,2                  ; @@@@@@@@ 1
304C 0001    UNDEFINED
304E 308E 0  LDCR R14,2                  ; @@@@@@@@ 1
3050 0001    UNDEFINED
3052 316A 1j LDCR @0001(R10),5 ; LABEL 1
3054 0001
3056 3174 1t LDCR *R4+,5       ; MOD 1
3058 0001    UNDEFINED
305A 317E 1~ LDCR *R14+,5      ; NOT 1
305C 0001    UNDEFINED
305E 3188 1  LDCR R8,6                   ; OF 2
3060 0002    UNDEFINED
3062 319C 1  LDCR *R12,6       ; PACKED 4
3064 0004    UNDEFINED
3066 308E 0  LDCR R14,2                  ; @@@@@@@@ 1
3068 0001    UNDEFINED
```

```
306A 31C4 1  LDCR R4,7                   ; RECORD 2
306C 0002    UNDEFINED
306E 31D8 1  LDCR *R8,7                  ; SEPARATE 3
3070 0003    UNDEFINED
3072 31F6 1  LDCR *R6+,7       ; THEN 3
3074 0003    UNDEFINED
3076 3214 2  LDCR *R4,8                  ; UNIT 3
3078 0003    UNDEFINED
307A 3232 22 LDCR *R2+,8       ; VAR 1
307C 0001    UNDEFINED
307E 323C 2< LDCR *R12+,8      ; WHILE 2
3080 0002    UNDEFINED
3082 308E 0  LDCR R14,2                  ; @@@@@@@@ 1
3084 0001    UNDEFINED
3086 308E 0  LDCR R14,2                  ; @@@@@@@@ 1
3088 0001    UNDEFINED
308A 308E 0  LDCR R14,2                  ; @@@@@@@@ 1
308C 0001    UNDEFINED

; Data for IDSEARCH
308E 4040 @@ SZC  R0,R1                  ; @@@@@@@@ 000F
3090 4040 @@ SZC  R0,R1
3092 4040 @@ SZC  R0,R1
3094 4040 @@ SZC  R0,R1
3096 000F    UNDEFINED
3098 414E AN SZC  R14,R5       ; AND 2702
309A 4420 D  SZC  @2020,*R0
309C 2020
309E 2020    COC  @2702,R0
30A0 2702 '
30A2 4152 AR SZC  *R2,R5       ; ARRAY 2C0F
30A4 5241 RA SZCB R1,R9
30A6 5920 Y  SZCB @2020,@2C0F(R4)
30A8 2020
30AA 2C0F ,
30AC 4245 BE SZC  R5,R9                  ; BEGIN 130F
30AE 4749 GI SZC  R9,*R13
30B0 4E20 N  SZC  @2020,*R8+
30B2 2020
30B4 130F    JEQ  30D4
30B6 4341 CA SZC  R1,R13       ; CASE 150F
30B8 5345 SE SZCB R5,R13
30BA 2020    COC  @2020,R0
30BC 2020
30BE 150F    JGT  30DE
30C0 434F CO SZC  R15,R13      ; CONST 1C0F
30C2 4E53 NS SZC  *R3,*R9+
30C4 5420 T  SZCB @2020,*R0
30C6 2020
30C8 1C0F    JOP  30E8
30CA 4445 DE SZC  R5,*R1       ; DEFINITI 000F
30CC 4649 FI SZC  R9,*R9
30CE 4E49 NI SZC  R9,*R9+
30D0 5449 TI SZCB R9,*R1
30D2 000F    UNDEFINED
30D4 4449 DI SZC  R9,*R1       ; DIV 2703
30D6 5620 V  SZCB @2020,*R8
30D8 2020
```

```
30DA 2020    COC  @2703,R0
30DC 2703 '
30DE 444F DO SZC  R15,*R1      ; DP 060F
30E0 2020    COC  @2020,R0
30E2 2020
30E4 2020    COC  @060F,R0
30E6 060F
30E8 444F DO SZC  R15,*R1      ; DOWNTO 080F
30EA 574E WN SZCB R14,*R13
30EC 544F TO SZCB R15,*R1
30EE 2020    COC  @080F,R0
30F0 080F
30F2 454C EL SZC  R12,*R5      ; ELSE 0D0F
30F4 5345 SE SZCB R5,R13
30F6 2020    COC  @2020,R0
30F8 2020
30FA 0D0F    UNDEFINED
30FC 454E EN SZC  R14,*R5      ; END 090F
30FE 4420 D  SZC  @2020,*R0
3100 2020
3102 2020    COC  @090F,R0
3104 090F
3106 4558 EX SZC  *R8,*R5      ; EXTERNAL 350F
3108 5445 TE SZCB R5,*R1
310A 524E RN SZCB R14,R9
310C 414C AL SZC  R12,R5
310E 350F 5  STCR R15,4
3110 464F FO SZC  R15,*R9      ; FOR 180F
3112 5220 R  SZCB @2020,R8
3114 2020
3116 2020    COC  @180F,R0
3118 180F
311A 4649 FI SZC  R9,*R9       ; FILE 2E0F
311C 4C45 LE SZC  R5,*R1+
311E 2020    COC  @2020,R0
3120 2020
3122 2E0F .  XOP  R15,8
3124 464F FO SZC  R15,*R9      ; FORWARD 220F
3126 5257 RW SZCB *R7,R9
3128 4152 AR SZC  *R2,R5
312A 4420 D  SZC  @220F,*R0
312C 220F "
312E 4655 FU SZC  *R5,*R9      ; FUNCTION 200F
3130 4E43 NC SZC  R3,*R9+
3132 5449 TI SZCB R9,*R1
3134 4F4E ON SZC  R14,*R13+
3136 200F    COC  R15,R0
3138 474F GO SZC  R15,*R13     ; GOTO 1A0F
313A 544F TO SZCB R15,*R1
313C 2020    COC  @2020,R0
313E 2020
3140 1A0F    JL   3160
3142 4946 IF SZC  R6,@2020(R5) ; IF 140F
3144 2020
3146 2020    COC  @2020,R0
3148 2020
314A 140F    JHE  316A
314C 494D IM SZC  R13,@504C(R5); IMPLEMEN 340F
```

```
314E 504C PL
3150 454D EM SZC  R13,*R5
3152 454E EN SZC  R14,*R5
3154 340F 4  STCR R15,0
3156 494E IN SZC  R14,@2020(R5); IN 290E
3158 2020
315A 2020    COC  @2020,R0
315C 2020
315E 290E )  XOR  R14,R4
3160 494E IN SZC  R14,@5445(R5); INTERFAC 330F
3162 5445 TE
3164 5246 RF SZCB R6,R9
3166 4143 AC SZC  R3,R5
3168 330F 3  LDCR R15,12
316A 4C41 LA SZC  R1,*R1+      ; LABEL 1B0F
316C 4245 BE SZC  R5,R9
316E 4C20 L  SZC  @2020,*R0+
3170 2020
3172 1B0F    JH   3192
3174 4D4F MO SZC  R15,*R5+     ; MOD 2704
3176 4420 D  SZC  @2020,*R0
3178 2020
317A 2020    COC  @2704,R0
317C 2704 '
317E 4E4F NO SZC  R15,*R9+     ; NOT 260F
3180 5420 T  SZCB @2020,*R0
3182 2020
3184 2020    COC  @260F,R0
3186 260F &
3188 4F46 OF SZC  R6,*R13+     ; OF 0B0F
318A 2020    COC  @2020,R0
318C 2020
318E 2020    COC  @0B0F,R0
3190 0B0F
3192 4F52 OR SZC  *R2,*R13+    ; OR 2807
3194 2020    COC  @2020,R0
3196 2020
3198 2020    COC  @2807,R0
319A 2807 (
319C 5041 PA SZCB R1,R1                  ; PACKED 2B0F
319E 434B CK SZC  R11,R13
31A0 4544 ED SZC  R4,*R5
31A2 2020    COC  @2B0F,R0
31A4 2B0F +
31A6 5052 PR SZCB *R2,R1       ; PROCEDUR 1F0F
31A8 4F43 OC SZC  R3,*R13+
31AA 4544 ED SZC  R4,*R5
31AC 5552 UR SZCB *R2,*R5
31AE 1F0F    TB   15
31B0 5052 PR SZCB *R2,R1       ; PROCESS 380F
31B2 4F43 OC SZC  R3,*R13+
31B4 4553 ES SZC  *R3,*R5
31B6 5320 S  SZCB @380F,R12
31B8 380F 8
31BA 5052 PR SZCB *R2,R1       ; PROGRAM 210F
31BC 4F47 OG SZC  R7,*R13+
31BE 5241 RA SZCB R1,R9
31C0 4D20 M  SZC  @210F,*R4+
```

```
31C2 210F !
31C4 5245 RE SZCB R5,R9                  ; RECORD 2D0F
31C6 434F CO SZC  R15,R13
31C8 5244 RD SZCB R4,R9
31CA 2020    COC  @2D0F,R0
31CC 2D0F -
31CE 5245 RE SZCB R5,R9                  ; REPEAT 160F
31D0 5045 PE SZCB R5,R1
31D2 4154 AT SZC  *R4,R5
31D4 2020    COC  @160F,R0
31D6 160F
31D8 5345 SE SZCB R5,R13       ; SEPARATE 360F
31DA 5041 PA SZCB R1,R1
31DC 5241 RA SZCB R1,R9
31DE 5445 TE SZCB R5,*R1
31E0 360F 6  STCR R15,8
31E2 5345 SE SZCB R5,R13       ; SET 2A0F
31E4 5420 T  SZCB @2020,*R0
31E6 2020
31E8 2020    COC  @2A0F,R0
31EA 2A0F *
31EC 5345 SE SZCB R5,R13       ; SEGMENT 210F
31EE 474D GM SZC  R13,*R13
31F0 454E EN SZC  R14,*R5
31F2 5420 T  SZCB @210F,*R0
31F4 210F !
31F6 5448 TH SZCB R8,*R1       ; THEN 0C0F
31F8 454E EN SZC  R14,*R5
31FA 2020    COC  @2020,R0
31FC 2020
31FE 0C0F    UNDEFINED
3200 544F TO SZCB R15,*R1      ; TO 070F
3202 2020    COC  @2020,R0
3204 2020
3206 2020    COC  @070F,R0
3208 070F
320A 5459 TY SZCB *R9,*R1      ; TYPE 1D0F
320C 5045 PE SZCB R5,R1
320E 2020    COC  @2020,R0
3210 2020
3212 1D0F    SBO  15
3214 554E UN SZCB R14,*R5      ; UNIT 320F
3216 4954 IT SZC  *R4,@2020(R5)
3218 2020
321A 2020    COC  @320F,R0
321C 320F 2
321E 554E UN SZCB R14,*R5      ; UNTIL 0A0F
3220 5449 TI SZCB R9,*R1
3222 4C20 L  SZC  @2020,*R0+
3224 2020
3226 0A0F    SLA  R15,0
3228 5553 US SZCB *R3,*R5      ; USES 310F
322A 4553 ES SZC  *R3,*R5
322C 2020    COC  @2020,R0
322E 2020
3230 310F 1  LDCR R15,4
3232 5641 VA SZCB R1,*R9       ; VAR 1E0F
3234 5220 R  SZCB @2020,R8
```

```
3236 2020
3238 2020    COC  @1E0F,R0
323A 1E0F
323C 5748 WH SZCB R8,*R13      ; WHILE 170F
323E 494C IL SZC  R12,@4520(R5)
3240 4520 E
3242 2020    COC  @170F,R0
3244 170F
3246 5749 WI SZCB R9,*R13      ; WITH 190F
3248 5448 TH SZCB R8,*R1
324A 2020    COC  @2020,R0
324C 2020
324E 190F    JNO  326E

; These two procedures are transferred to RAM PAD at 8300H when used, for
speed reasons.
; Code for PME instruction decode. Used when p-code is in mapped memory,
i.e. GROM or VDP
3250 0588    INC  R8
3252 D05D  ] MOVB *R13,R1      ;Fetch op-code
3254 0971  q SRL  R1,7
3256 C0A1    MOV  @OPCODE(R1),R2         ;Op-code table
3258 2DB2 -
325A C032  2 MOV  *R2+,R0      ;Branch address
325C 0450  P B    *R0          ;Branch to instruction, or parameter fetch
325E 0588    INC  R8
3260 04C4    CLR  R4
3262 D81D    MOVB *R13,@8389   ;Byte parameter to R4
3264 8389
3266 0588    INC  R8
3268 04C3    CLR  R3
326A D81D    MOVB *R13,@8387   ;Byte parameter to R3
326C 8387
326E 0452  R B    *R2                    ;Interpret instruction
3270 0588    INC  R8
3272 04C5    CLR  R5
3274 D81D    MOVB *R13,@838B   ;Byte parameter to R5
3276 838B
3278 0588    INC  R8
327A 04C4    CLR  R4
327C D81D    MOVB *R13,@8389   ;Byte parameter to R4
327E 8389
3280 04C3    CLR  R3
3282 D0DD    MOVB *R13,R3      ;Byte or word parameter to R3
3284 1103    JLT  328C
3286 06C3    SWPB R3
3288 0588    INC  R8
328A 0452  R B    *R2                    ;Interpret instruction if in
0..127 range
328C 0243  C ANDI R3,7F00
328E 7F00 
3290 D81D    MOVB *R13,@8387   ;Expand to 0..32767 range
3292 8387
3294 05C8    INCT R8
3296 0452  R B    *R2                    ;Interpret instruction

; Code for PME instruction decode. Used when p-code is in CPU RAM
3298 D078  x MOVB *R8+,R1      ;Fetch op-code
```

```
329A 0971  q SRL  R1,7
329C C0A1    MOV  @OPCODE(R1),R2         ;Op-code table
329E 2DB2 -
32A0 C032  2 MOV  *R2+,R0      ;Branch address
32A2 0450  P B    *R0          ;Branch to instruction, or parameter fetch
32A4 1000    NOP
32A6 04C4    CLR  R4
32A8 D838  8 MOVB *R8+,@8389   ;Byte parameter to R4
32AA 8389
32AC 1000    NOP
32AE 04C3    CLR  R3
32B0 D838  8 MOVB *R8+,@8387   ;Byte parameter to R3
32B2 8387
32B4 0452  R B    *R2                   ;Interpret instruction
32B6 1000    NOP
32B8 04C5    CLR  R5
32BA D838  8 MOVB *R8+,@838B   ;Byte parameter to R5
32BC 838B
32BE 1000    NOP
32C0 04C4    CLR  R4
32C2 D838  8 MOVB *R8+,@8389   ;Byte parameter to R4
32C4 8389
32C6 1000    NOP
32C8 04C3    CLR  R3
32CA D0F8    MOVB *R8+,R3      ;Byte or word parameter to R3
32CC 1102    JLT  32D2
32CE 06C3    SWPB R3
32D0 0452  R B    *R2                   ;Interpret instruction if in
0..127 range
32D2 0243  C ANDI R3,7F00
32D4 7F00 ⌧
32D6 D838  8 MOVB *R8+,@8387   ;Expand to 0..32767 range
32D8 8387
32DA 0452  R B    *R2                   ;Interpret instruction

; Code for simulated interrupts
; The p-system always runs with interrupts disabled. Interrupt service is
provided by reading the
; interrupt request as an ordinary IO-bit. If it's active, the code is
executed.
; The IO-bit is tested after every taken (p-code) JUMP instruction and
also during various output tasks.
; Both vertical blanking and external interrupts are serviced.
32DC 1D02    SBO  2
32DE D820    MOVB @8802,@281E
32E0 8802
32E2 281E (
32E4 05A0    INC  @TIMELOW     ; Increment TIME values
32E6 2BA4 +
32E8 1702    JNC  32EE
32EA 05A0    INC  @TIMEHI
32EC 2BA6 +
32EE 05E0    INCT @288E
32F0 288E (
32F2 1609    JNE  3306
32F4 4820 H  SZC  @2D9E,@VDPCOPY
32F6 2D9E -
32F8 2810 (
```

```
32FA C260  ` MOV  @VDPCOPY,R9
32FC 2810 (
32FE C2A0    MOV  @83B4,R10
3300 83B4
3302 06A0    BL   @47C8
3304 47C8 G
3306 04CC    CLR  R12
3308 1F01    TB   1
330A 131C    JEQ  3344
330C 1D01    SBO  1
330E 020C    LI   R12,1F00
3310 1F00
3312 1E00    SBZ  0
3314 04E0    CLR  @2984
3316 2984 )
3318 0202    LI   R2,2986
331A 2986 )
331C A0A0    A    @2984,R2
331E 2984 )
3320 C312    MOV  *R2,R12
3322 130D    JEQ  333E
3324 05E0    INCT @2984
3326 2984 )
3328 1D00    SBO  0
332A C0A0    MOV  @400C,R2
332C 400C @
332E C002    MOV  R2,R0
3330 C0A2    MOV  @0002(R2),R2
3332 0002
3334 0692    BL   *R2
3336 C090    MOV  *R0,R2
3338 16FA    JNE  332E
333A 1E00    SBZ  0
333C 10ED    JMP  3318
333E 020C    LI   R12,1F00
3340 1F00
3342 1D00    SBO  0
3344 06A0    BL   @43FA
3346 43FA C
3348 04C1    CLR  R1
334A 06A0    BL   @4556
334C 4556 EV
334E 0581    INC  R1
3350 0281    CI   R1,0004
3352 0004
3354 16FA    JNE  334A
3356 0620    DEC  @2CD4
3358 2CD4 ,
335A 1301    JEQ  335E
335C 1005    JMP  3368
335E C820    MOV  @2CD2,@2CD4
3360 2CD2 ,
3362 2CD4 ,
3364 0420    BLWP @KEYSCAN
3366 2AAE *
3368 C2E0    MOV  @8396,R11
336A 8396
336C 02E0    LWPI PASCALWS
```

```
336E 8380
3370 06A0    BL   @LOADAD1
3372 4144 AD
3374 02E0    LWPI 28B6
3376 28B6 (
3378 C80B    MOV  R11,@8396
337A 8396
337C 0380    RTWP

; Code used to find CRU base and entry point of a device. The p-system
finds the DSR for each device
; in advance and stores information about where to find it in the
Peripheral Control Block.
337E 06CA    SWPB R10
3380 098A    SRL  R10,8
3382 020C    LI   R12,1000
3384 1000
3386 1D00    SBO  0
3388 0203    LI   R3,4000
338A 4000 @
338C 9813    CB   *R3,@2CAC
338E 2CAC ,
3390 1309    JEQ  33A4
3392 1E00    SBZ  0
3394 022C  , AI   R12,0100
3396 0100
3398 028C    CI   R12,2000
339A 2000
339C 16F4    JNE  3386
339E 04C7    CLR  R7
33A0 04C9    CLR  R9
33A2 101F    JMP  33E2
33A4 C10A    MOV  R10,R4
33A6 06C4    SWPB R4
33A8 0984    SRL  R4,8
33AA A0C4    A    R4,R3
33AC C0D3    MOV  *R3,R3
33AE 13F1    JEQ  3392
33B0 C103    MOV  R3,R4
33B2 05C4    INCT R4
33B4 C1F4    MOV  *R4+,R7
33B6 9D08    CB   R8,*R4+
33B8 16F9    JNE  33AC
33BA C146  F MOV  R6,R5
33BC 0225  % AI   R5,000A
33BE 000A
33C0 0209    LI   R9,8C02
33C2 8C02
33C4 06C5    SWPB R5
33C6 D645  E MOVB R5,*R9
33C8 06C5    SWPB R5
33CA D645  E MOVB R5,*R9
33CC C148  H MOV  R8,R5
33CE 0985    SRL  R5,8
33D0 9D29  ) CB   @FBFE(R9),*R4+
33D2 FBFE
33D4 16EB    JNE  33AC
33D6 0605    DEC  R5
```

```
33D8 16FB    JNE  33D0
33DA 1E00    SBZ  0
33DC C24C  L MOV  R12,R9
33DE 022A  * AI   R10,0100
33E0 0100
33E2 020C    LI   R12,1F00
33E4 1F00
33E6 1D00    SBO  0
33E8 0420    BLWP @478E
33EA 478E G
33EC 2972 )r XOR  *R2+,R5
33EE 2960 )` XOR  @0008,R5
33F0 0008
33F2 1E00    SBZ  0
33F4 045B  [ B    *R11

; Peripheral device call. Goes directly to the correct CRU base and entry
address, if known.
; Otherwise the code above is called first, to search for the device.
33F6 2964 )d XOR  @33FA(R4),R5
33F8 33FA 3
33FA C1A0    MOV  @2960,R6
33FC 2960 )`
33FE 0226  & AI   R6,0008
3400 0008
3402 0420    BLWP @4774
3404 4774 Gt
3406 2960 )` XOR  @2972,R5
3408 2972 )r
340A 000A    UNDEFINED
340C 024B  K ANDI R11,001F
340E 001F
3410 A2E0    A    @2962,R11
3412 2962 )b
3414 0420    BLWP @478E
3416 478E G
3418 297A )z XOR  *R10+,R5
341A 2970 )p XOR  *R0+,R5
341C 0002    UNDEFINED
341E 020C    LI   R12,1F00
3420 1F00
3422 1E00    SBZ  0
3424 C1C7    MOV  R7,R7
3426 1602    JNE  342C
3428 06A0    BL   @337E
342A 337E 3~
342C C249  I MOV  R9,R9
342E 1602    JNE  3434
3430 0209    LI   R9,1F00
3432 1F00
3434 0205    LI   R5,83DA
3436 83DA
3438 C035  5 MOV  *R5+,R0
343A C075  u MOV  *R5+,R1
343C C0B5    MOV  *R5+,R2
343E C148  H MOV  R8,R5
3440 0985    SRL  R5,8
3442 C805    MOV  R5,@8354
```

```
3444 8354  T
3446 A146  F A    R6,R5
3448 0225  % AI   R5,000A
344A 000A
344C C805    MOV  R5,@8356
344E 8356  V
3450 C2CA    MOV  R10,R11
3452 098B    SRL  R11,8
3454 C80B    MOV  R11,@83E2
3456 83E2
3458 C807    MOV  R7,@83E4
345A 83E4
345C 131E    JEQ  349A
345E C809    MOV  R9,@83F8
3460 83F8
3462 02E0    LWPI GPLWS
3464 83E0
3466 1D00    SBO  0
3468 0692    BL   *R2
346A 101A    JMP  34A0
346C 1E00    SBZ  0
346E 02E0    LWPI 2964
3470 2964 )d
3472 0205    LI   R5,83DA
3474 83DA
3476 CD40  @ MOV  R0,*R5+
3478 CD41  A MOV  R1,*R5+
347A CD42  B MOV  R2,*R5+
347C 04C0    CLR  R0
347E 020C    LI   R12,1F00
3480 1F00
3482 1D00    SBO  0
3484 0420    BLWP @4774
3486 4774 Gt
3488 2970 )p XOR  *R0+,R5
348A 297A )z XOR  *R10+,R5
348C 0002    UNDEFINED
348E 024B  K ANDI R11,00E0
3490 00E0
3492 1302    JEQ  3498
3494 C020    MOV  @2CD8,R0
3496 2CD8 ,
3498 0380    RTWP
349A C020    MOV  @2CDA,R0
349C 2CDA ,
349E 10EF    JMP  347E
34A0 02E0    LWPI 2964
34A2 2964 )d
34A4 C309    MOV  R9,R12
34A6 06A0    BL   @3392
34A8 3392 3
34AA 10C9    JMP  343E

; Kernel procedure UNITREAD
34AC C820    MOV  @CONTRUE,@IOASM
34AE 40B6 @
34B0 29A6 )
34B2 C80B    MOV  R11,@2ADA
```

```
34B4 2ADA *
34B6 C820    MOV  @2BEC,@29A8  ; Normal entry
34B8 2BEC +
34BA 29A8 )
34BC 06A0    BL   @352C
34BE 352C 5,
34C0 0460  ` B    @509C
34C2 509C P

; Kernel procedure UNITWRITE
34C4 C820    MOV  @CONTRUE,@IOASM
34C6 40B6 @
34C8 29A6 )
34CA C80B    MOV  R11,@2ADA
34CC 2ADA *
34CE C820    MOV  @2BEE,@29A8  ; Normal entry
34D0 2BEE +
34D2 29A8 )
34D4 06A0    BL   @352C
34D6 352C 5,
34D8 0460  ` B    @509C
34DA 509C P

; Kernel procedure UNITCLEAR
34DC C820    MOV  @CONTRUE,@IOASM
34DE 40B6 @
34E0 29A6 )
34E2 C80B    MOV  R11,@2ADA
34E4 2ADA *
34E6 06A0    BL   @352C                  ; Normal entry
34E8 352C 5,
34EA 0460  ` B    @5196
34EC 5196 Q

; Kernel procedure UNITSTATUS
34EE C820    MOV  @CONTRUE,@IOASM
34F0 40B6 @
34F2 29A6 )
34F4 C80B    MOV  R11,@2ADA
34F6 2ADA *
34F8 06A0    BL   @352C                  ; Normal entry
34FA 352C 5,
34FC 0460  ` B    @51BC
34FE 51BC Q

; Kernel procedure UNITBUSY. Always return false.
3500 05CA    INCT SP
3502 C6A0    MOV  @CONFALS,*SP
3504 40A8 @
3506 0460  ` B    @LOADADDR
3508 4142 AB

; Kernel procedure UNITWAIT. Always returns immediately.
350A 05CA    INCT SP
350C 0460  ` B    @LOADADDR
350E 4142 AB

; Some code. Turns on normal page and  some other things.
```

```
3510 06A0    BL   @353C
3512 353C 5<
3514 06A0    BL   @LOADAD1
3516 4144 AD
3518 8820    C    @IOASM,@CONTRUE
351A 29A6 )
351C 40B6 @
351E 1301    JEQ  3522
3520 045C  \ B    *R12          ; PME fetch
3522 04E0    CLR  @IOASM
3524 29A6 )
3526 C2E0    MOV  @2ADA,R11
3528 2ADA *
352A 045B  [ B    *R11

; Code which switches in alternate p-code ROM page.
352C C80C    MOV  R12,@29A4
352E 29A4 )
3530 020C    LI   R12,1F80
3532 1F80
3534 1D00    SBO  0
3536 C320    MOV  @29A4,R12
3538 29A4 )
353A 045B  [ B    *R11

; Code which turns on normal p-code ROM page. Sometimes entered at 3540.
353C C80C    MOV  R12,@29A4
353E 29A4 )
3540 020C    LI   R12,1F80
3542 1F80
3544 1E00    SBZ  0
3546 C320    MOV  @29A4,R12
3548 29A4 )
354A 045B  [ B    *R11

; Kernel procedure RELOCSEGS
354C 06A0    BL   @352C
354E 352C 5,
3550 020C    LI   R12,3558
3552 3558 5X
3554 0460  ` B    @5F5E
3556 5F5E _^
3558 06A0    BL   @3540
355A 3540 5@
355C 045C  \ B    *R12

; Code enabling normal p-code ROM page and then issues the run-time error
"unknown instruction".
355E 06A0    BL   @3540
3560 3540 5@
3562 06A0    BL   @FAULGEN
3564 484A HJ
3566 000B    .WORD 11          ; Unimplemented instruction

; Code erasing the 80 column screen memory.
3568 C2E0    MOV  @23F8,R11
356A 23F8 #
356C C060  ` MOV  @2CBE,R1
```

```
356E 2CBE ,
3570 0202    LI   R2,0780
3572 0780
3574 CC60  ` MOV  @40A2,*R1+
3576 40A2 @
3578 0642  B DECT R2
357A 16FC    JNE  3574
357C 045B  [ B    *R11
357E 045B  [ B    *R11

; Kernel global data area.
3580 3580 5  STCR R0,6
3582 0000    UNDEFINED
3584 0000    UNDEFINED
3586 0000    UNDEFINED
3588 0000    UNDEFINED
358A 2B6E +n XOR  @A7B8(R14),R13
358C A7B8
358E A7FA    A    *R10+,*R15
3590 A83C  < A    *R12+,@0000
3592 0000
3594 0000    UNDEFINED
3596 0000    UNDEFINED
3598 A8BE    A    *R14+,@A87E(R2)
359A A87E  ~
359C 0000    UNDEFINED
359E 0000    UNDEFINED
35A0 0000    UNDEFINED
35A2 0000    UNDEFINED
35A4 0000    UNDEFINED
35A6 001B    UNDEFINED
35A8 0000    UNDEFINED
35AA 0000    UNDEFINED
35AC 0650  P DECT *R0
35AE 4153 AS SZC  *R3,R5
35B0 5359 SY SZCB *R9,R13
35B2 535B S[ SZCB *R11,R13
35B4 0650  P DECT *R0
35B6 4153 AS SZC  *R3,R5
35B8 5359 SY SZCB *R9,R13
35BA 535B S[ SZCB *R11,R13
35BC 0650  P DECT *R0
35BE 4153 AS SZC  *R3,R5
35C0 5359 SY SZCB *R9,R13
35C2 535B S[ SZCB *R11,R13
35C4 0000    UNDEFINED
35C6 FFFF    SOCB *R15+,*R15+
35C8 FFFF    SOCB *R15+,*R15+
35CA FFFF    SOCB *R15+,*R15+
35CC FFFF    SOCB *R15+,*R15+
35CE FFFF    SOCB *R15+,*R15+
35D0 FFFF    SOCB *R15+,*R15+
35D2 FFFF    SOCB *R15+,*R15+
35D4 0000    UNDEFINED
35D6 FFFF    SOCB *R15+,*R15+
35D8 FFFF    SOCB *R15+,*R15+
35DA FFFF    SOCB *R15+,*R15+
35DC FFFF    SOCB *R15+,*R15+
```

```
35DE FFFF    SOCB *R15+,*R15+
35E0 FFFF    SOCB *R15+,*R15+
35E2 FFFF    SOCB *R15+,*R15+
35E4 0000    UNDEFINED
35E6 FFFF    SOCB *R15+,*R15+
35E8 FFFF    SOCB *R15+,*R15+
35EA FFFF    SOCB *R15+,*R15+
35EC FFFF    SOCB *R15+,*R15+
35EE FFFF    SOCB *R15+,*R15+
35F0 FFFF    SOCB *R15+,*R15+
35F2 FFFF    SOCB *R15+,*R15+
35F4 BDE6    AB   @2BD0(R6),*R7+
35F6 2BD0 +
35F8 A778  x A    *R8+,*R13
35FA A83C  < A    *R12+,@A7FA
35FC A7FA
35FE A7B8    A    *R8+,*R14
3600 0223  # AI   R3,355C      ; Prefix volume name (string[7])
3602 355C 5\
3604 5359 SY SZCB *R9,R13
3606 535B S[ SZCB *R11,R13
3608 0650  P DECT *R0                    ; Root volume name (string[7])
360A 4153 AS SZC  *R3,R5
360C 5359 SY SZCB *R9,R13
360E 535B S[ SZCB *R11,R13
3610 33E3 3  LDCR @0005(R3),15 ; System date.
                                         ; packed record
                                         ;   month: 1..12;
                                         ;   date:  1..31;
                                         ;   year:  0..100;
                                         ; end;
3612 0005
3614 0001    UNDEFINED                             ; Heap_Info.lock
:semaphore;
3616 0000    UNDEFINED
3618 BDF2    AB   *R2+,*R7+              ; Heap_Info.HeapTop :MemPtr;
361A BDE6    AB   @0000(R6),*R7+         ; Heap_Info.TopMark :MemPtr;
361C 0000                                ; Task_Info.TaskDone:semaphore;
361E 0000    UNDEFINED
3620 0001    UNDEFINED                             ; Task_Info.Lock
:semaphore;
3622 0000    UNDEFINED
3624 0000    UNDEFINED                             ; Task_Info.N_Tasks
:integer;
; Decimal constants
3626 0001    .WORD 1
3628 000A    .WORD 10
362A 0064  d .WORD 100
362C 03E8    .WORD 1000
362E 2710 '  .WORD 10000
3630 0000    UNDEFINED
3632 0000    UNDEFINED
3634 0001    UNDEFINED
3636 0020    UNDEFINED
3638 0000    UNDEFINED
363A 0000    UNDEFINED
363C 0000    UNDEFINED
363E 0000    UNDEFINED
```

```
3640 0000    UNDEFINED
3642 03FF    LREX
3644 FFFF    SOCB *R15+,*R15+
3646 FFFF    SOCB *R15+,*R15+
3648 FFFF    SOCB *R15+,*R15+
364A FFFF    SOCB *R15+,*R15+
364C FFFF    SOCB *R15+,*R15+
364E FFFF    SOCB *R15+,*R15+
3650 FFFF    SOCB *R15+,*R15+
3652 FFFF    SOCB *R15+,*R15+
3654 FFFF    SOCB *R15+,*R15+
3656 FFFF    SOCB *R15+,*R15+
3658 FFFF    SOCB *R15+,*R15+
365A FFFF    SOCB *R15+,*R15+
365C FFFF    SOCB *R15+,*R15+
365E FFFF    SOCB *R15+,*R15+
3660 FFFF    SOCB *R15+,*R15+
3662 FFFF    SOCB *R15+,*R15+
3664 FFFF    SOCB *R15+,*R15+
3666 FFFF    SOCB *R15+,*R15+
3668 FFFF    SOCB *R15+,*R15+
366A FFFF    SOCB *R15+,*R15+
366C FFFF    SOCB *R15+,*R15+
366E FFFF    SOCB *R15+,*R15+
3670 FFFF    SOCB *R15+,*R15+
3672 FFFF    SOCB *R15+,*R15+
3674 FFFF    SOCB *R15+,*R15+
3676 FFFF    SOCB *R15+,*R15+
3678 FFFF    SOCB *R15+,*R15+
367A FFFF    SOCB *R15+,*R15+
367C FFFF    SOCB *R15+,*R15+
367E FFFF    SOCB *R15+,*R15+
3680 0000    UNDEFINED
3682 0000    UNDEFINED
3684 0000    UNDEFINED
3686 0000    UNDEFINED
3688 0000    UNDEFINED
368A 0080    UNDEFINED
368C 0000    UNDEFINED
368E 0000    UNDEFINED
3690 0000    UNDEFINED
3692 0000    UNDEFINED
3694 0000    UNDEFINED

; Table with name and type of units.
; record
;   volume_id   : string[7];
;   is_blocked  : boolean;
;   no_of_blocks: integer; (* maxint if not blocked *)
; end;

3696 0000    UNDEFINED                              ; #0
3698 0008    UNDEFINED
369A 0000    UNDEFINED
369C 0000    UNDEFINED
369E 0000    UNDEFINED
36A0 7FFF    SB   *R15+,*R15+
36A2 0743  C ABS  R3
```

```
36A4 4F4E ON SZC  R14,*R13+
36A6 534F SO SZCB R15,R13
36A8 4C45 LE SZC  R5,*R1+
36AA 0000    UNDEFINED
36AC 7FFF    SB   *R15+,*R15+
36AE 0753  S ABS  *R3
36B0 5953 YS SZCB *R3,@5445(R5)
36B2 5445 TE
36B4 524D RM SZCB R13,R9
36B6 0000    UNDEFINED
36B8 7FFF    SB   *R15+,*R15+
36BA 0000    UNDEFINED
36BC 0014    UNDEFINED
36BE 0000    UNDEFINED
36C0 FFFF    SOCB *R15+,*R15+
36C2 0000    UNDEFINED
36C4 7FFF    SB   *R15+,*R15+
36C6 0650  P DECT *R0
36C8 4153 AS SZC  *R3,R5
36CA 5359 SY SZCB *R9,R13
36CC 535B S[ SZCB *R11,R13
36CE 0001    UNDEFINED
36D0 02D0    STST R0
36D2 054D  M INV  R13
36D4 554C UL SZCB R12,*R5
36D6 5449 TI SZCB R9,*R1
36D8 2D46 -F XOP  R6,5
36DA 0001    UNDEFINED
36DC 02D0    STST R0
36DE 0750  P ABS  *R0
36E0 5249 RI SZCB R9,R9
36E2 4E54 NT SZC  *R4,*R9+
36E4 4552 ER SZC  *R2,*R5
36E6 0000    UNDEFINED
36E8 7FFF    SB   *R15+,*R15+
36EA 0552  R INV  *R2
36EC 454D EM SZC  R13,*R5
36EE 494E IN SZC  R14,@FFFF(R5)
36F0 FFFF
36F2 0000    UNDEFINED
36F4 7FFF    SB   *R15+,*R15+
36F6 0652  R DECT *R2
36F8 454D EM SZC  R13,*R5
36FA 4F55 OU SZC  *R5,*R13+
36FC 5452 TR SZCB *R2,*R1
36FE 0000    UNDEFINED
3700 7FFF    SB   *R15+,*R15+
3702 0649  I DECT R9
3704 4E56 NV SZC  *R6,*R9+
3706 454E EN SZC  R14,*R5
3708 5404 T  SZCB R4,*R0
370A 0001    UNDEFINED
370C 02D0    STST R0
370E 0750  P ABS  *R0
3710 534F SO SZCB R15,R13
3712 5552 UR SZCB *R2,*R5
3714 4345 CE SZC  R5,R13
3716 0001    UNDEFINED
```

```
3718 02D0    STST R0
371A 0000    UNDEFINED
371C 0000    UNDEFINED
371E 0000    UNDEFINED
3720 0000    UNDEFINED
3722 0001    UNDEFINED
3724 7FFF    SB   *R15+,*R15+
3726 0000    UNDEFINED
3728 0000    UNDEFINED
372A 0001    UNDEFINED
372C 0000    UNDEFINED
372E 0001    UNDEFINED
3730 7FFF    SB   *R15+,*R15+
3732 0000    UNDEFINED
3734 0000    UNDEFINED
3736 0000    UNDEFINED
3738 0000    UNDEFINED
373A 0001    UNDEFINED
373C 7FFF    SB   *R15+,*R15+
373E 024F  O ANDI R15,5300
3740 5300 S
3742 FFFF    SOCB *R15+,*R15+
3744 FFFF    SOCB *R15+,*R15+
3746 0001    UNDEFINED
3748 0068  h UNDEFINED
374A 0000    UNDEFINED
374C FFFF    SOCB *R15+,*R15+
374E FFFF    SOCB *R15+,*R15+
3750 FFFF    SOCB *R15+,*R15+
3752 0001    UNDEFINED
3754 7FFF    SB   *R15+,*R15+
3756 0000    UNDEFINED
3758 FFFF    SOCB *R15+,*R15+
375A FFFF    SOCB *R15+,*R15+
375C FFFF    SOCB *R15+,*R15+
375E 0001    UNDEFINED
3760 7FFF    SB   *R15+,*R15+
3762 0000    UNDEFINED
3764 FFFF    SOCB *R15+,*R15+
3766 FFFF    SOCB *R15+,*R15+
3768 FFFF    SOCB *R15+,*R15+
376A 0001    UNDEFINED
376C 7FFF    SB   *R15+,*R15+
376E 0000    UNDEFINED
3770 FFFF    SOCB *R15+,*R15+
3772 FFFF    SOCB *R15+,*R15+
3774 FFFF    SOCB *R15+,*R15+
3776 0001    UNDEFINED
3778 7FFF    SB   *R15+,*R15+
377A 0000    UNDEFINED
377C FFFF    SOCB *R15+,*R15+
377E FFFF    SOCB *R15+,*R15+
3780 0000    UNDEFINED
3782 0001    UNDEFINED
3784 7FFF    SB   *R15+,*R15+
3786 0000    UNDEFINED
3788 0000    UNDEFINED
378A 0000    UNDEFINED
```

```
378C 0000    UNDEFINED
378E 0001    UNDEFINED
3790 7FFF    SB   *R15+,*R15+
3792 0000    UNDEFINED
3794 0000    UNDEFINED
3796 0000    UNDEFINED
3798 0000    UNDEFINED
379A 0001    UNDEFINED
379C 7FFF    SB   *R15+,*R15+
379E 0000    UNDEFINED
37A0 0000    UNDEFINED
37A2 0000    UNDEFINED
37A4 0000    UNDEFINED
37A6 0001    UNDEFINED
37A8 7FFF    SB   *R15+,*R15+
37AA 0000    UNDEFINED
37AC 0000    UNDEFINED
37AE 0000    UNDEFINED
37B0 0000    UNDEFINED
37B2 0001    UNDEFINED
37B4 7FFF    SB   *R15+,*R15+
37B6 0000    UNDEFINED
37B8 0000    UNDEFINED
37BA 0000    UNDEFINED
37BC 0000    UNDEFINED
37BE 0001    UNDEFINED
37C0 7FFF    SB   *R15+,*R15+
37C2 0000    UNDEFINED
37C4 FFFF    SOCB *R15+,*R15+
37C6 FFFF    SOCB *R15+,*R15+
37C8 FFFF    SOCB *R15+,*R15+
37CA 0001    UNDEFINED
37CC 7FFF    SB   *R15+,*R15+
37CE 0000    UNDEFINED
37D0 FFFF    SOCB *R15+,*R15+
37D2 FFFF    SOCB *R15+,*R15+
37D4 FFFF    SOCB *R15+,*R15+
37D6 0001    UNDEFINED
37D8 7FFF    SB   *R15+,*R15+
37DA 0000    UNDEFINED
37DC FFFF    SOCB *R15+,*R15+
37DE FFFF    SOCB *R15+,*R15+
37E0 FFFF    SOCB *R15+,*R15+
37E2 0001    UNDEFINED
37E4 7FFF    SB   *R15+,*R15+
37E6 0000    UNDEFINED
37E8 FFFF    SOCB *R15+,*R15+
37EA FFFF    SOCB *R15+,*R15+
37EC FFFF    SOCB *R15+,*R15+
37EE 0001    UNDEFINED
37F0 7FFF    SB   *R15+,*R15+
37F2 0000    UNDEFINED
37F4 FFFF    SOCB *R15+,*R15+
37F6 FFFF    SOCB *R15+,*R15+
37F8 FFFF    SOCB *R15+,*R15+
37FA 0001    UNDEFINED
37FC 7FFF    SB   *R15+,*R15+
37FE 0000    UNDEFINED
```

```
3800 0000    UNDEFINED
3802 0000    UNDEFINED
3804 0000    UNDEFINED
3806 0001    UNDEFINED
3808 7FFF    SB   *R15+,*R15+
380A 0454  T B    *R4
380C 4150 AP SZC  *R0,R5
380E 4555 EU SZC  *R5,*R5
3810 0000    UNDEFINED
3812 0000    UNDEFINED
3814 7FFF    SB   *R15+,*R15+
3816 0000    UNDEFINED
3818 0000    UNDEFINED
381A 0000    UNDEFINED
381C 0000    UNDEFINED
381E 0000    UNDEFINED
3820 7FFF    SB   *R15+,*R15+

; File name for the assembler, string[23]
3822 1650  P JNE  38C4
3824 4153 AS SZC  *R3,R5
3826 5359 SY SZCB *R9,R13
3828 533A S: SZCB *R10+,R12
382A 5359 SY SZCB *R9,R13
382C 5354 ST SZCB *R4,R13
382E 454D EM SZC  R13,*R5
3830 2E41 .A XOP  R1,9
3832 5353 SS SZCB *R3,R13
3834 4D42 MB SZC  R2,*R5+
3836 4C45 LE SZC  R5,*R1+
3838 52BD R  SZCB *R13+,R10

; File name for the compiler, string[23]
383A 1650  P JNE  38DC
383C 4153 AS SZC  *R3,R5
383E 5359 SY SZCB *R9,R13
3840 533A S: SZCB *R10+,R12
3842 5359 SY SZCB *R9,R13
3844 5354 ST SZCB *R4,R13
3846 454D EM SZC  R13,*R5
3848 2E43 .C XOP  R3,9
384A 4F4D OM SZC  R13,*R13+
384C 5049 PI SZCB R9,R1
384E 4C45 LE SZC  R5,*R1+
3850 52BD R  SZCB *R13+,R10

; File name for the editor, string[23]
3852 1450  P JHE  38F4
3854 4153 AS SZC  *R3,R5
3856 5359 SY SZCB *R9,R13
3858 533A S: SZCB *R10+,R12
385A 5359 SY SZCB *R9,R13
385C 5354 ST SZCB *R4,R13
385E 454D EM SZC  R13,*R5
3860 2E45 .E XOP  R5,9
3862 4449 DI SZC  R9,*R1
3864 544F TO SZCB R15,*R1
3866 5245 RE SZCB R5,R9
```

```
3868 FFFF    SOCB *R15+,*R15+

; File name for the filer, string[23]
386A 1350  P JEQ  390C
386C 4153 AS SZC  *R3,R5
386E 5359 SY SZCB *R9,R13
3870 533A S: SZCB *R10+,R12
3872 5359 SY SZCB *R9,R13
3874 5354 ST SZCB *R4,R13
3876 454D EM SZC  R13,*R5
3878 2E46 .F XOP  R6,9
387A 494C IL SZC  R12,@4552(R5)
387C 4552 ER
387E FFFF    SOCB *R15+,*R15+
3880 0000    UNDEFINED

; File name for the linker,string[23]
3882 1450  P JHE  3924
3884 4153 AS SZC  *R3,R5
3886 5359 SY SZCB *R9,R13
3888 533A S: SZCB *R10+,R12
388A 5359 SY SZCB *R9,R13
388C 5354 ST SZCB *R4,R13
388E 454D EM SZC  R13,*R5
3890 2E4C .L XOP  R12,9
3892 494E IN SZC  R14,@4B45(R5)
3894 4B45 KE
3896 5245 RE SZCB R5,R9

3898 0000    UNDEFINED
389A 0000    UNDEFINED
389C 0000    UNDEFINED
389E 0000    UNDEFINED
38A0 0000    UNDEFINED
38A2 0000    UNDEFINED
38A4 0020    UNDEFINED
38A6 0000    UNDEFINED
38A8 0000    UNDEFINED
38AA 0001    UNDEFINED
38AC A360  ` A    @AD6A,R13
38AE AD6A  j
38B0 2A46 *F XOR  R6,R9
38B2 A8FE    A    *R14+,@0001(R3)
38B4 0001
38B6 0000    UNDEFINED
38B8 0000    UNDEFINED
38BA A360  ` A    @0000,R13
38BC 0000
38BE 0000    UNDEFINED
38C0 0000    UNDEFINED
38C2 0000    UNDEFINED
38C4 0000    UNDEFINED
38C6 0000    UNDEFINED
38C8 0000    UNDEFINED

; File name for the library text file, usually '*USERLIB.TEXT'
38CA 0D2A  * UNDEFINED
38CC 5553 US SZCB *R3,*R5
```

```
38CE 4552 ER SZC  *R2,*R5
38D0 4C49 LI SZC  R9,*R1+
38D2 422E B. SZC  @5445(R14),R8
38D4 5445 TE
38D6 5854 XT SZCB *R4,@FFFF(R1)
38D8 FFFF
38DA FFFF    SOCB *R15+,*R15+
38DC FFFF    SOCB *R15+,*R15+
38DE FFFF    SOCB *R15+,*R15+
38E0 FFFF    SOCB *R15+,*R15+
38E2 A45E  ^ A    *R14,*R1
38E4 C94C  L MOV  R12,@0000(R5)
38E6 0000
38E8 0000    UNDEFINED
38EA 0000    UNDEFINED
38EC 0000    UNDEFINED
38EE 0000    UNDEFINED
38F0 0000    UNDEFINED
38F2 0000    UNDEFINED
38F4 000A    UNDEFINED
38F6 0000    UNDEFINED
38F8 0000    UNDEFINED
38FA 3939 99 .ASCII '99'       ; '99' when running
                                    ; 'GO' when stopped because
e.g. stack overflow
                                    ; 'NO' when stopped by Halt
command
38FC 2F34 /4 .ASCII '/4'       ; '/4' when running
38FE 0000    UNDEFINED

; Used during startup
; Does not seem to have any meaningful use when in operation
3900 0000    UNDEFINED
3902 0000    UNDEFINED
3904 2BD0 +  XOR  *R0,R15
3906 0000    UNDEFINED
3908 0000    UNDEFINED
390A 0012    UNDEFINED
390C 0000    UNDEFINED
390E 0000    UNDEFINED
3910 0000    UNDEFINED
3912 0000    UNDEFINED
3914 0000    UNDEFINED
3916 0000    UNDEFINED
3918 0000    UNDEFINED
391A 0000    UNDEFINED
391C 0000    UNDEFINED
391E 0000    UNDEFINED
3920 0000    UNDEFINED
3922 0800    SRA  R0,0
3924 0000    UNDEFINED
3926 0030  0 UNDEFINED
3928 0000    UNDEFINED
392A 0000    UNDEFINED
392C 0000    UNDEFINED
392E 0000    UNDEFINED
3930 0000    UNDEFINED
3932 0000    UNDEFINED
```

```
3934 0001    UNDEFINED
3936 0000    UNDEFINED
3938 0000    UNDEFINED
393A 0000    UNDEFINED
393C 0000    UNDEFINED
393E 0000    UNDEFINED
3940 FFFF    SOCB *R15+,*R15+
3942 FFFF    SOCB *R15+,*R15+
3944 FFFF    SOCB *R15+,*R15+
3946 FFFF    SOCB *R15+,*R15+
3948 FFFF    SOCB *R15+,*R15+
394A FFFF    SOCB *R15+,*R15+
394C FFFF    SOCB *R15+,*R15+
394E FFFF    SOCB *R15+,*R15+
3950 FFFF    SOCB *R15+,*R15+
3952 FFFF    SOCB *R15+,*R15+
3954 FFFF    SOCB *R15+,*R15+
3956 FFFF    SOCB *R15+,*R15+
3958 FFFF    SOCB *R15+,*R15+
395A FFFF    SOCB *R15+,*R15+
395C FFFF    SOCB *R15+,*R15+
395E FFFF    SOCB *R15+,*R15+
3960 FFFF    SOCB *R15+,*R15+
3962 FFFF    SOCB *R15+,*R15+
3964 FFFF    SOCB *R15+,*R15+
3966 FFFF    SOCB *R15+,*R15+
3968 FFFF    SOCB *R15+,*R15+
396A FFFF    SOCB *R15+,*R15+
396C FFFF    SOCB *R15+,*R15+
396E FFFF    SOCB *R15+,*R15+
3970 FFFF    SOCB *R15+,*R15+
3972 FFFF    SOCB *R15+,*R15+
3974 FFFF    SOCB *R15+,*R15+
3976 FFFF    SOCB *R15+,*R15+
3978 FFFF    SOCB *R15+,*R15+
397A FFFF    SOCB *R15+,*R15+
397C FFEF    SOCB @FFFF(R15),*R15+
397E FFFF
3980 0000    UNDEFINED
3982 0000    UNDEFINED
3984 0000    UNDEFINED
3986 0000    UNDEFINED
3988 0000    UNDEFINED
398A 0000    UNDEFINED
398C 0000    UNDEFINED
398E 0000    UNDEFINED
3990 0000    UNDEFINED
3992 0000    UNDEFINED
3994 0000    UNDEFINED
3996 0000    UNDEFINED
3998 0000    UNDEFINED
399A 0000    UNDEFINED
399C 0080    UNDEFINED
399E 0040  @ UNDEFINED
39A0 0000    UNDEFINED
39A2 0000    UNDEFINED
39A4 0000    UNDEFINED
39A6 0000    UNDEFINED
```

```
39A8 0002    UNDEFINED
39AA 0040  @ UNDEFINED
39AC 0004    UNDEFINED
39AE 0040  @ UNDEFINED
39B0 0000    UNDEFINED
39B2 0000    UNDEFINED
39B4 0000    UNDEFINED
39B6 0000    UNDEFINED
39B8 0000    UNDEFINED
39BA 0000    UNDEFINED
39BC 0000    UNDEFINED
39BE 0000    UNDEFINED
39C0 FFFF    SOCB *R15+,*R15+
39C2 FFFF    SOCB *R15+,*R15+
39C4 FFFF    SOCB *R15+,*R15+
39C6 FFFF    SOCB *R15+,*R15+
39C8 FFFF    SOCB *R15+,*R15+
39CA FFFF    SOCB *R15+,*R15+
39CC FFFF    SOCB *R15+,*R15+
39CE FFFF    SOCB *R15+,*R15+
39D0 FFFF    SOCB *R15+,*R15+
39D2 FFFF    SOCB *R15+,*R15+
39D4 FFFF    SOCB *R15+,*R15+
39D6 FFFF    SOCB *R15+,*R15+
39D8 FFFF    SOCB *R15+,*R15+
39DA FFFF    SOCB *R15+,*R15+
39DC FFFF    SOCB *R15+,*R15+
39DE FFFF    SOCB *R15+,*R15+
39E0 FFFF    SOCB *R15+,*R15+
39E2 FFFF    SOCB *R15+,*R15+
39E4 FFFF    SOCB *R15+,*R15+
39E6 FFFF    SOCB *R15+,*R15+
39E8 FFFF    SOCB *R15+,*R15+
39EA FFFF    SOCB *R15+,*R15+
39EC FFFF    SOCB *R15+,*R15+
39EE FFFF    SOCB *R15+,*R15+
39F0 FFFF    SOCB *R15+,*R15+
39F2 FFFF    SOCB *R15+,*R15+
39F4 FFFF    SOCB *R15+,*R15+
39F6 FFFF    SOCB *R15+,*R15+
39F8 FFFF    SOCB *R15+,*R15+
39FA FFFF    SOCB *R15+,*R15+
39FC FFFF    SOCB *R15+,*R15+
39FE FFFF    SOCB *R15+,*R15+
3A00 0000    UNDEFINED
3A02 0000    UNDEFINED
3A04 0000    UNDEFINED
3A06 0000    UNDEFINED
3A08 0000    UNDEFINED
3A0A 0000    UNDEFINED
3A0C 0000    UNDEFINED
3A0E 0020    UNDEFINED
3A10 0000    UNDEFINED
3A12 0000    UNDEFINED
3A14 0000    UNDEFINED
3A16 0000    UNDEFINED
3A18 0000    UNDEFINED
3A1A 0080    UNDEFINED
```

```
3A1C 0081    UNDEFINED
3A1E 0004    UNDEFINED
3A20 0000    UNDEFINED
3A22 0000    UNDEFINED
3A24 0001    UNDEFINED
3A26 0020    UNDEFINED
3A28 0000    UNDEFINED
3A2A 0000    UNDEFINED
3A2C 0000    UNDEFINED
3A2E 0004    UNDEFINED
3A30 0000    UNDEFINED
3A32 0000    UNDEFINED
3A34 0040  @ UNDEFINED
3A36 0000    UNDEFINED
3A38 0000    UNDEFINED
3A3A 0000    UNDEFINED
3A3C 0000    UNDEFINED
3A3E 0000    UNDEFINED
3A40 FFFF    SOCB *R15+,*R15+
3A42 FFFF    SOCB *R15+,*R15+
3A44 FFFF    SOCB *R15+,*R15+
3A46 FFFF    SOCB *R15+,*R15+
3A48 FFFF    SOCB *R15+,*R15+
3A4A FFFF    SOCB *R15+,*R15+
3A4C FFFF    SOCB *R15+,*R15+
3A4E FFFF    SOCB *R15+,*R15+
3A50 FFFF    SOCB *R15+,*R15+
3A52 FFFF    SOCB *R15+,*R15+
3A54 FFFF    SOCB *R15+,*R15+
3A56 FFFF    SOCB *R15+,*R15+
3A58 FFFF    SOCB *R15+,*R15+
3A5A FFFF    SOCB *R15+,*R15+
3A5C FFFF    SOCB *R15+,*R15+
3A5E FFFF    SOCB *R15+,*R15+
3A60 FFFF    SOCB *R15+,*R15+
3A62 FFFF    SOCB *R15+,*R15+
3A64 FFFE    SOCB *R14+,*R15+
3A66 FFFF    SOCB *R15+,*R15+
3A68 FFFF    SOCB *R15+,*R15+
3A6A FFFF    SOCB *R15+,*R15+
3A6C FFFF    SOCB *R15+,*R15+
3A6E FFFF    SOCB *R15+,*R15+
3A70 FFFF    SOCB *R15+,*R15+
3A72 FFFF    SOCB *R15+,*R15+
3A74 FFFF    SOCB *R15+,*R15+
3A76 FFFF    SOCB *R15+,*R15+
3A78 FFFF    SOCB *R15+,*R15+
3A7A FFFF    SOCB *R15+,*R15+
3A7C FFFF    SOCB *R15+,*R15+
3A7E FFFF    SOCB *R15+,*R15+
3A80 0000    UNDEFINED
3A82 0000    UNDEFINED
3A84 0000    UNDEFINED
3A86 0000    UNDEFINED
3A88 0000    UNDEFINED
3A8A 0000    UNDEFINED
3A8C 0000    UNDEFINED
3A8E 0000    UNDEFINED
```

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3A90 0000    UNDEFINED
3A92 0000    UNDEFINED
3A94 0010    UNDEFINED
3A96 0000    UNDEFINED
3A98 0000    UNDEFINED
3A9A 0000    UNDEFINED
3A9C 0020    UNDEFINED
3A9E 0000    UNDEFINED
3AA0 0000    UNDEFINED
3AA2 0000    UNDEFINED
3AA4 0000    UNDEFINED
3AA6 0000    UNDEFINED
3AA8 0000    UNDEFINED
3AAA 0000    UNDEFINED
3AAC 0000    UNDEFINED
3AAE 0000    UNDEFINED
3AB0 0000    UNDEFINED
3AB2 0000    UNDEFINED
3AB4 0000    UNDEFINED
3AB6 0000    UNDEFINED
3AB8 0000    UNDEFINED
3ABA 0000    UNDEFINED
3ABC 0000    UNDEFINED
3ABE 0000    UNDEFINED
3AC0 FFFF    SOCB *R15+,*R15+
3AC2 FFFF    SOCB *R15+,*R15+
3AC4 FFFF    SOCB *R15+,*R15+
3AC6 FFFF    SOCB *R15+,*R15+
3AC8 FFFF    SOCB *R15+,*R15+
3ACA FFFF    SOCB *R15+,*R15+
3ACC FFFF    SOCB *R15+,*R15+
3ACE FFFF    SOCB *R15+,*R15+
3AD0 FFFF    SOCB *R15+,*R15+
3AD2 FFFF    SOCB *R15+,*R15+
3AD4 FFFF    SOCB *R15+,*R15+
3AD6 FFFF    SOCB *R15+,*R15+
3AD8 FFFF    SOCB *R15+,*R15+
3ADA FFFF    SOCB *R15+,*R15+
3ADC FFFF    SOCB *R15+,*R15+
3ADE FFFF    SOCB *R15+,*R15+
3AE0 FFFF    SOCB *R15+,*R15+
3AE2 FFFF    SOCB *R15+,*R15+
3AE4 FFFF    SOCB *R15+,*R15+
3AE6 FFFF    SOCB *R15+,*R15+
3AE8 FFFF    SOCB *R15+,*R15+
3AEA FFFF    SOCB *R15+,*R15+
3AEC FFFF    SOCB *R15+,*R15+
3AEE FFFF    SOCB *R15+,*R15+
3AF0 FFFF    SOCB *R15+,*R15+
3AF2 FFFF    SOCB *R15+,*R15+
3AF4 FFFF    SOCB *R15+,*R15+
3AF6 FFFF    SOCB *R15+,*R15+
3AF8 FFFF    SOCB *R15+,*R15+
3AFA FFFF    SOCB *R15+,*R15+
3AFC FFFF    SOCB *R15+,*R15+
3AFE FFFF    SOCB *R15+,*R15+
3B00 0000    UNDEFINED
3B02 0000    UNDEFINED
```

```
3B04 0000    UNDEFINED
3B06 0000    UNDEFINED
3B08 0008    UNDEFINED
3B0A 0010    UNDEFINED
3B0C 0000    UNDEFINED
3B0E 0000    UNDEFINED
3B10 0000    UNDEFINED
3B12 0000    UNDEFINED
3B14 0008    UNDEFINED
3B16 0000    UNDEFINED
3B18 0000    UNDEFINED
3B1A 0004    UNDEFINED
3B1C 0002    UNDEFINED
3B1E 0000    UNDEFINED
3B20 0000    UNDEFINED
3B22 0000    UNDEFINED
3B24 0000    UNDEFINED
3B26 0000    UNDEFINED
3B28 0000    UNDEFINED
3B2A 0000    UNDEFINED
3B2C 0004    UNDEFINED
3B2E 0000    UNDEFINED
3B30 0000    UNDEFINED
3B32 0000    UNDEFINED
3B34 0000    UNDEFINED
3B36 0000    UNDEFINED
3B38 0000    UNDEFINED
3B3A 0000    UNDEFINED
3B3C 0000    UNDEFINED
3B3E 0000    UNDEFINED
3B40 FFFF    SOCB *R15+,*R15+
3B42 FFFF    SOCB *R15+,*R15+
3B44 FFFF    SOCB *R15+,*R15+
3B46 FFFF    SOCB *R15+,*R15+
3B48 FFFF    SOCB *R15+,*R15+
3B4A FFFF    SOCB *R15+,*R15+
3B4C FFFF    SOCB *R15+,*R15+
3B4E FFFF    SOCB *R15+,*R15+
3B50 FFFF    SOCB *R15+,*R15+
3B52 FFFF    SOCB *R15+,*R15+
3B54 FFFF    SOCB *R15+,*R15+
3B56 FFFF    SOCB *R15+,*R15+
3B58 FFFF    SOCB *R15+,*R15+
3B5A FFFF    SOCB *R15+,*R15+
3B5C FFFF    SOCB *R15+,*R15+
3B5E FFFF    SOCB *R15+,*R15+
3B60 FFFF    SOCB *R15+,*R15+
3B62 FFFF    SOCB *R15+,*R15+
3B64 FFFF    SOCB *R15+,*R15+
3B66 FFFF    SOCB *R15+,*R15+
3B68 FFFF    SOCB *R15+,*R15+
3B6A FFFF    SOCB *R15+,*R15+
3B6C FFFF    SOCB *R15+,*R15+
3B6E FFFF    SOCB *R15+,*R15+
3B70 FFFF    SOCB *R15+,*R15+
3B72 FFFF    SOCB *R15+,*R15+
3B74 FFFF    SOCB *R15+,*R15+
3B76 FFFF    SOCB *R15+,*R15+
```

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3B78 FFFF    SOCB *R15+,*R15+
3B7A FFFF    SOCB *R15+,*R15+
3B7C FFFF    SOCB *R15+,*R15+
3B7E FFFF    SOCB *R15+,*R15+
3B80 0000    UNDEFINED
3B82 0000    UNDEFINED
3B84 0000    UNDEFINED
3B86 0000    UNDEFINED
3B88 0000    UNDEFINED
3B8A 0000    UNDEFINED
3B8C 0000    UNDEFINED
3B8E 0000    UNDEFINED
3B90 0000    UNDEFINED
3B92 0000    UNDEFINED
3B94 0000    UNDEFINED
3B96 0080    UNDEFINED
3B98 0000    UNDEFINED
3B9A 0000    UNDEFINED
3B9C 0000    UNDEFINED
3B9E 0000    UNDEFINED
3BA0 0000    UNDEFINED
3BA2 0000    UNDEFINED
3BA4 0000    UNDEFINED
3BA6 0000    UNDEFINED
3BA8 0000    UNDEFINED
3BAA 0004    UNDEFINED
3BAC 0000    UNDEFINED
3BAE 0000    UNDEFINED
3BB0 0000    UNDEFINED
3BB2 0000    UNDEFINED
3BB4 0000    UNDEFINED
3BB6 0000    UNDEFINED
3BB8 0000    UNDEFINED
3BBA 0000    UNDEFINED
3BBC 0000    UNDEFINED
3BBE 0000    UNDEFINED
3BC0 FFFF    SOCB *R15+,*R15+
3BC2 FFFF    SOCB *R15+,*R15+
3BC4 FFFF    SOCB *R15+,*R15+
3BC6 FFFF    SOCB *R15+,*R15+
3BC8 FFFF    SOCB *R15+,*R15+
3BCA FFF7    SOCB *R7+,*R15+
3BCC FFFF    SOCB *R15+,*R15+
3BCE FFFF    SOCB *R15+,*R15+
3BD0 FFFF    SOCB *R15+,*R15+
3BD2 FFFF    SOCB *R15+,*R15+
3BD4 FFFF    SOCB *R15+,*R15+
3BD6 FFFF    SOCB *R15+,*R15+
3BD8 FFFF    SOCB *R15+,*R15+
3BDA FFFF    SOCB *R15+,*R15+
3BDC FFFF    SOCB *R15+,*R15+
3BDE FFFF    SOCB *R15+,*R15+
3BE0 FFFF    SOCB *R15+,*R15+
3BE2 FFFF    SOCB *R15+,*R15+
3BE4 FFFF    SOCB *R15+,*R15+
3BE6 FFFF    SOCB *R15+,*R15+
3BE8 FFFF    SOCB *R15+,*R15+
3BEA FFFF    SOCB *R15+,*R15+
```

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3BEC FFFF    SOCB *R15+,*R15+
3BEE FFFF    SOCB *R15+,*R15+
3BF0 FFFF    SOCB *R15+,*R15+
3BF2 FFFF    SOCB *R15+,*R15+
3BF4 FFFF    SOCB *R15+,*R15+
3BF6 FFFF    SOCB *R15+,*R15+
3BF8 FFFF    SOCB *R15+,*R15+
3BFA FFFF    SOCB *R15+,*R15+
3BFC FFFF    SOCB *R15+,*R15+
3BFE FFFF    SOCB *R15+,*R15+
3C00 0000    UNDEFINED
3C02 0000    UNDEFINED
3C04 0000    UNDEFINED
3C06 0000    UNDEFINED
3C08 0001    UNDEFINED
3C0A 0000    UNDEFINED
3C0C 0000    UNDEFINED
3C0E 0000    UNDEFINED
3C10 0000    UNDEFINED
3C12 0000    UNDEFINED
3C14 0002    UNDEFINED
3C16 0000    UNDEFINED
3C18 0000    UNDEFINED
3C1A 0000    UNDEFINED
3C1C 0000    UNDEFINED
3C1E 0000    UNDEFINED
3C20 0000    UNDEFINED
3C22 0000    UNDEFINED
3C24 0000    UNDEFINED
3C26 0002    UNDEFINED
3C28 0000    UNDEFINED
3C2A 0000    UNDEFINED
3C2C 0000    UNDEFINED
3C2E 0000    UNDEFINED
3C30 0000    UNDEFINED
3C32 0000    UNDEFINED
3C34 0000    UNDEFINED
3C36 0040  @ UNDEFINED
3C38 0000    UNDEFINED
3C3A 0000    UNDEFINED
3C3C 0000    UNDEFINED
3C3E 0000    UNDEFINED
3C40 FFFF    SOCB *R15+,*R15+
3C42 FFFF    SOCB *R15+,*R15+
3C44 FFFF    SOCB *R15+,*R15+
3C46 FFFF    SOCB *R15+,*R15+
3C48 FFFF    SOCB *R15+,*R15+
3C4A FFFF    SOCB *R15+,*R15+
3C4C FFFF    SOCB *R15+,*R15+
3C4E FFFF    SOCB *R15+,*R15+
3C50 FFFF    SOCB *R15+,*R15+
3C52 FFFF    SOCB *R15+,*R15+
3C54 FFFF    SOCB *R15+,*R15+
3C56 FFFF    SOCB *R15+,*R15+
3C58 FFFF    SOCB *R15+,*R15+
3C5A FFFF    SOCB *R15+,*R15+
3C5C FFFF    SOCB *R15+,*R15+
3C5E FFFF    SOCB *R15+,*R15+
```

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3C60 FFFF    SOCB *R15+,*R15+
3C62 FFFF    SOCB *R15+,*R15+
3C64 FFFF    SOCB *R15+,*R15+
3C66 FFFF    SOCB *R15+,*R15+
3C68 FFFF    SOCB *R15+,*R15+
3C6A FFFF    SOCB *R15+,*R15+
3C6C FFFF    SOCB *R15+,*R15+
3C6E FFFF    SOCB *R15+,*R15+
3C70 FFFF    SOCB *R15+,*R15+
3C72 FFFF    SOCB *R15+,*R15+
3C74 FFFF    SOCB *R15+,*R15+
3C76 FFFF    SOCB *R15+,*R15+
3C78 FFFF    SOCB *R15+,*R15+
3C7A FFFF    SOCB *R15+,*R15+
3C7C FFFF    SOCB *R15+,*R15+
3C7E FFFF    SOCB *R15+,*R15+
3C80 0000    UNDEFINED
3C82 0000    UNDEFINED
3C84 0000    UNDEFINED
3C86 0000    UNDEFINED
3C88 0000    UNDEFINED
3C8A 0000    UNDEFINED
3C8C 0000    UNDEFINED
3C8E 0000    UNDEFINED
3C90 0000    UNDEFINED
3C92 0000    UNDEFINED
3C94 0000    UNDEFINED
3C96 0000    UNDEFINED
3C98 0000    UNDEFINED
3C9A 0000    UNDEFINED
3C9C 0000    UNDEFINED
3C9E 0000    UNDEFINED
3CA0 0000    UNDEFINED
3CA2 0000    UNDEFINED
3CA4 0000    UNDEFINED
3CA6 0000    UNDEFINED
3CA8 0000    UNDEFINED
3CAA 0000    UNDEFINED
3CAC 0000    UNDEFINED
3CAE 0000    UNDEFINED
3CB0 0000    UNDEFINED
3CB2 0000    UNDEFINED
3CB4 0000    UNDEFINED
3CB6 0000    UNDEFINED
3CB8 0000    UNDEFINED
3CBA 0000    UNDEFINED
3CBC 0000    UNDEFINED
3CBE 0000    UNDEFINED
3CC0 FFFF    SOCB *R15+,*R15+
3CC2 FFFF    SOCB *R15+,*R15+
3CC4 FFFF    SOCB *R15+,*R15+
3CC6 FFFF    SOCB *R15+,*R15+
3CC8 FFFF    SOCB *R15+,*R15+
3CCA FFFF    SOCB *R15+,*R15+
3CCC FFFF    SOCB *R15+,*R15+
3CCE FFFF    SOCB *R15+,*R15+
3CD0 FFFF    SOCB *R15+,*R15+
3CD2 FFFF    SOCB *R15+,*R15+
```

```
3CD4 FFFF    SOCB *R15+,*R15+
3CD6 FFFF    SOCB *R15+,*R15+
3CD8 FFFF    SOCB *R15+,*R15+
3CDA FFFF    SOCB *R15+,*R15+
3CDC FFFF    SOCB *R15+,*R15+
3CDE FFFF    SOCB *R15+,*R15+
3CE0 FFFF    SOCB *R15+,*R15+
3CE2 FFFF    SOCB *R15+,*R15+
3CE4 FFFF    SOCB *R15+,*R15+
3CE6 FFFF    SOCB *R15+,*R15+
3CE8 FFFF    SOCB *R15+,*R15+
3CEA FFFF    SOCB *R15+,*R15+
3CEC FFFF    SOCB *R15+,*R15+
3CEE FFFF    SOCB *R15+,*R15+
3CF0 FFFF    SOCB *R15+,*R15+
3CF2 FFFF    SOCB *R15+,*R15+
3CF4 FFFF    SOCB *R15+,*R15+
3CF6 FFFF    SOCB *R15+,*R15+
3CF8 FFFF    SOCB *R15+,*R15+
3CFA FFFF    SOCB *R15+,*R15+
3CFC FFFF    SOCB *R15+,*R15+
3CFE FFFF    SOCB *R15+,*R15+
3D00 0000    UNDEFINED
3D02 0000    UNDEFINED
3D04 0000    UNDEFINED
3D06 0004    UNDEFINED
3D08 0000    UNDEFINED
3D0A 0000    UNDEFINED
3D0C 0020    UNDEFINED
3D0E 0000    UNDEFINED
3D10 0000    UNDEFINED
3D12 0001    UNDEFINED
3D14 0000    UNDEFINED
3D16 0000    UNDEFINED
3D18 0000    UNDEFINED
3D1A 0000    UNDEFINED
3D1C 0000    UNDEFINED
3D1E 00C0    UNDEFINED
3D20 0000    UNDEFINED
3D22 0000    UNDEFINED
3D24 0003    UNDEFINED
3D26 0000    UNDEFINED
3D28 0000    UNDEFINED
3D2A 0000    UNDEFINED
3D2C 0000    UNDEFINED
3D2E 0000    UNDEFINED
3D30 0000    UNDEFINED
3D32 0000    UNDEFINED
3D34 0000    UNDEFINED
3D36 0000    UNDEFINED
3D38 0000    UNDEFINED
3D3A 0000    UNDEFINED
3D3C 0001    UNDEFINED
3D3E 0000    UNDEFINED
3D40 FFFF    SOCB *R15+,*R15+
3D42 FFFF    SOCB *R15+,*R15+
3D44 FFFF    SOCB *R15+,*R15+
3D46 FFFF    SOCB *R15+,*R15+
```

```
3D48 FFFF    SOCB *R15+,*R15+
3D4A FFFF    SOCB *R15+,*R15+
3D4C FFFF    SOCB *R15+,*R15+
3D4E FFFF    SOCB *R15+,*R15+
3D50 FFFF    SOCB *R15+,*R15+
3D52 FFFF    SOCB *R15+,*R15+
3D54 FFFF    SOCB *R15+,*R15+
3D56 FFFF    SOCB *R15+,*R15+
3D58 FFFF    SOCB *R15+,*R15+
3D5A FFFF    SOCB *R15+,*R15+
3D5C FFFF    SOCB *R15+,*R15+
3D5E FFFF    SOCB *R15+,*R15+
3D60 FFFF    SOCB *R15+,*R15+
3D62 FFFF    SOCB *R15+,*R15+
3D64 FFFF    SOCB *R15+,*R15+
3D66 FFFF    SOCB *R15+,*R15+
3D68 FFFF    SOCB *R15+,*R15+
3D6A FFFF    SOCB *R15+,*R15+
3D6C FFFF    SOCB *R15+,*R15+
3D6E FFFF    SOCB *R15+,*R15+
3D70 FFFF    SOCB *R15+,*R15+
3D72 FFFF    SOCB *R15+,*R15+
3D74 FFFF    SOCB *R15+,*R15+
3D76 FFFF    SOCB *R15+,*R15+
3D78 FFFF    SOCB *R15+,*R15+
3D7A FFFF    SOCB *R15+,*R15+
3D7C FFFF    SOCB *R15+,*R15+
3D7E FFFF    SOCB *R15+,*R15+
3D80 0000    UNDEFINED
3D82 0000    UNDEFINED
3D84 0000    UNDEFINED
3D86 0000    UNDEFINED
3D88 0028  ( UNDEFINED
3D8A 0000    UNDEFINED
3D8C 0000    UNDEFINED
3D8E 0000    UNDEFINED
3D90 0000    UNDEFINED
3D92 0000    UNDEFINED
3D94 0000    UNDEFINED
3D96 0000    UNDEFINED
3D98 0000    UNDEFINED
3D9A 0000    UNDEFINED
3D9C 0000    UNDEFINED
3D9E 0000    UNDEFINED
3DA0 0000    UNDEFINED
3DA2 0000    UNDEFINED
3DA4 0000    UNDEFINED
3DA6 0000    UNDEFINED
3DA8 0000    UNDEFINED
3DAA 0000    UNDEFINED
3DAC 0000    UNDEFINED
3DAE 0004    UNDEFINED
3DB0 0000    UNDEFINED
3DB2 0000    UNDEFINED
3DB4 0000    UNDEFINED
3DB6 0000    UNDEFINED
3DB8 0000    UNDEFINED
3DBA 0000    UNDEFINED
```

```
3DBC 0000    UNDEFINED
3DBE 0000    UNDEFINED
3DC0 FFFF    SOCB *R15+,*R15+
3DC2 FFFF    SOCB *R15+,*R15+
3DC4 FFFF    SOCB *R15+,*R15+
3DC6 FFFF    SOCB *R15+,*R15+
3DC8 FFFF    SOCB *R15+,*R15+
3DCA FFFF    SOCB *R15+,*R15+
3DCC FFFF    SOCB *R15+,*R15+
3DCE FFFF    SOCB *R15+,*R15+
3DD0 FFFF    SOCB *R15+,*R15+
3DD2 FFFF    SOCB *R15+,*R15+
3DD4 FFFF    SOCB *R15+,*R15+
3DD6 FFFF    SOCB *R15+,*R15+
3DD8 FFFF    SOCB *R15+,*R15+
3DDA FFFF    SOCB *R15+,*R15+
3DDC FFFF    SOCB *R15+,*R15+
3DDE FFFF    SOCB *R15+,*R15+
3DE0 FFFF    SOCB *R15+,*R15+
3DE2 FFFF    SOCB *R15+,*R15+
3DE4 FFFF    SOCB *R15+,*R15+
3DE6 FFFF    SOCB *R15+,*R15+
3DE8 FFFF    SOCB *R15+,*R15+
3DEA FFFF    SOCB *R15+,*R15+
3DEC FFFF    SOCB *R15+,*R15+
3DEE FFFF    SOCB *R15+,*R15+
3DF0 FFFF    SOCB *R15+,*R15+
3DF2 FFFF    SOCB *R15+,*R15+
3DF4 FFFF    SOCB *R15+,*R15+
3DF6 FFFF    SOCB *R15+,*R15+
3DF8 FFFF    SOCB *R15+,*R15+
3DFA FFFF    SOCB *R15+,*R15+
3DFC FFFF    SOCB *R15+,*R15+
3DFE FFFF    SOCB *R15+,*R15+
3E00 0000    UNDEFINED
3E02 0000    UNDEFINED
3E04 0000    UNDEFINED
3E06 0000    UNDEFINED
3E08 0000    UNDEFINED
3E0A 0000    UNDEFINED
3E0C 0000    UNDEFINED
3E0E 0000    UNDEFINED
3E10 0000    UNDEFINED
3E12 0000    UNDEFINED
3E14 0000    UNDEFINED
3E16 0000    UNDEFINED
3E18 0010    UNDEFINED
3E1A 0000    UNDEFINED
3E1C 0000    UNDEFINED
3E1E 0000    UNDEFINED
3E20 0010    UNDEFINED
3E22 0000    UNDEFINED
3E24 0040  @ UNDEFINED
3E26 0000    UNDEFINED
3E28 0000    UNDEFINED
3E2A 0004    UNDEFINED
3E2C 000C    UNDEFINED
3E2E 0000    UNDEFINED
```

```
3E30 0020    UNDEFINED
3E32 0000    UNDEFINED
3E34 0000    UNDEFINED
3E36 0000    UNDEFINED
3E38 0000    UNDEFINED
3E3A 0000    UNDEFINED
3E3C 0000    UNDEFINED
3E3E 0000    UNDEFINED
3E40 FFFF    SOCB *R15+,*R15+
3E42 FFFF    SOCB *R15+,*R15+
3E44 FFFF    SOCB *R15+,*R15+
3E46 FFFF    SOCB *R15+,*R15+
3E48 FFFF    SOCB *R15+,*R15+
3E4A FFFF    SOCB *R15+,*R15+
3E4C FFFF    SOCB *R15+,*R15+
3E4E FFFF    SOCB *R15+,*R15+
3E50 FFFF    SOCB *R15+,*R15+
3E52 FFFF    SOCB *R15+,*R15+
3E54 FFFF    SOCB *R15+,*R15+
3E56 FFFF    SOCB *R15+,*R15+
3E58 FFFF    SOCB *R15+,*R15+
3E5A FFFF    SOCB *R15+,*R15+
3E5C FFFF    SOCB *R15+,*R15+
3E5E FFFF    SOCB *R15+,*R15+
3E60 FFFF    SOCB *R15+,*R15+
3E62 FFFF    SOCB *R15+,*R15+
3E64 FFFF    SOCB *R15+,*R15+
3E66 FFFF    SOCB *R15+,*R15+
3E68 FFFF    SOCB *R15+,*R15+
3E6A FFFF    SOCB *R15+,*R15+
3E6C FFFF    SOCB *R15+,*R15+
3E6E FFFF    SOCB *R15+,*R15+
3E70 FFFF    SOCB *R15+,*R15+
3E72 FFFF    SOCB *R15+,*R15+
3E74 FFFF    SOCB *R15+,*R15+
3E76 FFFF    SOCB *R15+,*R15+
3E78 FFFF    SOCB *R15+,*R15+
3E7A FFFF    SOCB *R15+,*R15+
3E7C FFFF    SOCB *R15+,*R15+
3E7E FFFF    SOCB *R15+,*R15+
3E80 0000    UNDEFINED
3E82 0000    UNDEFINED
3E84 0000    UNDEFINED
3E86 0000    UNDEFINED
3E88 0000    UNDEFINED
3E8A 0000    UNDEFINED
3E8C 0000    UNDEFINED
3E8E 0000    UNDEFINED
3E90 0000    UNDEFINED
3E92 0000    UNDEFINED
3E94 0000    UNDEFINED
3E96 0000    UNDEFINED
3E98 0000    UNDEFINED
3E9A 0000    UNDEFINED
3E9C 0000    UNDEFINED
3E9E 0000    UNDEFINED
3EA0 0010    UNDEFINED
3EA2 0000    UNDEFINED
```

```
3EA4 0000    UNDEFINED
3EA6 0000    UNDEFINED
3EA8 0000    UNDEFINED
3EAA 0000    UNDEFINED
3EAC 0000    UNDEFINED
3EAE 0000    UNDEFINED
3EB0 0000    UNDEFINED
3EB2 0000    UNDEFINED
3EB4 0000    UNDEFINED
3EB6 0000    UNDEFINED
3EB8 0000    UNDEFINED
3EBA 0000    UNDEFINED
3EBC 0000    UNDEFINED
3EBE 0000    UNDEFINED
3EC0 FFFF    SOCB *R15+,*R15+
3EC2 FFFF    SOCB *R15+,*R15+
3EC4 FFFF    SOCB *R15+,*R15+
3EC6 FFFF    SOCB *R15+,*R15+
3EC8 FFFF    SOCB *R15+,*R15+
3ECA FFFF    SOCB *R15+,*R15+
3ECC FFFF    SOCB *R15+,*R15+
3ECE FFFF    SOCB *R15+,*R15+
3ED0 FFFF    SOCB *R15+,*R15+
3ED2 FFFF    SOCB *R15+,*R15+
3ED4 FFFF    SOCB *R15+,*R15+
3ED6 FFFF    SOCB *R15+,*R15+
3ED8 FFFF    SOCB *R15+,*R15+
3EDA FFFF    SOCB *R15+,*R15+
3EDC FFFF    SOCB *R15+,*R15+
3EDE FFFF    SOCB *R15+,*R15+
3EE0 FFFF    SOCB *R15+,*R15+
3EE2 FFFF    SOCB *R15+,*R15+
3EE4 FFFF    SOCB *R15+,*R15+
3EE6 FFFF    SOCB *R15+,*R15+
3EE8 FFFF    SOCB *R15+,*R15+
3EEA FFFF    SOCB *R15+,*R15+
3EEC FFFF    SOCB *R15+,*R15+
3EEE FFFF    SOCB *R15+,*R15+
3EF0 FFFF    SOCB *R15+,*R15+
3EF2 FFFF    SOCB *R15+,*R15+
3EF4 FFFF    SOCB *R15+,*R15+
3EF6 FFFF    SOCB *R15+,*R15+
3EF8 FFFF    SOCB *R15+,*R15+
3EFA FFFF    SOCB *R15+,*R15+
3EFC FFFF    SOCB *R15+,*R15+
3EFE FFFF    SOCB *R15+,*R15+
3F00 0000    UNDEFINED
3F02 0000    UNDEFINED
3F04 0000    UNDEFINED
3F06 0000    UNDEFINED
3F08 0000    UNDEFINED
3F0A 0000    UNDEFINED
3F0C 0000    UNDEFINED
3F0E 0000    UNDEFINED
3F10 0000    UNDEFINED
3F12 0000    UNDEFINED
3F14 0000    UNDEFINED
3F16 0000    UNDEFINED
```

```
3F18 0000    UNDEFINED
3F1A 0000    UNDEFINED
3F1C 0000    UNDEFINED
3F1E 0000    UNDEFINED
3F20 0000    UNDEFINED
3F22 0000    UNDEFINED
3F24 0000    UNDEFINED
3F26 0000    UNDEFINED
3F28 0000    UNDEFINED
3F2A 0000    UNDEFINED
3F2C 0000    UNDEFINED
3F2E 0008    UNDEFINED
3F30 0020    UNDEFINED
3F32 0020    UNDEFINED
3F34 0040  @ UNDEFINED
3F36 0000    UNDEFINED
3F38 0000    UNDEFINED
3F3A 0000    UNDEFINED
3F3C 0000    UNDEFINED
3F3E 0000    UNDEFINED
3F40 FFFF    SOCB *R15+,*R15+
3F42 FFFF    SOCB *R15+,*R15+
3F44 FFFF    SOCB *R15+,*R15+
3F46 FFFF    SOCB *R15+,*R15+
3F48 FFFF    SOCB *R15+,*R15+
3F4A FFFF    SOCB *R15+,*R15+
3F4C FFFF    SOCB *R15+,*R15+
3F4E FFFF    SOCB *R15+,*R15+
3F50 FFFF    SOCB *R15+,*R15+
3F52 FFFF    SOCB *R15+,*R15+
3F54 FFFF    SOCB *R15+,*R15+
3F56 FFFF    SOCB *R15+,*R15+
3F58 FFFF    SOCB *R15+,*R15+
3F5A FFFF    SOCB *R15+,*R15+
3F5C FFFF    SOCB *R15+,*R15+
3F5E FFFF    SOCB *R15+,*R15+
3F60 FFFF    SOCB *R15+,*R15+
3F62 FFFF    SOCB *R15+,*R15+
3F64 FFFF    SOCB *R15+,*R15+
3F66 FFFF    SOCB *R15+,*R15+
3F68 FFFF    SOCB *R15+,*R15+
3F6A FFFF    SOCB *R15+,*R15+
3F6C FFFF    SOCB *R15+,*R15+
3F6E FFFF    SOCB *R15+,*R15+
3F70 FFFF    SOCB *R15+,*R15+
3F72 FFFF    SOCB *R15+,*R15+
3F74 FFFF    SOCB *R15+,*R15+
3F76 FFFF    SOCB *R15+,*R15+
3F78 FFFF    SOCB *R15+,*R15+
3F7A FFFF    SOCB *R15+,*R15+
3F7C FFFF    SOCB *R15+,*R15+
3F7E FFFF    SOCB *R15+,*R15+
3F80 0000    UNDEFINED
3F82 0000    UNDEFINED
3F84 0000    UNDEFINED
3F86 0000    UNDEFINED
3F88 0000    UNDEFINED
3F8A 0000    UNDEFINED
```

```
3F8C 0000    UNDEFINED
3F8E 0000    UNDEFINED
3F90 0000    UNDEFINED
3F92 0000    UNDEFINED
3F94 0000    UNDEFINED
3F96 0000    UNDEFINED
3F98 0000    UNDEFINED
3F9A 0000    UNDEFINED
3F9C 0000    UNDEFINED
3F9E 0000    UNDEFINED
3FA0 0000    UNDEFINED
3FA2 0000    UNDEFINED
3FA4 0000    UNDEFINED
3FA6 0000    UNDEFINED
3FA8 0000    UNDEFINED
3FAA 0002    UNDEFINED
3FAC 0000    UNDEFINED
3FAE 0000    UNDEFINED
3FB0 0000    UNDEFINED
3FB2 0000    UNDEFINED
3FB4 0010    UNDEFINED
3FB6 0008    UNDEFINED
3FB8 0000    UNDEFINED
3FBA 0000    UNDEFINED
3FBC 0000    UNDEFINED
3FBE 0000    UNDEFINED
3FC0 FFFF    SOCB *R15+,*R15+
3FC2 FFFF    SOCB *R15+,*R15+
3FC4 FFFF    SOCB *R15+,*R15+
3FC6 FFFF    SOCB *R15+,*R15+
3FC8 FFFF    SOCB *R15+,*R15+
3FCA FFFF    SOCB *R15+,*R15+
3FCC FFFF    SOCB *R15+,*R15+
3FCE FFFF    SOCB *R15+,*R15+
3FD0 FFFF    SOCB *R15+,*R15+
3FD2 FFFF    SOCB *R15+,*R15+
3FD4 FFFF    SOCB *R15+,*R15+
3FD6 FFFF    SOCB *R15+,*R15+
3FD8 FFFF    SOCB *R15+,*R15+
3FDA FFFF    SOCB *R15+,*R15+
3FDC FFFF    SOCB *R15+,*R15+
3FDE FFFF    SOCB *R15+,*R15+
3FE0 FFFF    SOCB *R15+,*R15+
3FE2 FFFF    SOCB *R15+,*R15+
3FE4 FFFF    SOCB *R15+,*R15+
3FE6 FFFF    SOCB *R15+,*R15+
3FE8 FFFF    SOCB *R15+,*R15+
3FEA FFFF    SOCB *R15+,*R15+
3FEC FFFF    SOCB *R15+,*R15+
3FEE FFFF    SOCB *R15+,*R15+
3FF0 FFFF    SOCB *R15+,*R15+
3FF2 FFFF    SOCB *R15+,*R15+
3FF4 FFFF    SOCB *R15+,*R15+
3FF6 FFFF    SOCB *R15+,*R15+
3FF8 FFFF    SOCB *R15+,*R15+
3FFA FFFD    SOCB *R13+,*R15+
3FFC FFFF    SOCB *R15+,*R15+
3FFE FFFF    SOCB *R15+,*R15+
```

Here starts the p-code card

```
          4000 AA01    .WORD AA01H       ;Header
          4002 0000    .WORD 0
          4004 4010 @  .WORD PWRUP       ;Power up link
          4006 0000    .WORD 0
          4008 0000    .WORD 0
          400A 0000    .WORD 0
          400C 0000    .WORD 0
          400E 0000    .WORD 0
PWRUP     4010 0000    .WORD 0
          4012 4038 @8 .WORD PWRBEG
          4014 0000    .WORD 0
4016 FFE0    SOCB @0006,*R15+
4018 0006
401A 484A HJ SZC  R10,@2D6E(R1)
401C 2D6E -n
401E 2B6E +n XOR  @2AAA(R14),R13
4020 2AAA *
4022 2DB2 -  XOP  *R2+,6
4024 34DC 4  STCR *R12,3
4026 34C4 4  STCR R4,3
4028 34AC 4  STCR @34EE(R12),2
402A 34EE 4
402C 2BF4 +  XOR  *R4+,R15
402E 2C20 ,  XOP  @29DC,0
4030 29DC )
4032 2BFC +  XOR  *R12+,R15
4034 47D8 G  SZC  *R8,*R15
4036 2780 '  CZC  R0,R14

;Beginning of power up routine
PWRBEG    4038 0201    LI   R1,3580H     ;Kernel global data area
          403A 3580 5
          403C 8860  ` C    @TXTNO,@037AH(R1)      ;Check if 'NO', then
don't start
          403E 406C @l
          4040 037A  z
          4042 1327  ' JEQ  NOSTART
          4044 02E0    LWPI PASCALWS
          4046 8380
          4048 0204    LI   R4,GROMWA    ;End of p-code card ROM
          404A 5FFE _
          404C D0A4    MOVB @-1026(R4),R2;From GROMRD
          404E FBFE
          4050 04E0    CLR  @3900H
          4052 3900 9
          4054 0202    LI   R2,E804      ;Code repository in GROM
          4056 E804
          4058 D502    MOVB R2,*R4       ;To GROMWA
          405A 06C2    SWPB R2
          405C D502    MOVB R2,*R4
          405E 06A0    BL   @4070                ;Fetch code from GROM
repository
          4060 4070 @p
          4062 06A0    BL   @2004                ;Branch to that
temporary code
```

```
          4064 2004
          4066 020C    LI   R12,8300     ;PME fetch address
          4068 8300
          406A 045C  \ B    *R12         ;PME fetch

TXTNO     406C 4E4F NO .ASCII 'NO'
TXTGO     406E 474F GO .ASCII 'GO'

          4070 0203    LI   R3,0004
          4072 0004
          4074 0201    LI   R1,838A      ;R5 Pascal WS
          4076 838A
LOOP01    4078 DC64  d MOVB @-1026(R4),*R1+        ;From GROM data
          407A FBFE
          407C 0603    DEC  R3
          407E 16FC    JNE  LOOP01
          4080 6185 a  S    R5,R6
          4082 CD45  E MOV  R5,*R5+
          4084 0646  F DECT R6
          4086 CD46  F MOV  R6,*R5+
          4088 0646  F DECT R6
LOOP02    408A DD64  d MOVB @-1026(R4),*R5+        ;From GROM data
          408C FBFE
          408E 0606    DEC  R6
          4090 16FC    JNE  LOOP02

; Returns to console if not p-code start, otherwise returns to p-system
boot sequence
NOSTART   4092 069B    BL   *R11

4094 415A AZ SZC  *R10,R5
4096 3039 09 LDCR *R9+,0
4098 617A az S    *R10+,R5
409A 5F24 _$ SZCB @200D(R4),*R12+
409C 200D
409E 0A10    SLA  R0,1
40A0 0C00    UNDEFINED
40A2 2020    COC  @000A,R0
40A4 000A
40A6 0800    SRA  R0,0
CONFALS   40A8 0000    UNDEFINED
40AA 0000    UNDEFINED         ; Constant used to compare with
40AC 0000    UNDEFINED
          40AE FFFF    SOCB *R15+,*R15+
CON001    40B0 0001    UNDEFINED
          40B2 0200    LI   R0,0006
          40B4 0006
CONTRUE   40B6 0001    UNDEFINED         ; Byte sex current segment
          40B8 FFFF    SOCB *R15+,*R15+
CON016    40BA 0010    ; Constant 16?
40BC 0000    UNDEFINED

; Table with bit masks used to handle packed fields
MASKTAB   40BE 0000    .WORD 0H
          40C0 0001    .WORD 1H
          40C2 0003    .WORD 3H
          40C4 0007    .WORD 7H
          40C6 000F    .WORD FH
```

```
           40C8 001F    .WORD 1FH
           40CA 003F    .WORD 3FH
           40CC 007F    .WORD 7FH
           40CE 00FF    .WORD FFH
           40D0 01FF    .WORD 1FFH
           40D2 03FF    .WORD 3FFH
           40D4 07FF    .WORD 7FFH
           40D6 0FFF    .WORD FFFH
           40D8 1FFF    .WORD 1FFFH
           40DA 3FFF    .WORD 3FFFH
           40DC 7FFF    .WORD 7FFFH
           40DE FFFF    .WORD FFFFH

; Tables with shift instructions for handling packed fields
SHLEFT     40E0 1000    NOP
           40E2 0A11    SLA  R1,1
           40E4 0A21  ! SLA  R1,2
           40E6 0A31  1 SLA  R1,3
           40E8 0A41  A SLA  R1,4
           40EA 0A51  Q SLA  R1,5
           40EC 0A61  a SLA  R1,6
           40EE 0A71  q SLA  R1,7
           40F0 0A81    SLA  R1,8
           40F2 0A91    SLA  R1,9
           40F4 0AA1    SLA  R1,10
           40F6 0AB1    SLA  R1,11
           40F8 0AC1    SLA  R1,12
           40FA 0AD1    SLA  R1,13
           40FC 0AE1    SLA  R1,14
           40FE 0AF1    SLA  R1,15
SHRIGHT    4100 1000    NOP
           4102 0913    SRL  R3,1
           4104 0923  # SRL  R3,2
           4106 0933  3 SRL  R3,3
           4108 0943  C SRL  R3,4
           410A 0953  S SRL  R3,5
           410C 0963  c SRL  R3,6
           410E 0973  s SRL  R3,7
           4110 0983    SRL  R3,8
           4112 0993    SRL  R3,9
           4114 09A3    SRL  R3,10
           4116 09B3    SRL  R3,11
           4118 09C3    SRL  R3,12
           411A 09D3    SRL  R3,13
           411C 09E3    SRL  R3,14
           411E 09F3    SRL  R3,15
SHCIRC     4120 1000    JMP  4122
           4122 0BF0    SRC  R0,15
           4124 0BE0    SRC  R0,14
           4126 0BD0    SRC  R0,13
           4128 0BC0    SRC  R0,12
           412A 0BB0    SRC  R0,11
           412C 0BA0    SRC  R0,10
           412E 0B90    SRC  R0,9
           4130 0B80    SRC  R0,8
           4132 0B70  p SRC  R0,7
           4134 0B60  ` SRC  R0,6
           4136 0B50  P SRC  R0,5
```

```
          4138 0B40  @ SRC  R0,4
          413A 0B30  0 SRC  R0,3
          413C 0B20    SRC  R0,2
          413E 0B10    SRC  R0,1
          4140 1000    NOP

; Load p-code address pointer
LOADADDR 4142 C2CC   MOV  R12,R11        ; PME fetch
LOADAD1   4144 C3CF    MOV  R15,R15      ; PME memory type
          4146 1508    JGT  GROMCODE     ; 4158 In GROM?
          4148 1306    JEQ  CPUCODE      ; 4156 In CPU RAM
; Code in VDP RAM
          414A DB60  ` MOVB @8391,@0402(R13)       ; Lsby R8 IPC
          414C 8391
          414E 0402
          4150 1000    NOP
          4152 DB48  H MOVB R8,@0402(R13)
          4154 0402
CPUCODE   4156 045B  [ B    *R11                   ; Done, return to PME
GROMCODE 4158 DB48  H MOVB R8,@0402(R13)
          415A 0402
          415C 1000    NOP
          415E DB60  ` MOVB @8391,@0402(R13)
          4160 8391
          4162 0402
          4164 045B  [ B    *R11

; Fetch data and flip bytes if needed. Return in R2
GETDAT    4166 C3CF    MOV  R15,R15
          4168 160B    JNE  NOTCPU
          416A C0B1    MOV  *R1+,R2
          416C C820    MOV  @PMEFLIP,@PMEFLIP      ; 2AE8
          416E 2AE8 *
          4170 2AE8 *
          4172 1601    JNE  NOFLIP       ; 4176
          4174 06C2    SWPB R2
NOFLIP    4176 045B  [ B    *R11

; Fetch data, no flipping. Return in R2
GETDAT2   4178 C3CF    MOV  R15,R15      ;Code type
          417A 1602    JNE  NOTCPU       ;Not CPU RAM
          417C C0B1    MOV  *R1+,R2
          417E 045B  [ B    *R11
NOTCPU    4180 1507    JGT  INGROM
          4182 DB60  ` MOVB @8383,@0402(R13)       ;Load address to VDP
RAM
          4184 8383
          4186 0402
          4188 1000    NOP
          418A DB41  A MOVB R1,@0402(R13)
          418C 0402
          418E 1006    JMP  CHKFLIP
INGROM    4190 DB41  A MOVB R1,@0402(R13)          ;Load address to GROM
          4192 0402
          4194 1000    NOP
          4196 DB60  ` MOVB @8383,@0402(R13)       ;Load address to GROM
          4198 8383
          419A 0402
```

```
CHKFLIP   419C C0A0    MOV  @PMEFLIP,R2
          419E 2AE8 *
          41A0 1605    JNE  NOTFLIP
          41A2 D81D    MOVB *R13,@8385   ;Code data
          41A4 8385
          41A6 05C1    INCT R1
          41A8 D09D    MOVB *R13,R2      ;Code data
          41AA 045B  [ B    *R11
NOTFLIP   41AC D09D    MOVB *R13,R2      ;Code data
          41AE 05C1    INCT R1
          41B0 D81D    MOVB *R13,@8385   ;Code data
          41B2 8385
          41B4 045B  [ B    *R11

41B6 2896 (  XOR  *R6,R2
41B8 41BA A  SZC  *R10+,R6
41BA 06A0    BL   @42C2
41BC 42C2 B
41BE 04C0    CLR  R0
41C0 D020    MOVB @8375,R0
41C2 8375  u
41C4 9800    CB   R0,@40B8
41C6 40B8 @
41C8 1360  ` JEQ  428A
41CA D060  ` MOVB @GPLST,R1
41CC 837C  |
41CE 1304    JEQ  41D8
41D0 C820    MOV  @2DA4,@2CD0
41D2 2DA4 -
41D4 2CD0 ,
41D6 1006    JMP  41E4
41D8 0620    DEC  @2CD0
41DA 2CD0 ,
41DC 1656  V JNE  428A
41DE C820    MOV  @2D9E,@2CD0
41E0 2D9E -
41E2 2CD0 ,
41E4 04E0    CLR  @288E
41E6 288E (
41E8 E820    SOC  @2D9E,@VDPCOPY
41EA 2D9E -
41EC 2810 (
41EE C260  ` MOV  @VDPCOPY,R9
41F0 2810 (
41F2 06A0    BL   @47C4
41F4 47C4 G
41F6 0280    CI   R0,8100
41F8 8100
41FA 135A  Z JEQ  42B0
41FC 0280    CI   R0,8600
41FE 8600
4200 1348  H JEQ  4292
4202 9020    CB   @2BC0,R0
4204 2BC0 +
4206 1605    JNE  4212
4208 0560  ` INV  @2894
420A 2894 (
420C 04E0    CLR  @2890
```

```
420E 2890 (
4210 103C  < JMP  428A
4212 9020    CB   @2BC2,R0
4214 2BC2 +
4216 1603    JNE  421E
4218 0560  ` INV  @2890
421A 2890 (
421C 1036  6 JMP  428A
421E 9020    CB   @2BC3,R0
4220 2BC3 +
4222 160C    JNE  423C
4224 04E0    CLR  @2894
4226 2894 (
4228 04E0    CLR  @2890
422A 2890 (
422C C820    MOV  @2D98,@2B70
422E 2D98 -
4230 2B70 +p
4232 0201    LI   R1,4840
4234 4840 H@
4236 C801    MOV  R1,@8398
4238 8398
423A 1027  ' JMP  428A
423C 9020    CB   @2BCA,R0
423E 2BCA +
4240 1603    JNE  4248
4242 0560  ` INV  @2892
4244 2892 (
4246 1021  ! JMP  428A
4248 21A0 !  COC  @2DA0,R6
424A 2DA0 -
424C 1603    JNE  4254
424E 06A0    BL   @480E
4250 480E H
4252 101B    JMP  428A
4254 C260  ` MOV  @2892,R9
4256 2892 (
4258 1308    JEQ  426A
425A 9800    CB   R0,@4098
425C 4098 @
425E 1A05    JL   426A
4260 9800    CB   R0,@4099
4262 4099 @
4264 1B02    JH   426A
4266 7020 p  SB   @409C,R0
4268 409C @
426A C247  G MOV  R7,R9
426C A260  ` A    @288A,R9
426E 288A (
4270 06A0    BL   @47C4
4272 47C4 G
4274 DA80    MOVB R0,@-2(SP)
4276 FFFE
4278 0587    INC  R7
427A 0247  G ANDI R7,001F
427C 001F
427E 8207    C    R7,R8
4280 1602    JNE  4286
```

```
4282 E1A0    SOC  @2DA0,R6
4284 2DA0 -
4286 41A0 A  SZC  @2D9E,R6
4288 2D9E -
428A C020    MOV  @2890,R0
428C 2890 (
428E 1695    JNE  41BA
4290 0380    RTWP
4292 0202    LI   R2,0050
4294 0050  P
4296 60A0 `  S    @2CC6,R2
4298 2CC6 ,
429A 04C1    CLR  R1
429C C020    MOV  @2CC6,R0
429E 2CC6 ,
42A0 0910    SRL  R0,1
42A2 3C40 <@ DIV  R0,R1
42A4 8801    C    R1,@2CCA
42A6 2CCA ,
42A8 1308    JEQ  42BA
42AA 05A0    INC  @2CCA
42AC 2CCA ,
42AE 1005    JMP  42BA
42B0 C020    MOV  @2CCA,R0
42B2 2CCA ,
42B4 1302    JEQ  42BA
42B6 0620    DEC  @2CCA
42B8 2CCA ,
42BA 06A0    BL   @47D2
42BC 47D2 G
42BE 0460  ` B    @41BA
42C0 41BA A
42C2 D820    MOVB @40A2,@GPLST
42C4 40A2 @
42C6 837C  |
42C8 D260  ` MOVB @8374,R9
42CA 8374  t
42CC 1628  ( JNE  431E
42CE 0200    LI   R0,83C6
42D0 83C6
42D2 020C    LI   R12,2AA0
42D4 2AA0 *
42D6 C050  P MOV  *R0,R1
42D8 CC3C  < MOV  *R12+,*R0+
42DA C090    MOV  *R0,R2
42DC CC3C  < MOV  *R12+,*R0+
42DE C0D0    MOV  *R0,R3
42E0 CC3C  < MOV  *R12+,*R0+
42E2 0200    LI   R0,83D4
42E4 83D4
42E6 C130  0 MOV  *R0+,R4
42E8 C170  p MOV  *R0+,R5
42EA C270  p MOV  *R0+,R9
42EC D820    MOVB @2811,@83D4
42EE 2811 (
42F0 83D4
42F2 02E0    LWPI GPLWS
42F4 83E0
```

```
42F6 06A0    BL   @000E
42F8 000E
42FA 02E0    LWPI 2896
42FC 2896 (
42FE 0200    LI   R0,83C6
4300 83C6
4302 020C    LI   R12,2AA0
4304 2AA0 *
4306 CF10    MOV  *R0,*R12+
4308 CC01    MOV  R1,*R0+
430A CF10    MOV  *R0,*R12+
430C CC02    MOV  R2,*R0+
430E CF10    MOV  *R0,*R12+
4310 CC03    MOV  R3,*R0+
4312 0200    LI   R0,83D4
4314 83D4
4316 CC04    MOV  R4,*R0+
4318 CC05    MOV  R5,*R0+
431A CC09    MOV  R9,*R0+
431C 045B  [ B    *R11
431E 04C9    CLR  R9
4320 04C0    CLR  R0
4322 0701    SETO R1
4324 020C    LI   R12,0024
4326 0024  $
4328 06C9    SWPB R9
432A 30C9 0  LDCR R9,3
432C 06C9    SWPB R9
432E 020C    LI   R12,0006
4330 0006
4332 3601 6  STCR R1,8
4334 06C1    SWPB R1
4336 0541  A INV  R1
4338 1310    JEQ  435A
433A C249  I MOV  R9,R9
433C 162C  , JNE  4396
433E 2060  ` COC  @2D9C,R1
4340 2D9C -
4342 1604    JNE  434C
4344 05C0    INCT R0
4346 4060 @` SZC  @2D9C,R1
4348 2D9C -
434A 1004    JMP  4354
434C 2060  ` COC  @2D9A,R1
434E 2D9A -
4350 1622  " JNE  4396
4352 0580    INC  R0
4354 4060 @` SZC  @2D9A,R1
4356 2D9A -
4358 161E    JNE  4396
435A 0589    INC  R9
435C 0289    CI   R9,0005
435E 0005
4360 11E0    JLT  4322
4362 D1A0    MOVB @40B8,R6
4364 40B8 @
4366 D806    MOVB R6,@8375
4368 8375  u
```

```
436A 0640  @ DECT R0
436C 1608    JNE  437E
436E 25A0 %  CZC  @2D9A,R6
4370 2D9A -
4372 1602    JNE  4378
4374 41A0 A  SZC  @2D9C,R6
4376 2D9C -
4378 E1A0    SOC  @2D9A,R6
437A 2D9A -
437C 10CF    JMP  431C
437E 21A0 !  COC  @2D9A,R6
4380 2D9A -
4382 16CC    JNE  431C
4384 41A0 A  SZC  @2D9A,R6
4386 2D9A -
4388 25A0 %  CZC  @2D9C,R6
438A 2D9C -
438C 16C7    JNE  431C
438E D820    MOVB @40A2,@8375
4390 40A2 @
4392 8375  u
4394 045B  [ B    *R11
4396 06C1    SWPB R1
4398 070C    SETO R12
439A 058C    INC  R12
439C 0A11    SLA  R1,1
439E 17FD    JNC  439A
43A0 028C    CI   R12,0004
43A2 0004
43A4 1105    JLT  43B0
43A6 0509    NEG  R9
43A8 0229  ) AI   R9,0009
43AA 0009
43AC 022C  , AI   R12,FFFC
43AE FFFC
43B0 0201    LI   R1,000A
43B2 000A
43B4 384C 8L MPY  R12,R1
43B6 A242  B A    R2,R9
43B8 06C9    SWPB R9
43BA 9189    CB   R9,R6
43BC 1607    JNE  43CC
43BE D820    MOVB @40BC,@GPLST
43C0 40BC @
43C2 837C  |
43C4 C006    MOV  R6,R0
43C6 0240  @ ANDI R0,0003
43C8 0003
43CA 1004    JMP  43D4
43CC 0246  F ANDI R6,FFFC
43CE FFFC
43D0 E180    SOC  R0,R6
43D2 D189    MOVB R9,R6
43D4 06C9    SWPB R9
43D6 0201    LI   R1,0028
43D8 0028  (
43DA 3840 8@ MPY  R0,R1
43DC A242  B A    R2,R9
```

```
43DE A260  ` A    @288C,R9
43E0 288C (
43E2 06C9    SWPB R9
43E4 D689    MOVB R9,*SP
43E6 06C9    SWPB R9
43E8 D689    MOVB R9,*SP
43EA 0640  @ DECT R0
43EC 1602    JNE  43F2
43EE 0266  f ORI  R6,0030
43F0 0030  0
43F2 D82A  * MOVB @-1026(R10),@8375
43F4 FBFE
43F6 8375  u
43F8 045B  [ B    *R11

43FA C020    MOV  @2808,R0
43FC 2808 (
43FE 1304    JEQ  4408
4400 0280    CI   R0,7FFF
4402 7FFF
4404 1308    JEQ  4416
4406 1005    JMP  4412
4408 C020    MOV  @281E,R0
440A 281E (
440C 0240  @ ANDI R0,2000
440E 2000
4410 1302    JEQ  4416
4412 05A0    INC  @2808
4414 2808 (
4416 C80B    MOV  R11,@280E
4418 280E (
441A 04CC    CLR  R12
441C 0201    LI   R1,0001
441E 0001
4420 020A    LI   R10,280A
4422 280A (
4424 C69A    MOV  *R10,*R10
4426 130A    JEQ  443C
4428 100E    JMP  4446
442A 058C    INC  R12
442C 0A11    SLA  R1,1
442E 170B    JNC  4446
4430 024C  L ANDI R12,001F
4432 001F
4434 1603    JNE  443C
4436 C2E0    MOV  @280E,R11
4438 280E (
443A 045B  [ B    *R11
443C 05CA    INCT R10
443E C69A    MOV  *R10,*R10
4440 13FA    JEQ  4436
4442 0201    LI   R1,0001
4444 0001
4446 C09A    MOV  *R10,R2
4448 2081    COC  R1,R2
444A 16EF    JNE  442A
444C C00C    MOV  R12,R0
444E 0A30  0 SLA  R0,3
```

```
4450 0220    AI   R0,0C80
4452 0C80
4454 0420    BLWP @4774
4456 4774 Gt
4458 28B6 (  XOR  *R6+,R2
445A 28BA (  XOR  *R10+,R2
445C 0008    UNDEFINED
445E C00C    MOV  R12,R0
4460 0A20    SLA  R0,2
4462 0220    AI   R0,0C00
4464 0C00
4466 0420    BLWP @4774
4468 4774 Gt
446A 28B6 (  XOR  *R6+,R2
446C 28C2 (  XOR  R2,R3
446E 0004    UNDEFINED
4470 06A0    BL   @4538
4472 4538 E8
4474 0286    CI   R6,C0FF
4476 C0FF
4478 120A    JLE  448E
447A 0286    CI   R6,E000
447C E000
447E 1B07    JH   448E
4480 D0C3    MOVB R3,R3
4482 1305    JEQ  448E
4484 1502    JGT  448A
4486 0226  & AI   R6,C000
4488 C000
448A 0226  & AI   R6,2000
448C 2000
448E 06C2    SWPB R2
4490 06C3    SWPB R3
4492 06C6    SWPB R6
4494 06A0    BL   @4538
4496 4538 E8
4498 06C2    SWPB R2
449A 06C3    SWPB R3
449C 06C6    SWPB R6
449E C104    MOV  R4,R4
44A0 133B  ; JEQ  4518
44A2 0604    DEC  R4
44A4 1639  9 JNE  4518
44A6 C145  E MOV  R5,R5
44A8 1604    JNE  44B2
44AA 4681 F  SZC  R1,*R10
44AC 04C6    CLR  R6
44AE 04C7    CLR  R7
44B0 1033  3 JMP  4518
44B2 C215    MOV  *R5,R8
44B4 0B18    SRC  R8,1
44B6 1702    JNC  44BC
44B8 D1E5    MOVB @0003(R5),R7
44BA 0003
44BC 0B18    SRC  R8,1
44BE 1708    JNC  44D0
44C0 C247  G MOV  R7,R9
44C2 0249  I ANDI R9,0080
```

```
44C4 0080
44C6 06C7    SWPB R7
44C8 D1E5    MOVB @0005(R5),R7
44CA 0005
44CC 06C7    SWPB R7
44CE E1C9    SOC  R9,R7
44D0 0B18    SRC  R8,1
44D2 1708    JNC  44E4
44D4 C265  e MOV  @0006(R5),R9
44D6 0006
44D8 1303    JEQ  44E0
44DA E1E0    SOC  @2CAE,R7
44DC 2CAE ,
44DE 1002    JMP  44E4
44E0 41E0 A  SZC  @2CAE,R7
44E2 2CAE ,
44E4 0B18    SRC  R8,1
44E6 1702    JNC  44EC
44E8 D1A5    MOVB @0009(R5),R6
44EA 0009
44EC 0B18    SRC  R8,1
44EE 1704    JNC  44F8
44F0 06C6    SWPB R6
44F2 D1A5    MOVB @000B(R5),R6
44F4 000B
44F6 06C6    SWPB R6
44F8 0B18    SRC  R8,1
44FA 1702    JNC  4500
44FC D0E5    MOVB @000D(R5),R3
44FE 000D
4500 0B18    SRC  R8,1
4502 1704    JNC  450C
4504 06C3    SWPB R3
4506 D0E5    MOVB @000F(R5),R3
4508 000F
450A 06C3    SWPB R3
450C C125  % MOV  @0010(R5),R4
450E 0010
4510 C265  e MOV  @0014(R5),R9
4512 0014
4514 C165  e MOV  @0012(R5),R5
4516 0012
4518 0420    BLWP @478E
451A 478E G
451C 28C2 (  XOR  R2,R3
451E 28B6 (  XOR  *R6+,R2
4520 0004    UNDEFINED
4522 C00C    MOV  R12,R0
4524 0A30  0 SLA  R0,3
4526 0220    AI   R0,0C80
4528 0C80
452A 0420    BLWP @478E
452C 478E G
452E 28BA (  XOR  *R10+,R2
4530 28B6 (  XOR  *R6+,R2
4532 0008    UNDEFINED
4534 0460  ` B    @442A
4536 442A D*
```

```
4538 04C9    CLR  R9
453A 04C8    CLR  R8
453C D242  B MOVB R2,R9
453E 0849  I SRA  R9,4
4540 D203    MOVB R3,R8
4542 0848  H SRA  R8,4
4544 A248  H A    R8,R9
4546 04C8    CLR  R8
4548 D206    MOVB R6,R8
454A A209    A    R9,R8
454C D188    MOVB R8,R6
454E 06C8    SWPB R8
4550 0948  H SRL  R8,4
4552 D088    MOVB R8,R2
4554 045B  [ B    *R11
4556 C081    MOV  R1,R2
4558 0A12    SLA  R2,1
455A 0622  " DEC  @278C(R2)
455C 278C '
455E 160A    JNE  4574
4560 C022  " MOV  @27AC(R2),R0
4562 27AC '
4564 0280    CI   R0,7FFF
4566 7FFF
4568 1302    JEQ  456E
456A 05A2    INC  @27AC(R2)
456C 27AC '
456E C8A2    MOV  @2784(R2),@278C(R2)
4570 2784 '
4572 278C '
4574 C022  " MOV  @2794(R2),R0
4576 2794 '
4578 130C    JEQ  4592
457A 0622  " DEC  @2794(R2)
457C 2794 '
457E 160C    JNE  4598
4580 C022  " MOV  @279C(R2),R0
4582 279C '
4584 0280    CI   R0,0001
4586 0001
4588 130C    JEQ  45A2
458A D821  ! MOVB @2C98(R1),@8400
458C 2C98 ,
458E 8400
4590 1008    JMP  45A2
4592 C022  " MOV  @279C(R2),R0
4594 279C '
4596 1605    JNE  45A2
4598 045B  [ B    *R11
459A D821  ! MOVB @27DC(R1),@8400
459C 27DC '
459E 8400
45A0 045B  [ B    *R11
45A2 0622  " DEC  @279C(R2)
45A4 279C '
45A6 16F8    JNE  4598
45A8 C0E2    MOV  @27A4(R2),R3
45AA 27A4 '
```

```
45AC D033  3 MOVB *R3+,R0
45AE 2020    COC  @2CA2,R0
45B0 2CA2 ,
45B2 1602    JNE  45B8
45B4 0460  ` B    @46BE
45B6 46BE F
45B8 2020    COC  @2DB0,R0
45BA 2DB0 -
45BC 1344  D JEQ  4646
45BE 2020    COC  @2DAE,R0
45C0 2DAE -
45C2 1326  & JEQ  4610
45C4 06C0    SWPB R0
45C6 D033  3 MOVB *R3+,R0
45C8 06C0    SWPB R0
45CA 0281    CI   R1,0003
45CC 0003
45CE 1317    JEQ  45FE
45D0 0204    LI   R4,0001
45D2 0001
45D4 C140  @ MOV  R0,R5
45D6 0915    SRL  R5,1
45D8 0225  % AI   R5,B503
45DA B503
45DC 1701    JNC  45E0
45DE 0584    INC  R4
45E0 3D00 =  DIV  R0,R4
45E2 C144  D MOV  R4,R5
45E4 0244  D ANDI R4,03FF
45E6 03FF
45E8 0A44  D SLA  R4,4
45EA 06C5    SWPB R5
45EC 0245  E ANDI R5,0F00
45EE 0F00
45F0 B161  a AB   @2C90(R1),R5
45F2 2C90 ,
45F4 D805    MOVB R5,@8400
45F6 8400
45F8 D804    MOVB R4,@8400
45FA 8400
45FC 10D7    JMP  45AC
45FE 0280    CI   R0,0007
4600 0007
4602 1BD4    JH   45AC
4604 06C0    SWPB R0
4606 B021  ! AB   @2C90(R1),R0
4608 2C90 ,
460A D800    MOVB R0,@8400
460C 8400
460E 10CE    JMP  45AC
4610 C883    MOV  R3,@27A4(R2)
4612 27A4 '
4614 C100    MOV  R0,R4
4616 06C4    SWPB R4
4618 0244  D ANDI R4,001F
461A 001F
461C 1602    JNE  4622
461E 0204    LI   R4,0020
```

```
4620 0020
4622 3922 9" MPY  @2784(R2),R4
4624 2784 '
4626 C105    MOV  R5,R4
4628 2020    COC  @2DAC,R0
462A 2DAC -
462C 160A    JNE  4642
462E 0225  % AI   R5,0004
4630 0004
4632 0935  5 SRL  R5,3
4634 6105 a  S    R5,R4
4636 C884    MOV  R4,@2794(R2)
4638 2794 '
463A 0585    INC  R5
463C C885    MOV  R5,@279C(R2)
463E 279C '
4640 10AC    JMP  459A
4642 04C5    CLR  R5
4644 10F8    JMP  4636
4646 9020    CB   @2CA5,R0
4648 2CA5 ,
464A 1316    JEQ  4678
464C 2020    COC  @2DAC,R0
464E 2DAC -
4650 1307    JEQ  4660
4652 2020    COC  @2DAA,R0
4654 2DAA -
4656 132D  - JEQ  46B2
4658 9020    CB   @2CA4,R0
465A 2CA4 ,
465C 1310    JEQ  467E
465E 10A6    JMP  45AC
4660 D133  3 MOVB *R3+,R4
4662 9020    CB   @2CAA,R0
4664 2CAA ,
4666 16A2    JNE  45AC
4668 0544  D INV  R4
466A 0244  D ANDI R4,0F00
466C 0F00
466E B121  ! AB   @2C94(R1),R4
4670 2C94 ,
4672 D844  D MOVB R4,@27DC(R1)
4674 27DC '
4676 109A    JMP  45AC
4678 D0E1    MOVB @27B4(R1),R3
467A 27B4 '
467C 1609    JNE  4690
467E D861  a MOVB @2C98(R1),@27DC(R1)
4680 2C98 ,
4682 27DC '
4684 04E2    CLR  @27E0(R2)
4686 27E0 '
4688 D862  b MOVB @27E0(R2),@27B4(R1)
468A 27E0 '
468C 27B4 '
468E 1085    JMP  459A
4690 0983    SRL  R3,8
4692 C103    MOV  R3,R4
```

```
4694 0A22  " SLA  R2,2
4696 A0C2    A    R2,R3
4698 0922  " SRL  R2,2
469A C8A3    MOV  @27B6(R3),@27A4(R2)
469C 27B6 '
469E 27A4 '
46A0 C8A3    MOV  @27E6(R3),@27E0(R2)
46A2 27E6 '
46A4 27E0 '
46A6 0644  D DECT R4
46A8 06C4    SWPB R4
46AA D844  D MOVB R4,@27B4(R1)
46AC 27B4 '
46AE 0460  ` B    @45A8
46B0 45A8 E
46B2 0240  @ ANDI R0,0F00
46B4 0F00
46B6 D840  @ MOVB R0,@27D8(R1)
46B8 27D8 '
46BA 0460  ` B    @45AC
46BC 45AC E
46BE 2020    COC  @2DAC,R0
46C0 2DAC -
46C2 1353  S JEQ  476A
46C4 9020    CB   @2CA2,R0
46C6 2CA2 ,
46C8 160D    JNE  46E4
46CA D033  3 MOVB *R3+,R0
46CC 06C0    SWPB R0
46CE D033  3 MOVB *R3+,R0
46D0 06C0    SWPB R0
46D2 C8A2    MOV  @27E0(R2),@27A4(R2)
46D4 27E0 '
46D6 27A4 '
46D8 A8A0    A    @4772,@27A4(R2)
46DA 4772 Gr
46DC 27A4 '
46DE A880    A    R0,@27A4(R2)
46E0 27A4 '
46E2 1041  A JMP  4766
46E4 9020    CB   @2CA8,R0
46E6 2CA8 ,
46E8 160F    JNE  4708
46EA D033  3 MOVB *R3+,R0
46EC 06C0    SWPB R0
46EE D033  3 MOVB *R3+,R0
46F0 06C0    SWPB R0
46F2 C8A2    MOV  @27E0(R2),@27A4(R2)
46F4 27E0 '
46F6 27A4 '
46F8 A8A0    A    @4772,@27A4(R2)
46FA 4772 Gr
46FC 27A4 '
46FE A880    A    R0,@27A4(R2)
4700 27A4 '
4702 C022  " MOV  @27E0(R2),R0
4704 27E0 '
4706 101D    JMP  4742
```

```
4708 9020    CB   @2CA6,R0
470A 2CA6 ,
470C 160B    JNE  4724
470E D8B3    MOVB *R3+,@27A4(R2)
4710 27A4 '
4712 D8B3    MOVB *R3+,@27A5(R2)
4714 27A5 '
4716 C8A2    MOV  @27A4(R2),@27E0(R2)
4718 27A4 '
471A 27E0 '
471C A8A0    A    @4772,@27A4(R2)
471E 4772 Gr
4720 27A4 '
4722 1021  ! JMP  4766
4724 9020    CB   @2C9D,R0
4726 2C9D ,
4728 1621  ! JNE  476C
472A D8B3    MOVB *R3+,@27A4(R2)
472C 27A4 '
472E D8B3    MOVB *R3+,@27A5(R2)
4730 27A5 '
4732 C022  " MOV  @27E0(R2),R0
4734 27E0 '
4736 C8A2    MOV  @27A4(R2),@27E0(R2)
4738 27A4 '
473A 27E0 '
473C A8A0    A    @4772,@27A4(R2)
473E 4772 Gr
4740 27A4 '
4742 D121  ! MOVB @27B4(R1),R4
4744 27B4 '
4746 0984    SRL  R4,8
4748 0284    CI   R4,0008
474A 0008
474C 1301    JEQ  4750
474E 05C4    INCT R4
4750 0A22  " SLA  R2,2
4752 C144  D MOV  R4,R5
4754 A102    A    R2,R4
4756 0922  " SRL  R2,2
4758 C903    MOV  R3,@27B6(R4)
475A 27B6 '
475C C900    MOV  R0,@27E6(R4)
475E 27E6 '
4760 06C5    SWPB R5
4762 D845  E MOVB R5,@27B4(R1)
4764 27B4 '
4766 0460  ` B    @45A8
4768 45A8 E
476A 0583    INC  R3
476C 05C3    INCT R3
476E 0460  ` B    @45AC
4770 45AC E
4772 0004    UNDEFINED
4774 83A0    C    @4778,R14
4776 4778 Gx
4778 C07E  ~ MOV  *R14+,R1
477A C0BE    MOV  *R14+,R2
```

```
477C C0FE    MOV  *R14+,R3
477E C251  Q MOV  *R1,R9
4780 06A0    BL   @47C8
4782 47C8 G
4784 DCAA    MOVB @-1026(R10),*R2+
4786 FBFE
4788 0603    DEC  R3
478A 16FC    JNE  4784
478C 0380    RTWP

478E 83A0    C    @4792,R14
4790 4792 G
4792 C07E  ~ MOV  *R14+,R1
4794 C0BE    MOV  *R14+,R2
4796 C0FE    MOV  *R14+,R3
4798 C252  R MOV  *R2,R9
479A 06A0    BL   @47C4
479C 47C4 G
479E DAB1    MOVB *R1+,@-2(R10)
47A0 FFFE
47A2 0603    DEC  R3
47A4 16FC    JNE  479E
47A6 0380    RTWP

47A8 83A0    C    @47AC,R14
47AA 47AC G
47AC C07E  ~ MOV  *R14+,R1
47AE C051  Q MOV  *R1,R1
47B0 10F1    JMP  4794
47B2 83A0    C    @47B6,R14
47B4 47B6 G
47B6 C07E  ~ MOV  *R14+,R1
47B8 C0BE    MOV  *R14+,R2
47BA C0FE    MOV  *R14+,R3
47BC C051  Q MOV  *R1,R1
47BE C0D3    MOV  *R3,R3
47C0 13F2    JEQ  47A6
47C2 10EA    JMP  4798
47C4 0269  i ORI  R9,4000
47C6 4000 @
47C8 06C9    SWPB R9
47CA D689    MOVB R9,*R10
47CC 06C9    SWPB R9
47CE D689    MOVB R9,*R10
47D0 045B  [ B    *R11

47D2 0420    BLWP @47D8
47D4 47D8 G
47D6 045B  [ B    *R11

47D8 83A0    C    @47DC,R14
47DA 47DC G
47DC C0E0    MOV  @2CC6,R3
47DE 2CC6 ,
47E0 0913    SRL  R3,1
47E2 38E0 8  MPY  @2CCA,R3
47E4 2CCA ,
47E6 A120    A    @2CBE,R4
```

```
47E8 2CBE ,
47EA 0209    LI   R9,0800
47EC 0800
47EE 06A0    BL   @47C4
47F0 47C4 G
47F2 C020    MOV  @2CC8,R0
47F4 2CC8 ,
47F6 C084    MOV  R4,R2
47F8 C0E0    MOV  @2CC6,R3
47FA 2CC6 ,
47FC DAB2    MOVB *R2+,@-2(R10)
47FE FFFE
4800 0603    DEC  R3
4802 16FC    JNE  47FC
4804 0224  $ AI   R4,0050
4806 0050  P
4808 0600    DEC  R0
480A 16F5    JNE  47F6
480C 0380    RTWP
480E 0203    LI   R3,4834
4810 4834 H4
4812 D133  3 MOVB *R3+,R4
4814 0984    SRL  R4,8
4816 D833  3 MOVB *R3+,@8400
4818 8400
481A 0604    DEC  R4
481C 16FC    JNE  4816
481E D0B3    MOVB *R3+,R2
4820 1308    JEQ  4832
4822 0982    SRL  R2,8
4824 0201    LI   R1,8000
4826 8000
4828 0601    DEC  R1
482A 16FE    JNE  4828
482C 0602    DEC  R2
482E 16FA    JNE  4824
4830 10F0    JMP  4812
4832 045B  [ B    *R11
4834 06BF    BL   *R15+
4836 DFFF    MOVB *R15+,*R15+
4838 8005    C    R5,R0
483A 9201    CB   R1,R8
483C 019F    UNDEFINED
483E 0000    UNDEFINED
4840 020C    LI   R12,8300     ;PME fetch address
4842 8300
4844 C060  ` MOV  @2B70,R1
4846 2B70 +p
4848 1003    JMP  4850

;Fault handling
FAULGEN   484A C05B  [ MOV  *R11,R1      ;Data after call instruction
          484C 04E0    CLR  @2AD6
          484E 2AD6 *
          4850 0281    CI   R1,0080      ;Segment fault code
          4852 0080
          4854 1605    JNE  NOSEGFLT
          4856 C805    MOV  R5,@FAEREC
```

```
           4858 2B82 +
           485A 04E0    CLR  @FAWORD
           485C 2B84 +
           485E 100B    JMP  CONT01

NOSEGFLT   4860 0281    CI   R1,0081      ;Stack fault
           4862 0081
           4864 1614    JNE  NOSTKFLT
           4866 C820    MOV  @CUREREC,@FAEREC
           4868 2AE0 *
           486A 2B82 +
           486C C0A0    MOV  @STKMISS,R2  ;Missing bytes (?) on stack
           486E 2AF0 *
           4870 0912    SRL  R2,1
           4872 C802    MOV  R2,@FAWORD
           4874 2B84 +

CONT01     4876 C820    MOV  @CURTSK,@FATIB
           4878 2ABA *
           487A 2B80 +
           487C C801    MOV  R1,@FATYPE
           487E 2B86 +
           4880 0201    LI   R1,2B7C
           4882 2B7C +|
           4884 064A  J DECT SP
           4886 C681    MOV  R1,*SP
           4888 0201    LI   R1,56B6
           488A 56B6 V
           488C 0451  Q B    *R1

NOSTKFLT   488E 064A  J DECT SP ; Not a stack fault
           4890 04DA    CLR  *SP
           4892 064A  J DECT SP
           4894 04DA    CLR  *SP
           4896 05A0    INC  @FLTCNT
4898 2AEA *
489A 064A  J DECT SP
489C C681    MOV  R1,*SP
489E C80E    MOV  R14,@GLOBPNT
48A0 2AEC *
48A2 0201    LI   R1,2A46
48A4 2A46 *F
48A6 C161  a MOV  @0002(R1),R5
48A8 0002
48AA 0203    LI   R3,0002
48AC 0002
48AE 020B    LI   R11,4D6E
48B0 4D6E Mn
48B2 0460  ` B    @4ECC
48B4 4ECC N

; Enter native code
NAT        48B6 48B8    .WORD NAT+2
           48B8 0588
           48BA 0248  H ANDI R8,FFFE      ; Make even
           48BC FFFE
           48BE 0698    BL   *R8                    ; Enter assembly
           48C0 C20B    MOV  R11,R8       ; Update IPC
```

```
          48C2 045C  \ B    *R12         ; PME Fetch

; Native code information
NAT_INFO  48C4 8316    .WORD LDUB
          48C6 A203    A    R3,R8                  ; Jump over native
information
          48C8 0460  ` B    @LOADADDR
          48CA 4142 AB

;Unused opcodes
UNUSED    48CC 48CE   .WORD UNUSED+2
          48CE 06A0   BL   @FAULGEN
          48D0 484A HJ
          48D2 000B    .WORD 11          ; Unimplemented instruction
; Breakpoint
BPT       48D4 48D6    .WORD BPT+2
          48D6 06A0    BL   @FAULGEN
          48D8 484A HJ
          48DA 0010    .WORD 16          ; Breakpoint code

; True jump
TJP       48DC 8316    .WORD LDUB
          48DE C07A  z MOV  *SP+,R1
          48E0 0911    SRL  R1,1
          48E2 1806    JOC  DOJUMP
          48E4 045C  \ B    *R12         ;PME fetch

; False jump
FJP       48E6 8316    .WORD LDUB
          48E8 C07A  z MOV  *SP+,R1
          48EA 0911    SRL  R1,1
          48EC 1701    JNC  DOJUMP
          48EE 045C  \ B    *R12         ;PME fetch

; Short jump, check which direction
DOJUMP    48F0 0283    CI   R3,0080H
          48F2 0080
          48F4 1102    JLT  JMPFWD
JMPREV    48F6 0263  c ORI  R3,FF00
          48F8 FF00
JMPFWD    48FA A203    A    R3,R8
          48FC 100F    JMP  POLLINT

; Unconditional jump
UJP       48FE 8316    .WORD LDUB
          4900 11FA    JLT  JMPREV
          4902 10FB    JMP  JMPFWD

; Equal false jump
EFJ       4904 8316    .WORD LDUB
          4906 6EBA n  S    *SP+,*SP+
          4908 16F3    JNE  DOJUMP
          490A 045C  \ B    *R12         ;PME fetch

; Not equal false jump
NFJ       490C 8316    .WORD LDUB
          490E 6EBA n  S    *SP+,*SP+
          4910 13EF    JEQ  DOJUMP
```

```
          4912 045C  \ B    *R12                    ; PME fetch

; Unconditional long jump
JPL       4914 830E    .WORD LDUBB       ; Get immediate value
          4916 D120    MOVB @8387,R4     ; Make offset -> word
          4918 8387
          491A A204    A    R4,R8                  ; Take jump
; Interrupt poll
POLLINT   491C C18C    MOV  R12,R6       ; Save fetch address
          491E 04CC    CLR  R12
          4920 1F02    TB   2
          4922 1304    JEQ  NOVPINT      ; 492C
          4924 0420    BLWP @VDPINT      ; 2AAA
          4926 2AAA *
          4928 C30C    MOV  R12,R12      ;Fetch or CRU value?
          492A 1601    JNE  492E
NOVPINT   492C C306    MOV  R6,R12       ; Restore fetch address
          492E 0460  ` B    @LOADADDR    ; 4142
          4930 4142 AB

; False long jump
FJPL      4932 830E    .WORD LDUBB       ; Get immediate value
          4934 C07A  z MOV  *SP+,R1      ; Pop boolean
          4936 0911    SRL  R1,1                  ; Move bit to carry
          4938 17EE    JNC  JPL+2                 ; 4916 False?
          493A 045C  \ B    *R12                   ; Return to fetch

; Case jump
; Parameter is base of jump table. Table starts with min and max index,
then the jump addresses
XJP       493C 8330 .WORD LDBIG
          493E 0A13    SLA  R3,1
          4940 A0E0    A    @SEGCNAB,R3  ;Current segment constant table
absolute
          4942 2ABE *
          4944 C043  C MOV  R3,R1
          4946 06A0    BL   @GETDAT
          4948 4166 Af
          494A C002    MOV  R2,R0
          494C 06A0    BL   @GETDAT
          494E 4166 Af
          4950 C0FA    MOV  *SP+,R3      ; Get index
          4952 8083    C    R3,R2                 ; Check if index in
range
          4954 1507    JGT  NOCASE
          4956 60C0 `  S    R0,R3
          4958 1105    JLT  NOCASE
          495A 0A13    SLA  R3,1
          495C A043  C A    R3,R1
          495E 06A0    BL   @GETDAT
          4960 4166 Af
          4962 A202    A    R2,R8                 ; Add the case table
jump offset to IPC
NOCASE    4964 0460  ` B    @LOADADDR
          4966 4142 AB

; Short load word constant
SLDC      4968 496A    .WORD SLDCX
```

```
SLDCX     496A 0911    SRL  R1,1                    ; Restore original
opcode
          496C 064A  J DECT SP
          496E C681    MOV  R1,*SP       ; Push the opcode
          4970 045C  \ B    *R12                    ; PME fetch

; Short load local
SLDL      4972 4974 .WORD SLDL+2
          4974 A049  I A    R9,R1         ;Frame pointer added to opcode
          4976 064A  J DECT SP
          4978 C6A1    MOV  @-54(R1),*SP ; Compensate for opcode
          497A FFCA
          497C 045C  \ B    *R12          ;PME fetch

; Short load global
SLDO      497E 4980 .WORD SLDO+2
          4980 A04E  N A   R14,R1         ;Global frame pointer added to
opcode
          4982 064A  J DECT SP
          4984 C6A1    MOV  @-86(R1),*SP
          4986 FFAA
          4988 045C  \ B    *R12          ;PME fetch

; Short index
SIND      498A 498C .WORD SIND+2
          498C 0221  ! AI   R1,-240
          498E FF10
          4990 A05A  Z A    *SP,R1
          4992 C691    MOV  *R1,*SP
          4994 045C  \ B    *R12          ;PME fetch

;Reserved
RESERVE6  4996 8300    .WORD FETCH
NOP       .EQU      RESERVE6

; Load byte
LDB       4998 499A .WORD LDB+2
          499A C07A  z MOV  *SP+,R1
          499C A05A  Z A    *SP,R1
          499E 04DA    CLR  *SP
          49A0 DA91    MOVB *R1,@0001(SP)
          49A2 0001
          49A4 045C  \ B    *R12          ;PME fetch

; Store byte
STB       49A6 49A8 .WORD STB+2
          49A8 C06A  j MOV  @2(SP),R1
          49AA 0002
          49AC A06A  j A    @4(SP),R1
          49AE 0004
          49B0 D46A  j MOVB @1(SP),*R1
          49B2 0001
          49B4 022A  * AI   SP,0006
          49B6 0006
          49B8 045C  \ B    *R12          ;PME fetch

; Index
IND       49BA 8330  0 .WORD LDBIG
```

```
          49BC 0A13    SLA  R3,1
          49BE A0DA    A    *SP,R3
          49C0 C693    MOV  *R3,*SP
          49C2 045C  \ B    *R12         ;PME fetch

; Increment
INC       49C4 8330  0 .WORD LDBIG
          49C6 0A13    SLA  R3,1
          49C8 A683    A    R3,*SP
          49CA 045C  \ B    *R12         ;PME fetch

; Index Array
IXA       49CC 8330  0 .WORD LDBIG
          49CE C03A  : MOV  *SP+,R0
          49D0 1303    JEQ  ZEROIX       ; Index is zero
          49D2 3803 8  MPY  R3,R0
          49D4 0A11    SLA  R1,1
          49D6 A681    A    R1,*SP
ZEROIX    49D8 045C  \ B    *R12         ;PME fetch

; Load constant NIL
LDCN      49DA 49DC I  SZC  .WORD LDCN+2
          49DC 064A  J
          49DE 04DA    CLR  *SP
          49E0 045C  \ B    *R12         ;PME fetch


;Load constant integer
LDCI      49E2 830E    .WORD LDUBB
          49E4 D120    MOVB @8387,R4     ;Assemble two bytes to one word
          49E6 8387
          49E8 064A  J DECT SP
          49EA C684    MOV  R4,*SP
          49EC 045C  \ B    *R12                 ;PME fetch

; Load local
          49EE 8330  .WORD LDBIG
          49F0 0A13    SLA  R3,1
          49F2 A0C9    A    R9,R3                ;Frame pointer
          49F4 064A  J DECT SP
          49F6 C6A3    MOV  @0008(R3),*SP
          49F8 0008
          49FA 045C  \ B    *R12         ;PME fetch

;Load local address
LLA       49FC 8330  0 .WORD LDBIG
          49FE 0A13    SLA  R3,1
          4A00 0223  # AI   R3,0008      ;Subtract 8, since base is zero
here
          4A02 0008
          4A04 C043  C MOV  R3,R1
          4A06 1005    JMP  ADRLOAD

;Short load local address
SLLA      4A08 4A0A    .WORD SLLAX
SLLAX     4A0A 0221  ! AI   R1,-182      ;Opcode *2 is 192 for SLLA1
          4A0C FF4A  J
          4A0E 0241  A ANDI R1,FFFE
```

```
          4A10 FFFE
ADRLOAD   4A12 A049  I A    R9,R1        ;Add frame pointer
          4A14 064A  J DECT SP
          4A16 C681    MOV  R1,*SP
          4A18 045C  \ B    *R12         ;PME Fetch

; Store local
STL       4A1A 8330  0 .WORD LDBIG
          4A1C 0A13    SLA  R3,1
          4A1E C043  C MOV  R3,R1
          4A20 1003    JMP  STOLOC

;Short store local
SSTL      4A22 4A24    .WORD SSTLX
SSTLX     4A24 0221    LI        R1,-206
          4A26 FF32  2
STOLOC    4A28 A049  I A    R9,R1        ;Frame pointer
          4A2A C87A  z MOV  *SP+,@8(R1)
          4A2C 0008
          4A2E 045C  \ B    *R12         ;PME fetch

;Load global
LDO       4A30 8330 .WORD LDIBG
          4A32 0A13    SLA  R3,1
          4A34 A0CE    A    R14,R3       ;Global frame pointer
          4A36 064A  J DECT SP
          4A38 C6A3    MOV  @8(R3),*SP
          4A3A 0008
4A3C 045C  \ B    *R12         ;PME fetch

; Load address global
LAO       4A3E 8330  .WORD LDBIG
          4A40 0A13    SLA  R3,1
          4A42 A0CE    A    R14,R3       ;Global data frame pointer
          4A44 0223  # AI   R3,8
          4A46 0008
          4A48 064A  J DECT SP
          4A4A C683    MOV  R3,*SP
          4A4C 045C  \ B    *R12         ;PME fetch

; Store global
SRO       4A4E 8330 .WORD LDBIG
          4A50 0A13    SLA  R3,1
          4A52 A0CE    A    R14,R3       ;Global data frame pointer
          4A54 C8FA    MOV  *SP+,@8(R3)
          4A56 0008
          4A58 045C  \ B    *R12         ;PME fetch

; Short load intermediate
SLOD1     4A5A 8330 .WORD LDBIG
          4A5C 0204    LI   R4,0001
          4A5E 0001
          4A60 1005    JMP  LODX
; Short load intermediate
SLOD2     4A62 8330  .WORD LDBIG
          4A64 0204    LI   R4,0002
          4A66 0002
          4A68 1001    JMP  LODX
```

```
; Load intermediate
LOD       4A6A 8328  .WORD LDUBIG
LODX      4A6C C089    MOV  R9,R2                  ;Local frame pointer
          4A6E C092    MOV  *R2,R2
          4A70 0604    DEC  R4
          4A72 15FD    JGT  4A6E
          4A74 0A13    SLA  R3,1
          4A76 A0C2    A    R2,R3
          4A78 064A  J DECT SP
          4A7A C6A3    MOV  @0008(R3),*SP
          4A7C 0008
          4A7E 045C  \ B    *R12         ;PME fetch

; Load intermediate address
LDA       4A80 8328  .WORD LDUBIG
          4A82 C089    MOV  R9,R2                  ;Local frame pointer
LOOP06    4A84 C092    MOV  *R2,R2       ; Traverse list of activation
records
          4A86 0604    DEC  R4
          4A88 15FD    JGT  LOOP06
          4A8A 0A13    SLA  R3,1
          4A8C 0223  # AI   R3,8
          4A8E 0008
          4A90 A0C2    A    R2,R3
          4A92 064A  J DECT SP
          4A94 C683    MOV  R3,*SP
          4A96 045C  \ B    *R12         ;PME fetch

; Store intermediate
STR       4A98 8328  .WORD LDUBIG
          4A9A C089    MOV  R9,R2                  ; Local frame
LOOP05    4A9C C092    MOV  *R2,R2       ; Traverse list of activation
records
          4A9E 0604    DEC  R4
          4AA0 15FD    JGT  LOOP05
          4AA2 0A13    SLA  R3,1
          4AA4 A0C2    A    R2,R3
          4AA6 C8FA    MOV  *SP+,@8(R3)
          4AA8 0008
          4AAA 045C  \ B    *R12         ;PME fetch

; Store
STO       4AAC 4AAE .WORD STO+2
          4AAE C06A  j MOV  @2(SP),R1
          4AB0 0002
          4AB2 C47A  z MOV  *SP+,*R1
          4AB4 05CA    INCT SP
          4AB6 045C  \ B    *R12                  ;PME fetch

;Find address with offset R3 in global activation record of segment R4
FNDOFFS   4AB8 0A14    SLA  R4,1                   ;Segment number to
word
          4ABA A120    A    @EVECP,R4    ;Point into Environment vector
          4ABC 2AB8 *
          4ABE C114    MOV  *R4,R4       ;Get pointer to pointer...
          4AC0 C124  $ MOV  @0(R4),R4
          4AC2 0000
```

```
          4AC4 0A13    SLA  R3,1
          4AC6 A103    A    R3,R4                  ;Add offset
          4AC8 045B  [ B    *R11

; Load extended
LDE       4ACA 8328    .WORD LDUBIG
          4ACC 06A0    BL   @FNDOFFS
          4ACE 4AB8 J
          4AD0 064A  J DECT SP
          4AD2 C6A4    MOV  @8(R4),*SP
          4AD4 0008
          4AD6 045C  \ B    *R12          ; PME fetch

;Load extended address
LAE       4AD8 8328  ( .WORD LDUBIG
          4ADA 06A0    BL   @FNDOFFS
          4ADC 4AB8 J
          4ADE 0224  $ AI   R4,8                   ;MCSW offset
          4AE0 0008
          4AE2 064A  J DECT SP
          4AE4 C684    MOV  R4,*SP
          4AE6 045C  \ B    *R12                   ;PME fetch

; Store extended
STE       4AE8 8328     .WORD LDUBIG
          4AEA 06A0    BL   @FNDOFFS
          4AEC 4AB8 J
          4AEE C93A  : MOV  *SP+,@8(R4)
          4AF0 0008
          4AF2 045C  \ B    *R12                   ;PME fetch

; Load multiple
LDM       4AF4 8316    .WORD LDUB
          4AF6 C808    MOV  R8,@IPCSAVE
          4AF8 2ACC *
          4AFA 0660  ` DECT @IPCSAVE
          4AFC 2ACC *
          4AFE 0A13    SLA  R3,1
          4B00 C803    MOV  R3,@STKMISS
          4B02 2AF0 *
          4B04 C820    MOV  @ERECP,@CUREREC
          4B06 2AB4 *
          4B08 2AE0 *
          4B0A 06A0    BL   @CHKSTK
          4B0C 4E98 N
          4B0E C0BA    MOV  *SP+,R2
          4B10 A0A0    A    @STKMISS,R2
          4B12 2AF0 *
          4B14 0642  B DECT R2
          4B16 0283    CI   R3,0000
          4B18 0000
          4B1A 1305    JEQ  EXLDM
LOOP19    4B1C 064A  J DECT SP
          4B1E C692    MOV  *R2,*SP
          4B20 0642  B DECT R2
          4B22 0643  C DECT R3
          4B24 15FB    JGT  LOOP19
EXLDM     4B26 045C  \ B    *R12          ;PME fetch
```

```
;Store multiple
STM       4B28 8316 .WORD LDUB
          4B2A 1307    JEQ  ZCNT01       ; Zero count
          4B2C C04A  J MOV  SP,R1
          4B2E A043  C A    R3,R1
          4B30 A043  C A    R3,R1
          4B32 C051  Q MOV  *R1,R1
LOOP07    4B34 CC7A  z MOV  *SP+,*R1+
          4B36 0603    DEC  R3
          4B38 15FD    JGT  LOOP07
ZCNT01    4B3A 05CA    INCT SP
          4B3C 045C  \ B    *R12         ;PME fetch

; Load static link
LSL       4B3E 8316 .WORD LDUB
          4B40 C089    MOV  R9,R2        ; Local frame pointer
          4B42 C0C3    MOV  R3,R3
          4B44 1303    JEQ  DONE01
LOOP08    4B46 C092    MOV  *R2,R2
          4B48 0603    DEC  R3
          4B4A 15FD    JGT  LOOP08
DONE01    4B4C 064A  J DECT SP
          4B4E C682    MOV  R2,*SP
          4B50 045C  \ B    *R12         ;PME fetch

;Load constant byte
LDCB      4B52 8316    .WORD LDUB
          4B54 064A  J DECT SP
          4B56 C683    MOV  R3,*SP
          4B58 045C  \ B    *R12                  ;PME fetch

; Move
MOV       4B5A 8328 .WORD LDUBIG
          4B5C C043  C MOV  R3,R1                 ; Words to move
          4B5E 1317    JEQ  ZCNT02
          4B60 C0FA    MOV  *SP+,R3       ; Address or offset
          4B62 C0BA    MOV  *SP+,R2       ; Address
          4B64 C144  D MOV  R4,R5                 ; Type
          4B66 1303    JEQ  BLOCK                 ; Block address
          4B68 A0E0    A    @SEGBASE,R3   ; Constant block in current
segment
          4B6A 2AE2 *
          4B6C 1011    JMP  LDCON4
BLOCK     4B6E 8820    C    @PMEFLIP,@CONTRUE      ; Check flip
indicator for current segment
          4B70 2AE8 *
          4B72 40B6 @
          4B74 1309    JEQ  LOOP09        ; Same sex, just continue
          4B76 0285    CI   R5,0002       ; Byte flipping if 2
          4B78 0002
          4B7A 1606    JNE  LOOP09
LOOP10    4B7C CCB3    MOV  *R3+,*R2+     ; Opposite sex, do byte
flipping
          4B7E 06E2    SWPB @FFFE(R2)
          4B80 FFFE
          4B82 0601    DEC  R1
          4B84 15FB    JGT  LOOP10
```

```
          4B86 045C  \ B    *R12          ;PME fetch
LOOP09    4B88 CCB3    MOV  *R3+,*R2+     ; Just move, no flipping
          4B8A 0601    DEC  R1
          4B8C 15FD    JGT  LOOP09
ZCNT02    4B8E 045C  \ B    *R12          ;PME fetch

LDCON4    4B90 C3CF    MOV  R15,R15       ;Code type flag
          4B92 13ED    JEQ  4B6E          ;CPU RAM
          4B94 C108    MOV  R8,R4
          4B96 C203    MOV  R3,R8
          4B98 06A0    BL   @LOADAD1
          4B9A 4144 AD
          4B9C C020    MOV  @PMEFLIP,R0
          4B9E 2AE8 *
          4BA0 160C    JNE  LOOP04
          4BA2 0285    CI   R5,0002
          4BA4 0002
          4BA6 1609    JNE  LOOP04
LOOP03    4BA8 D01D    MOVB *R13,R0       ;Code read data
          4BAA 06C0    SWPB R0
          4BAC D01D    MOVB *R13,R0       ;Code read data
          4BAE CC80    MOV  R0,*R2+
          4BB0 0601    DEC  R1
          4BB2 15FA    JGT  LOOP03
LDCON5    4BB4 C204    MOV  R4,R8
          4BB6 0460  ` B    @LOADADDR
          4BB8 4142 AB

LOOP04    4BBA D01D    MOVB *R13,R0       ;Code read data
          4BBC 1000    NOP
          4BBE D81D    MOVB *R13,@8381    ;Code read data
          4BC0 8381
          4BC2 CC80    MOV  R0,*R2+
          4BC4 0601    DEC  R1
          4BC6 15F9    JGT  LOOP04
          4BC8 10F5    JMP  LDCON5

CON008    4BCA 0008    .WORD 8

; Load constant real
LDCRL     4BCC 4BCE K  .WORD LDCRL+2
          4BCE C808    MOV  R8,@IPCSAVE   ;Set up for stack fault check
          4BD0 2ACC *
          4BD2 0620    DEC  @IPCSAVE
          4BD4 2ACC *
          4BD6 0202    LI   R2,CONT02
          4BD8 4BDE K
          4BDA 0460  ` B    @LDBIG
          4BDC 8330  0
CONT02    4BDE 0201    LI   R1,0004
          4BE0 0004
          4BE2 04C5    CLR  R5
          4BE4 C820    MOV  @CON008,@STKMISS
          4BE6 4BCA K
          4BE8 2AF0 *
          4BEA 1016    JMP  LDCON3

;Load constant
```

```
LDC       4BEC 4BEE    .WORD LDC+2
          4BEE C808    MOV  R8,@IPCSAVE  ;Save in case of stack fault
          4BF0 2ACC *
          4BF2 0620    DEC  @IPCSAVE
          4BF4 2ACC *
          4BF6 0202    LI   R2,LDCCON1
          4BF8 4BFE K
          4BFA 0460  ` B    @LDUBIG      ;Mode and word offset
          4BFC 8328  (
LDCCON1   4BFE C144  D MOV  R4,R5
          4C00 C103    MOV  R3,R4
          4C02 0202    LI   R2,LDCCON2
          4C04 4C0A L
          4C06 0460  ` B    @LDUB                  ;Byte count
          4C08 8316
LDCCON2   4C0A C043  C MOV  R3,R1
          4C0C 1601    JNE  4C10
          4C0E 045C  \ B    *R12                   ;PME fetch
          4C10 0A13    SLA  R3,1
          4C12 C803    MOV  R3,@STKMISS  ;To stack fault data missing
          4C14 2AF0 *
          4C16 C0C4    MOV  R4,R3
LDCON3    4C18 C820    MOV  @ERECP,@CUREREC        ;E_Rec_P to current
environment
          4C1A 2AB4 *
          4C1C 2AE0 *
          4C1E 06A0    BL   @CHKSTK
          4C20 4E98 N
          4C22 6281 b  S    R1,R10
          4C24 6281 b  S    R1,R10
          4C26 C08A    MOV  R10,R2
          4C28 0A13    SLA  R3,1
          4C2A A0E0    A    @SEGCNAB,R3
          4C2C 2ABE *
          4C2E 10B0    JMP  LDCON4

;Load constant offset
LCO       4C30 8330    .WORD LDBIG
          4C32 0A13    SLA  R3,1
          4C34 A0E0    A    @2AC0,R3     ;Current segment constant pool
pointer (relative)
          4C36 2AC0 *
          4C38 064A  J DECT SP
          4C3A C683    MOV  R3,*SP
          4C3C 045C  \ B    *R12         ; PME fetch

; Swap at top of stack
SWAP      4C3E 4C40    .WORD SWAP+2
          4C40 C05A  Z MOV  *SP,R1
          4C42 C6AA    MOV  @0002(SP),*SP
          4C44 0002
          4C46 CA81    MOV  R1,@0002(SP)
          4C48 0002
          4C4A 045C  \ B    *R12         ; PME fetch

; Duplicate one word
DUPI      4C4C 4C4E    .WORD DUPI+2
          4C4E 064A  J DECT SP
```

```
          4C50 C6AA    MOV  @0002(SP),*SP
          4C52 0002
          4C54 045C  \ B    *R12         ; PME fetch

; Duplicate real
DUPR      4C56 4C58    .WORD DUPR+2
          4C58 C04A  J MOV  SP,R1
          4C5A 022A  * AI   SP,-8
          4C5C FFF8
          4C5E C08A    MOV  SP,R2
          4C60 CCB1    MOV  *R1+,*R2+
          4C62 CCB1    MOV  *R1+,*R2+
          4C64 CCB1    MOV  *R1+,*R2+
          4C66 C491    MOV  *R1,*R2
          4C68 045C  \ B    *R12         ; PME fetch

;Load processor register
; Loads/stores in following registers
; -1 CURTASK
; -2 EVEC
; -3 READYQ
; If top of stack isn't negative access is into TIB

LPR       4C6A 4C6C .WORD LPRX
LPRX      4C6C C07A  z MOV  *SP+,R1
          4C6E 0A11    SLA  R1,1
          4C70 1107    JLT  REGNEG1
          4C72 A060  ` A    @CURTSK,R1
          4C74 2ABA *
          4C76 06A0    BL   @UPDTIB      ; Update register copies in
Task Info Block
          4C78 579A W
          4C7A 064A  J DECT SP
          4C7C C691    MOV  *R1,*SP
          4C7E 045C  \ B    *R12         ; PME fetch
REGNEG1   4C80 064A  J DECT SP
          4C82 C6A1    MOV  @CURTSK+2(R1),*SP
          4C84 2ABC *
          4C86 045C  \ B    *R12         ; PME fetch

; Store processor register
SPR       4C88 4C8A    .WORD SPR+2
          4C8A C03A  : MOV  *SP+,R0
          4C8C C07A  z MOV  *SP+,R1
          4C8E 0A11    SLA  R1,1
          4C90 1109    JLT  REGNEG2
          4C92 06A0    BL   @UPDTIB      ; Update register copies in
Task Info Block
          4C94 579A W
          4C96 A060  ` A    @CURTSK,R1
          4C98 2ABA *
          4C9A C440  @ MOV  R0,*R1
          4C9C 020B    LI   R11,4142     ; Fake return address
          4C9E 4142 AB
          4CA0 0460  ` B    @57BE
          4CA2 57BE W
REGNEG2   4CA4 C840  @ MOV  R0,@CURTSK+2(R1)
          4CA6 2ABC *
```

```
          4CA8 045C  \ B    *R12         ; PME fetch

; Short call intermediate procedure
SCPI1     4CAA 8316    .WORD LDUB
          4CAC 0204    LI   R4,0001
          4CAE 0001
          4CB0 1003    JMP  SCPIX
SCPI2     4CB2 8316    .WORD LDUB
          4CB4 0204    LI   R4,0002
          4CB6 0002
SCPIX     4CB8 C808    MOV  R8,@IPCSAVE
          4CBA 2ACC *
          4CBC 0660  ` DECT @IPCSAVE
          4CBE 2ACC *
          4CC0 020B    LI   R11,CLP02
          4CC2 4CF2 L
          4CC4 0460  ` B    @TRAVLOC     ; Traverse activation records
          4CC6 4DB4 M

; Call intermediate procedure
CIP       4CC8 830E    .WORD LDUBB
          4CCA C088    MOV  R8,R2
          4CCC 0222  " AI   R2,-3
          4CCE FFFD
          4CD0 C802    MOV  R2,@IPCSAVE
          4CD2 2ACC *
          4CD4 020B    LI   R11,CLP02
          4CD6 4CF2 L
          4CD8 0460  ` B    @TRAVLOC     ; Traverse activation records
          4CDA 4DB4 M

; Call local procedure
CLP       4CDC 8316    .WORD LDUB
          4CDE C809    MOV  R9,@GLOBPNT  ; Local frame pointer
          4CE0 2AEC *
          4CE2 1003    JMP  CLP01

; Call global procedure
CGP       4CE4 8316    .WORD LDUB
          4CE6 C80E    MOV  R14,@GLOBPNT ; Global frame pointer
          4CE8 2AEC *
CLP01     4CEA C808    MOV  R8,@IPCSAVE  ; Save IPC
          4CEC 2ACC *
          4CEE 0660  ` DECT @IPCSAVE
          4CF0 2ACC *
CLP02     4CF2 C120    MOV  @DICTPNT,R4  ; Dictionary pointer
          4CF4 2AE4 *
          4CF6 C160  ` MOV  @ERECP,R5
          4CF8 2AB4 *
          4CFA C1A0    MOV  @SEGBASE,R6
          4CFC 2AE2 *
          4CFE 06A0    BL   @4DC6
          4D00 4DC6 M
          4D02 0460  ` B    @LOADADDR    ; Jump in p-code
          4D04 4142 AB

; Call external intermediate
CXI       4D06 8316    .WORD LDUB
```

```
          4D08 C808    MOV  R8,@IPCSAVE
          4D0A 2ACC *
          4D0C 0660  ` DECT @IPCSAVE
          4D0E 2ACC *
          4D10 C143  C MOV  R3,R5
          4D12 0202    LI   R2,CONT03
          4D14 4D1A M
          4D16 0460  ` B    @LDUBB
          4D18 830E
CONT03    4D1A 020B    LI   R11,CALX1
          4D1C 4D6A Mj
          4D1E 0460  ` B    @TRAVLOC     ; Traverse activation records
          4D20 4DB4 M

; Call external local
CXL       4D22 830E    .WORD LDUBB
          4D24 C809    MOV  R9,@GLOBPNT  ; Local frame pointer
          4D26 2AEC *
          4D28 101A    JMP  CALX2

;Short call external global 1
SCXG1     4D2A 8316    .WORD LDUB
          4D2C 06A0    BL   @4DA0
          4D2E 4DA0 M
          4D30 C160  ` MOV  @CON001,R5
          4D32 40B0 @
          4D34 1005    JMP  CALX3

;Short call external global
SCXG      4D36 8316    .WORD LDUB
          4D38 0911    SRL  R1,1
          4D3A 0221  ! AI   R1,-111
          4D3C FF91
          4D3E C141  A MOV  R1,R5
CALX3     4D40 C808    MOV  R8,@IPCSAVE
          4D42 2ACC *
          4D44 0660  ` DECT @IPCSAVE
          4D46 2ACC *
          4D48 C80E    MOV  R14,@GLOBPNT
          4D4A 2AEC *
          4D4C 100E    JMP  CALX1

; Call external global
CXG       4D4E 830E    .WORD LDUBB
          4D50 0284    CI   R4,0001
          4D52 0001
          4D54 1602    JNE  4D5A
          4D56 06A0    BL   @4DA0
          4D58 4DA0 M
          4D5A C80E    MOV  R14,@GLOBPNT ; Global frame pointer
          4D5C 2AEC *
CALX2     4D5E C088    MOV  R8,R2
          4D60 0222  " AI   R2,FFFD
          4D62 FFFD
          4D64 C802    MOV  R2,@IPCSAVE
          4D66 2ACC *
          4D68 C144  D MOV  R4,R5
CALX1     4D6A 06A0    BL   @4EC4
```

```
          4D6C 4EC4 N
CALX4     4D6E C184    MOV  R4,R6
          4D70 06A0    BL   @4FCA
          4D72 4FCA O
          4D74 06A0    BL   @4DC6
          4D76 4DC6 M
          4D78 06A0    BL   @4E6C
          4D7A 4E6C Nl
          4D7C 0460  ` B    @LOADADDR
          4D7E 4142 AB

; Call formal procedure
CFP       4D80 4D82    .WORD CFP+2
          4D82 C808    MOV  R8,@IPCSAVE
          4D84 2ACC *
          4D86 0620    DEC  @IPCSAVE
          4D88 2ACC *
          4D8A 05A0    INC  @2ACE
          4D8C 2ACE *
          4D8E C16A  j MOV  @0002(SP),R5 ; Environment record pointer
          4D90 0002
          4D92 06A0    BL   @4ECC
          4D94 4ECC N
          4D96 C0DA    MOV  *SP,R3       ; Procedure number
          4D98 C82A  * MOV  @4(SP),@GLOBPNT
          4D9A 0004
          4D9C 2AEC *
          4D9E 10E7    JMP  CALX4

          4DA0 0283    CI   R3,38                  ; Number of Kernel
procedures
          4DA2 0026  &
          4DA4 1506    JGT  RET01
          4DA6 0A13    SLA  R3,1
          4DA8 C0A3    MOV  @KERNPROC-2(R3),R2     ; Kernel procedure
table index
          4DAA 2FB0 /
          4DAC 1301    JEQ  NOLINK
          4DAE 0452  R B    *R2                    ; Call Kernel
procedure
NOLINK    4DB0 0813    SRA  R3,1
RET01     4DB2 045B  [ B    *R11

; Traverse stack activation records
TRAVLOC   4DB4 C089    MOV  R9,R2                  ; Local frame pointer
          4DB6 C104    MOV  R4,R4                  ; Level count
          4DB8 1303    JEQ  ENDLEV
LOOP11    4DBA C092    MOV  *R2,R2
          4DBC 0604    DEC  R4
          4DBE 15FD    JGT  LOOP11
ENDLEV    4DC0 C802    MOV  R2,@GLOBPNT
          4DC2 2AEC *
          4DC4 045B  [ B    *R11

; Called when Calling a procedure
          4DC6 C80B    MOV  R11,@RETSAV2
          4DC8 2ADC *
          4DCA C043  C MOV  R3,R1
```

```
          4DCC 0A11    SLA  R1,1
          4DCE 0501    NEG  R1
          4DD0 A044  D A    R4,R1
          4DD2 06A0    BL   @GETDAT2
          4DD4 4178 Ax
          4DD6 0A12    SLA  R2,1
          4DD8 A086    A    R6,R2
          4DDA C042  B MOV  R2,R1
          4DDC 06A0    BL   @GETDAT2
          4DDE 4178 Ax
          4DE0 C801    MOV  R1,@SAVPACK
          4DE2 2AF2 *
          4DE4 04E0    CLR  @CURTSK+2
4DE6 2ABC *
4DE8 0A12    SLA  R2,1
4DEA 1504    JGT  4DF4
4DEC 1303    JEQ  4DF4
4DEE 0502    NEG  R2
4DF0 05A0    INC  @CURTSK+2
4DF2 2ABC *
4DF4 0222  " AI   R2,000A
4DF6 000A
4DF8 C020    MOV  @FLTCNT,R0
4DFA 2AEA *
4DFC 1606    JNE  4E0A
4DFE C802    MOV  R2,@STKMISS
4E00 2AF0 *
4E02 C805    MOV  R5,@CURE REC
4E04 2AE0 *
4E06 06A0    BL   @CHKSTK
4E08 4E98 N
4E0A 04E0    CLR  @FLTCNT
4E0C 2AEA *
4E0E C020    MOV  @2ACE,R0
4E10 2ACE *
4E12 1304    JEQ  4E1C
4E14 022A  * AI   R10,0006
4E16 0006
4E18 04E0    CLR  @2ACE
4E1A 2ACE *
4E1C C020    MOV  @CURTSK+2,R0
4E1E 2ABC *
4E20 130B    JEQ  4E38
4E22 04E0    CLR  @CURTSK+2
4E24 2ABC *
4E26 0691    BL   *R1
4E28 C160  ` MOV  @ERECP,R5
4E2A 2AB4 *
4E2C 06A0    BL   @4ECC
4E2E 4ECC N
4E30 06A0    BL   @4FCA
4E32 4FCA O
4E34 0460  ` B    @LOADADDR
4E36 4142 AB
4E38 6282 b  S    R2,R10
4E3A 6220 b  S    @SEGBASE,R8
4E3C 2AE2 *
4E3E CAA0    MOV  @CURPRNO-1,@0008(R10)
```

```
4E40 2AC2 *
4E42 0008
4E44 CAA0    MOV  @ERECP,@0006(R10)
4E46 2AB4 *
4E48 0006
4E4A CA88    MOV  R8,@0004(R10)
4E4C 0004
4E4E CA89    MOV  R9,@0002(R10)
4E50 0002
4E52 CAA0    MOV  @GLOBPNT,@0000(R10)
4E54 2AEC *
4E56 0000
4E58 C201    MOV  R1,R8
4E5A C803    MOV  R3,@CURPRNO-1
4E5C 2AC2 *
4E5E C24A  J MOV  R10,R9
4E60 064A  J DECT R10
4E62 C6A0    MOV  @SAVPACK,*R10
4E64 2AF2 *
4E66 C2E0    MOV  @RETSAV2,R11
4E68 2ADC *
4E6A 045B  [ B    *R11
4E6C C805    MOV  R5,@ERECP
4E6E 2AB4 *
4E70 C825  % MOV  @0002(R5),@EVECP
4E72 0002
4E74 2AB8 *
4E76 C3A5    MOV  @0000(R5),R14
4E78 0000
4E7A C165  e MOV  @0004(R5),R5
4E7C 0004
4E7E C805    MOV  R5,@CURSIB
4E80 2AB2 *
4E82 C806    MOV  R6,@SEGBASE
4E84 2AE2 *
4E86 C0A0    MOV  @CURSIB,R2
4E88 2AB2 *
4E8A A8A0    A    @40B4,@0004(R2)
4E8C 40B4 @
4E8E 0004
4E90 1902    JNO  4E96
4E92 0460  ` B    @5A60
4E94 5A60 Z`
4E96 045B  [ B    *R11

;Stack fault check
CHKSTK    4E98 C120    MOV  @CURTSK,R4   ;Current TIB
          4E9A 2ABA *
          4E9C C124  $ MOV  @0004(R4),R4
          4E9E 0004
          4EA0 0224  $ AI   R4,0050      ;80 bytes, 40 words
          4EA2 0050  P
          4EA4 1805    JOC  4EB0
          4EA6 A120    A    @STKMISS,R4
          4EA8 2AF0 *
          4EAA 1802    JOC  4EB0
          4EAC 8284    C    R4,R10
          4EAE 1A09    JL   4EC2
```

```
          4EB0 0204    LI   R4,0050      ;80 bytes, 40 words
          4EB2 0050  P
          4EB4 A804    A    R4,@STKMISS
          4EB6 2AF0 *
          4EB8 C220    MOV  @IPCSAVE,R8
          4EBA 2ACC *
          4EBC 06A0    BL   @FAULGEN
          4EBE 484A HJ
          4EC0 0081    .WORD 81H         ;Data status stack fault
          4EC2 045B  [ B    *R11

4EC4 0A15    SLA  R5,1
4EC6 A160  ` A    @EVECP,R5
4EC8 2AB8 *
4ECA C155  U MOV  *R5,R5

; Called when calling a procedure
4ECC C00F    MOV  R15,R0
4ECE C065  e MOV  @0004(R5),R1 ; R5 is Environment record
4ED0 0004
4ED2 C111    MOV  *R1,R4
4ED4 1302    JEQ  4EDA
4ED6 04C2    CLR  R2
4ED8 105D  ] JMP  4F94
4EDA C0A1    MOV  @0020(R1),R2
4EDC 0020
4EDE 1305    JEQ  4EEA
4EE0 C220    MOV  @IPCSAVE,R8
4EE2 2ACC *
4EE4 06A0    BL   @FAULGEN
4EE6 484A HJ
4EE8 0080    .WORD 80H         ;Segment fault
4EEA C121  ! MOV  @0016(R1),R4
4EEC 0016
4EEE C114    MOV  *R4,R4
4EF0 0284    CI   R4,000D
4EF2 000D
4EF4 1103    JLT  4EFC
4EF6 0284    CI   R4,001F
4EF8 001F
4EFA 112A  * JLT  4F50
4EFC C121  ! MOV  @0002(R1),R4
4EFE 0002
4F00 1624  $ JNE  4F4A
4F02 C0A0    MOV  @CURSIB,R2
4F04 2AB2 *
4F06 C803    MOV  R3,@SAVPACK
4F08 2AF2 *
4F0A 04C4    CLR  R4
4F0C C0E2    MOV  @0002(R2),R3
4F0E 0002
4F10 1302    JEQ  4F16
4F12 C122  " MOV  @0012(R2),R4
4F14 0012
4F16 C0A0    MOV  @2B88,R2
4F18 2B88 +
4F1A 1308    JEQ  4F2C
4F1C C0E2    MOV  @0008(R2),R3
```

```
4F1E 0008
4F20 1302    JEQ  4F26
4F22 A122  " A    @0012(R2),R4
4F24 0012
4F26 C0A2    MOV  @001C(R2),R2
4F28 001C
4F2A 16F8    JNE  4F1C
4F2C C0E0    MOV  @SAVPACK,R3
4F2E 2AF2 *
4F30 A121  ! A    @0012(R1),R4
4F32 0012
4F34 C0A0    MOV  @2782,R2
4F36 2782 '
4F38 60A0 `  S    @2780,R2
4F3A 2780 '
4F3C 0912    SRL  R2,1
4F3E 8084    C    R4,R2
4F40 15CF    JGT  4EE0
4F42 C08B    MOV  R11,R2
4F44 06A0    BL   @5A9E
4F46 5A9E Z
4F48 C2C2    MOV  R2,R11
4F4A 0202    LI   R2,0002
4F4C 0002
4F4E 1022  " JMP  4F94
4F50 0224  $ AI   R4,FFF4
4F52 FFF4
4F54 0A14    SLA  R4,1
4F56 C084    MOV  R4,R2
4F58 C121  ! MOV  @0014(R1),R4
4F5A 0014
4F5C 05C4    INCT R4
4F5E 0A14    SLA  R4,1
4F60 A122  " A    @2D6C(R2),R4
4F62 2D6C -l
4F64 C062  b MOV  @2CFA(R2),R1
4F66 2CFA ,
4F68 C822  " MOV  @2D20(R2),@SAVPACK
4F6A 2D20 -
4F6C 2AF2 *
4F6E 1107    JLT  4F7E
4F70 D844  D MOVB R4,@0402(R1)
4F72 0402
4F74 1000    JMP  4F76
4F76 D860  ` MOVB @8389,@0402(R1)
4F78 8389
4F7A 0402
4F7C 1007    JMP  4F8C
4F7E D860  ` MOVB @8389,@0402(R1)
4F80 8389
4F82 0402
4F84 1000    JMP  4F86
4F86 D844  D MOVB R4,@0402(R1)
4F88 0402
4F8A 1000    JMP  4F8C
4F8C D111    MOVB *R1,R4
4F8E 1000    JMP  4F90
4F90 D811    MOVB *R1,@8389
```

```
4F92 8389
4F94 C060  ` MOV  @2AD6,R1
4F96 2AD6 *
4F98 1307    JEQ  4FA8
4F9A C062  b MOV  @2CFA(R2),R1
4F9C 2CFA ,
4F9E C022  " MOV  @2D20(R2),R0
4FA0 2D20 -
4FA2 04E0    CLR  @2AD6
4FA4 2AD6 *
4FA6 045B  [ B    *R11
4FA8 C362  b MOV  @2CFA(R2),R13
4FAA 2CFA ,
4FAC C3E2    MOV  @2D20(R2),R15
4FAE 2D20 -
4FB0 280F (  XOR  R15,R0
4FB2 0910    SRL  R0,1
4FB4 1709    JNC  4FC8
4FB6 C062  b MOV  @2D46(R2),R1
4FB8 2D46 -F
4FBA 0202    LI   R2,8300
4FBC 8300
4FBE 0200    LI   R0,0025
4FC0 0025  %
4FC2 CCB1    MOV  *R1+,*R2+
4FC4 0600    DEC  R0
4FC6 15FD    JGT  4FC2
4FC8 045B  [ B    *R11

4FCA C044  D MOV  R4,R1
4FCC 0221  ! AI   R1,000C
4FCE 000C
4FD0 05A0    INC  @PMEFLIP
4FD2 2AE8 *
4FD4 C00B    MOV  R11,R0
4FD6 06A0    BL   @GETDAT2
4FD8 4178 Ax
4FDA 04E0    CLR  @PMEFLIP
4FDC 2AE8 *
4FDE 0282    CI   R2,0001
4FE0 0001
4FE2 1602    JNE  4FE8
4FE4 05A0    INC  @PMEFLIP
4FE6 2AE8 *
4FE8 06A0    BL   @GETDAT2
4FEA 4178 Ax
4FEC 0A12    SLA  R2,1
4FEE C802    MOV  R2,@2AC0
4FF0 2AC0 *
4FF2 A084    A    R4,R2
4FF4 C802    MOV  R2,@SEGCNAB
4FF6 2ABE *
4FF8 C044  D MOV  R4,R1
4FFA 1002    JMP  PAGE2A
4FFC 0000    UNDEFINED
4FFE AE03    A    R3,*R8+

; Beginning of the paged ROM. This is standard page.
```

```
PAGE2A    5000 06A0    BL   @GETDAT2
          5002 4178 Ax
5004 0A12    SLA  R2,1
5006 A102    A    R2,R4
5008 C804    MOV  R4,@DICTPNT
500A 2AE4 *
500C 0450  P B    *R0

; Return from procedure
RPU       500E 5010    .WORD RPU+2
          5010 C808    MOV  R8,@IPCSAVE
          5012 2ACC *
          5014 0620    DEC  @IPCSAVE
          5016 2ACC *
          5018 0202    LI   R2,CONT04
          501A 5020 P
          501C 0460  ` B    @LDBIG
          501E 8330  0
CONT04    5020 8829  ) C    @0006(R9),@ERECP       ; Indexed by local
frame pointer
          5022 0006
          5024 2AB4 *
          5026 1309    JEQ  503A
          5028 C169  i MOV  @6(R9),R5    ; Indexed by local frame
pointer

          502A 0006
          502C 06A0    BL   @4ECC
          502E 4ECC N
          5030 C184    MOV  R4,R6
          5032 06A0    BL   @4FCA
          5034 4FCA O
          5036 06A0    BL   @4E6C
          5038 4E6C Nl
          503A C289    MOV  R9,SP
          503C C82A  * MOV  @8(SP),@CURPRNO-1
          503E 0008
          5040 2AC2 *
          5042 1103    JLT  504A
          5044 C22A  * MOV  @4(SP),R8
          5046 0004
          5048 1011    JMP  506C
          504A C060  ` MOV  @CURPRNO-1,R1
          504C 2AC2 *
          504E 0520    NEG  @CURPRNO-1
          5050 2AC2 *
          5052 0A11    SLA  R1,1
          5054 A060  ` A    @DICTPNT,R1
          5056 2AE4 *
          5058 06A0    BL   @GETDAT2
          505A 4178 Ax
          505C 0A12    SLA  R2,1
          505E A0A0    A    @SEGBASE,R2
          5060 2AE2 *
          5062 C042  B MOV  R2,R1
          5064 0641  A DECT R1
          5066 06A0    BL   @GETDAT2
          5068 4178 Ax
```

```
           506A C202    MOV  R2,R8
           506C C26A  j MOV  @2(SP),R9
           506E 0002
           5070 A220    A    @SEGBASE,R8
           5072 2AE2 *
           5074 022A  * AI   SP,10
           5076 000A
           5078 0A13    SLA  R3,1
           507A A283    A    R3,SP
           507C 0460  ` B    @LOADADDR
           507E 4142 AB

; Called by copy array/string parameter
           5080 C1DA    MOV  *SP,R7
           5082 C067  g MOV  @2(R7),R1
           5084 0002
           5086 8817    C    *R7,@40AA
           5088 40AA @
           508A 1604    JNE  5094
           508C 05CA    INCT SP
           508E C0BA    MOV  *SP+,R2
           5090 04C0    CLR  R0
           5092 045B  [ B    *R11

           5094 C157  W MOV  *R7,R5
           5096 C1C1    MOV  R1,R7
           5098 C18B    MOV  R11,R6
           509A 05A0    INC  @2AD6
           509C 2AD6 *
           509E 06A0    BL   @4ECC
           50A0 4ECC N
           50A2 A107    A    R7,R4
           50A4 05CA    INCT SP
           50A6 C0BA    MOV  *SP+,R2
           50A8 C000    MOV  R0,R0
           50AA 1503    JGT  GROM1
           50AC 1109    JLT  VDP1
           50AE C044  D MOV  R4,R1                  ; CPU RAM address
           50B0 0456  V B    *R6
GROM1      50B2 D844  D MOVB R4,@0402(R1) ; Write address
           50B4 0402
           50B6 1000    NOP
           50B8 D860  ` MOVB @8389,@0402(R1)
           50BA 8389
           50BC 0402
           50BE 0456  V B    *R6
VDP1       50C0 D860  ` MOVB @8389,@0402(R1)        ; Write address
           50C2 8389
           50C4 0402
           50C6 1000    NOP
           50C8 D844  D MOVB R4,@0402(R1)
           50CA 0402
           50CC 0456  V B    *R6

; Copy array parameter
CAP        50CE 50D0    .WORD CAP+2
           50D0 C808    MOV  R8,@IPCSAVE
           50D2 2ACC *
```

```
          50D4 0620    DEC  @IPCSAVE
          50D6 2ACC *
          50D8 0202    LI   R2,CONT05
          50DA 50E0 P
          50DC 0460  ` B    @LDBIG
          50DE 8330  0
CONT05    50E0 C0C3    MOV  R3,R3
          50E2 1603    JNE  50EA
          50E4 022A  * AI   SP,0004
          50E6 0004
          50E8 045C  \ B    *R12
          50EA 06A0    BL   @5080
          50EC 5080 P
50EE C000    MOV  R0,R0
50F0 1307    JEQ  5100
50F2 DC91    MOVB *R1,*R2+
50F4 1000    JMP  50F6
50F6 DC91    MOVB *R1,*R2+
50F8 0603    DEC  R3
50FA 15FB    JGT  50F2
50FC 0460  ` B    @LOADADDR
50FE 4142 AB
5100 CCB1    MOV  *R1+,*R2+
5102 0603    DEC  R3
5104 15FD    JGT  5100
5106 0460  ` B    @LOADADDR
5108 4142 AB

; Copy string parameter
CSP       510A 8316    .WORD LDUB
          510C C808    MOV  R8,@IPCSAVE
          510E 2ACC *
          5110 0660  ` DECT @IPCSAVE
          5112 2ACC *
          5114 06A0    BL   @5080                  ; Something used by
both array and string parameter
          5116 5080 P
          5118 C000    MOV  R0,R0
          511A 130F    JEQ  CSP01
          511C D011    MOVB *R1,R0
          511E 9800    CB   R0,@8387
          5120 8387
          5122 1B08    JH   STRBIG
          5124 DC80    MOVB R0,*R2+
          5126 1304    JEQ  CSP02
          5128 0980    SRL  R0,8
LOOP12    512A DC91    MOVB *R1,*R2+
          512C 0600    DEC  R0
          512E 15FD    JGT  LOOP12
CSP02     5130 0460  ` B    @LOADADDR
          5132 4142 AB
STRBIG    5134 06A0    BL   @FAULGEN
          5136 484A HJ
          5138 000D    .WORD 13          ; String overflow

CSP01     513A D011    MOVB *R1,R0
          513C 9800    CB   R0,@8387
          513E 8387
```

```
            5140 1BF9    JH   5134
            5142 0990    SRL  R0,9
            5144 0580    INC  R0
            5146 CCB1    MOV  *R1+,*R2+
            5148 0600    DEC  R0
            514A 15FD    JGT  5146
            514C 0460  ` B    @LOADADDR
            514E 4142 AB

; Load packed
LDP         5150 5152    .WORD LDP+2
            5152 C07A  z MOV  *SP+,R1
            5154 0A11    SLA  R1,1
            5156 C0BA    MOV  *SP+,R2
            5158 0A12    SLA  R2,1
            515A C0FA    MOV  *SP+,R3
            515C C0D3    MOV  *R3,R3
            515E 04A1    X    @SHRIGHT(R1)
            5160 4100 A
            5162 C122  " MOV  @MASKTAB(R2),R4
            5164 40BE @
            5166 0544  D INV  R4
            5168 40C4 @  SZC  R4,R3
            516A 064A  J DECT SP
            516C C683    MOV  R3,*SP
            516E 045C  \ B    *R12         ;PME fetch

; Store packed
STP         5170 5172    .WORD STP+2
            5172 C07A  z MOV  *SP+,R1
            5174 C0BA    MOV  *SP+,R2
            5176 0A12    SLA  R2,1
            5178 C0FA    MOV  *SP+,R3
            517A 0A13    SLA  R3,1
            517C C123  # MOV  @MASKTAB(R3),R4
            517E 40BE @
            5180 0544  D INV  R4
            5182 4044 @D SZC  R4,R1
            5184 C001    MOV  R1,R0
            5186 04A2    X    @SHLEFT(R2)
            5188 40E0 @
            518A C801    MOV  R1,@SAVPACK
            518C 2AF2 *
            518E C063  c MOV  @MASKTAB(R3),R1
            5190 40BE @
            5192 0541  A INV  R1
            5194 E001    SOC  R1,R0
            5196 0541  A INV  R1
            5198 4001 @  SZC  R1,R0
            519A 04A2    X    @SHCIRC(R2)
            519C 4120 A
            519E C07A  z MOV  *SP+,R1
            51A0 0540  @ INV  R0
            51A2 4440 D@ SZC  R0,*R1
            51A4 E460  ` SOC  @SAVPACK,*R1
            51A6 2AF2 *
            51A8 045C  \ B    *R12         ;PME fetch
```

```
; Index packed array
IXP       51AA 830E    .WORD LDUBB
          51AC 04C0    CLR  R0
          51AE C07A  z MOV  *SP+,R1
          51B0 3C04 <  DIV  R4,R0
          51B2 0A10    SLA  R0,1
          51B4 A680    A    R0,*SP
          51B6 064A  J DECT SP
          51B8 C683    MOV  R3,*SP
          51BA C001    MOV  R1,R0
          51BC 3803 8  MPY  R3,R0
          51BE 064A  J DECT SP
          51C0 C681    MOV  R1,*SP
          51C2 045C  \ B    *R12         ;PME fetch

; Logical AND
LAND      51C4 51C6 Q  .WORD LAND+2
          51C6 055A  Z INV  *SP
          51C8 46BA F  SZC  *SP+,*SP
          51CA 045C  \ B    *R12         ;PME fetch

; Logical OR
LOR       51CC 51CE Q  .WORD LOR+2
          51CE E6BA    SOC  *SP+,*SP
          51D0 045C  \ B    *R12         ;PME fetch

; Logical not
LNOT      51D2 51D4    .WORD LNOT+2
          51D4 055A  Z INV  *SP
          51D6 045C  \ B    *R12         ;PME fetch

; Boolean NOT
BNOT      51D8 51DA Q  .WORD BNOT+2
          51DA C05A  Z MOV  *SP,R1
          51DC 0541  A INV  R1
          51DE 0241  A ANDI R1,0001
          51E0 0001
          51E2 C681    MOV  R1,*SP
          51E4 045C  \ B    *R12         ;PME fetch

; Increment integer
INCI      51E6 51E8 Q  .WORD INCI+2
          51E8 059A    INC  *SP
          S51EA 045C  \ B    *R12        ;PME fetch

; Decrement integer
DECI      51EC 51EE Q  .WORD DECI+2
          51EE 061A    DEC *SP
          51F0 045C  \ B    *R12         ;PME fetch

; Absolute value integer
ABI       51F2 51F4 Q  .WORD ABI+2
          51F4 075A  Z ABS  *SP
          51F6 045C  \ B    *R12         ;PME fetch

; Add integer
ADI       51F8 51FA Q  .WORD ADI+2
          51FA A6BA    A    *SP+,*SP
```

```
          51FC 045C  \ B    *R12         ;PME fetch

; Negate integer
NGI       51FE 5200 R  .WORD NGI+2
          5200 051A    NEG  *SP
          5202 045C  \ B    *R12         ;PME fetch

; Subtract integer
SBI       5204 5206 R  .WORD SBI+2
          5206 66BA f  S    *SP+,*SP
          5208 045C  \ B    *R12         ;PME fetch

; Modulo integer
MODI      520A 520C R  .WORD MODI+2
          520C C0FA    MOV  *SP+,R3
          520E 131B    JEQ  FAUDIVZ      ;Fault, divide by zero
          5210 0743  C ABS  R3
          5212 C09A    MOV  *SP,R2
          5214 0742  B ABS  R2
          5216 04C1    CLR  R1
          5218 3C43 <C DIV  R3,R1
          521A C682    MOV  R2,*SP       ; Reminder
          521C 045C  \ B    *R12         ;PME fetch

; Divide integer
DVI       521E 5220 R  .WORD DVI+2
          5220 04C0    CLR  R0           ; Flags
          5222 04C4    CLR  R4
          5224 C0FA    MOV  *SP+,R3
          5226 130F    JEQ  FAUDIVZ
          5228 1502    JGT  POS1
          522A 0704    SETO R4           ; Flag negative
          522C 0743  C ABS  R3
POS1      522E C09A    MOV  *SP,R2
          5230 1309    JEQ  ZERORES
          5232 1502    JGT  POS2
          5234 0700    SETO R0           ; Flag negative
          5236 0742  B ABS  R2
POS2      5238 04C1    CLR  R1
          523A 3C43 <C DIV  R3,R1
          523C 2900 )  XOR  R0,R4
          523E 1301    JEQ  POSRES
          5240 0501    NEG  R1
POSRES    5242 C681    MOV  R1,*SP
ZERORES   5244 045C  \ B    *R12         ;PME fetch

FAUDIVZ   5246 06A0    BL   @FAULGEN     ; Create a fault
          5248 484A HJ
          524A 0006    .WORD 6           ; Divide by zero

; Multiply integer. Returns low 16 bits
MPI       524C 524E RN .WORD MPI+2
          524E C07A  z MOV  *SP+,R1
          5250 385A 8Z MPY  *SP,R1
          5252 C682    MOV  R2,*SP
          5254 045C  \ B    *R12         ;PME fetch

; Check subrange limit
```

```
CHK        5256 5258    .WORD CHK+2
           5258 C07A  z MOV  *SP+,R1
           525A C0BA    MOV  *SP+,R2
           525C 805A  Z C    *SP,R1
           525E 1503    JGT  RGFAUL
           5260 809A    C    *SP,R2
           5262 1101    JLT  RGFAUL
           5264 045C  \ B    *R12         ;PME fetch

RGFAUL     5266 06A0    BL   @FAULGEN
           5268 484A HJ
           526A 0001    .WORD 1 ; Value range error, invalid index

526C 8106    C    R6,R4
526E 1502    JGT  5274
5270 C0C6    MOV  R6,R3
5272 1001    JMP  5276
5274 C0C4    MOV  R4,R3
5276 1325  % JEQ  CTTEST
5278 C165  e MOV  @527E(R5),R5
527A 527E R~
527C 0455  U B    *R5

527E 5286 R  SZCB R6,SP
5280 5290 R  SZCB *R0,SP
5282 529A R  SZCB *SP,SP
5284 52A4 R  SZCB @9C32(R4),SP

5286 9C32  2
5288 161D    JNE  52C4
528A 0603    DEC  R3
528C 15FC    JGT  5286
528E 1019    JMP  CTTEST

5290 9772  r CB   *R2+,*R13
5292 1618    JNE  52C4
5294 0603    DEC  R3
5296 15FC    JGT  5290
5298 1014    JMP  CTTEST

529A 9C1D    CB   *R13,*R0+
529C 1613    JNE  52C4
529E 0603    DEC  R3
52A0 15FC    JGT  529A
52A2 100F    JMP  CTTEST

52A4 C148  H MOV  R8,R5
52A6 C202    MOV  R2,R8
52A8 06A0    BL   @LOADAD1
52AA 4144 AD
52AC 0582    INC  R2
52AE D1DD    MOVB *R13,R7
52B0 C200    MOV  R0,R8
52B2 06A0    BL   @LOADAD1
52B4 4144 AD
52B6 0580    INC  R0
52B8 C205    MOV  R5,R8
52BA 9747  G CB   R7,*R13
```

```
52BC 1603    JNE  52C4
52BE 0603    DEC  R3
52C0 15F1    JGT  52A4
CTTEST    52C2 8106    C    R6,R4                  ; Compare, then test

TESTS     52C4 0491    X    *R1                    ; Jumps to 52CE if
test is true (condition false)
          52C6 C6A0    MOV  @CONTRUE,*SP ; Condition is true
          52C8 40B6 @
          52CA 0460  ` B    @LOADADDR
          52CC 4142 AB
          52CE 04DA    CLR  *SP                    ; Condition is false
          52D0 0460  ` B    @LOADADDR
          52D2 4142 AB

; Instruction table for test instructions
; They are inverted due to the design above
          52D4 1604    JNE  52DE         ; EQUI
          52D6 1304    JEQ  52E0         ; NEQI
          52D8 1504    JGT  52E2         ; LEQI
          52DA 1104    JLT  52E4         ; GEQI
          52DC 1B04    JH   52E6         ; LEUSW
          52DE 1A04    JL   52E8         ; GEUSW
          52E0 1604    JNE  52EA
          52E2 1504    JGT  52EC
          52E4 1104    JLT  52EE

; Common entry for
; integer =, <>, <=, >= and unsigned <= and >=
; Opcodes B0 to B5
EQUI      52E6 52E8    .WORD EQUI+2
          52E8 0221  ! AI   R1,5174      ; To make 2*opcode point to the
right entry in the table above
          52EA 5174 Qt
          52EC 8EAA    C    @0002(SP),*SP+
          52EE 0002
          52F0 0460  ` B    @TESTS
          52F2 52C4 R

; Equal byte array, less than or equal or greater than or equal byte
array
EQBYT     52F4 8320     .WORD LDUBBG
          52F6 0221  !  AI   R1,516E     ; Opcode EQBYT is B9 (185) *2
          52F8 516E Qn
          52FA C183    MOV  R3,R6
          52FC 04C3    CLR  R3
          52FE C03A  : MOV  *SP+,R0
          5300 C09A    MOV  *SP,R2
          5302 C145  E MOV  R5,R5
          5304 130A    JEQ  531A
          5306 A020    A    @SEGBASE,R0
          5308 2AE2 *
          530A C3CF    MOV  R15,R15
          530C 1306    JEQ  531A
          530E 05C3    INCT R3
          5310 C1C8    MOV  R8,R7
          5312 C200    MOV  R0,R8
          5314 06A0    BL   @LOADAD1
```

```
            5316 4144 AD
            5318 C207    MOV  R7,R8
            531A C104    MOV  R4,R4
            531C 130B    JEQ  5334
            531E A0A0    A    @SEGBASE,R2
            5320 2AE2 *
            5322 C3CF    MOV  R15,R15
            5324 1307    JEQ  5334
            5326 0223  # AI   R3,0004
            5328 0004
            532A C1C8    MOV  R8,R7
            532C C202    MOV  R2,R8
            532E 06A0    BL   @LOADAD1
            5330 4144 AD
            5332 C207    MOV  R7,R8
            5334 C143  C MOV  R3,R5
            5336 C106    MOV  R6,R4
            5338 0460  ` B    @526C
            533A 526C Rl

; Equal, less than or equal and greater than or equal string
EQSTR       533C 830E    .WORD LDUBB
            533E 0221  ! AI   R1,5110      ; Opcode EQSTR is E8 (232) *2
            5340 5110 Q
            5342 04C5    CLR  R5
            5344 C03A  : MOV  *SP+,R0
            5346 C09A    MOV  *SP,R2
            5348 C104    MOV  R4,R4
            534A 130F    JEQ  536A
            534C A020    A    @SEGBASE,R0
            534E 2AE2 *
            5350 C3CF    MOV  R15,R15
            5352 130B    JEQ  536A                  ; CPU RAM
            5354 05C5    INCT R5
            5356 C1C8    MOV  R8,R7
            5358 C200    MOV  R0,R8
            535A 06A0    BL   @LOADAD1
            535C 4144 AD
            535E 04C4    CLR  R4
            5360 D81D    MOVB *R13,@8389
            5362 8389
            5364 0580    INC  R0
            5366 C207    MOV  R7,R8
            5368 1003    JMP  5370

            536A 04C4    CLR  R4
            536C D830  0 MOVB *R0+,@8389
            536E 8389
            5370 C0C3    MOV  R3,R3
            5372 1311    JEQ  5396
            5374 A0A0    A    @SEGBASE,R2
            5376 2AE2 *
            5378 C3CF    MOV  R15,R15
            537A 130D    JEQ  5396                  ; CPU RAM
            537C 0225  % AI   R5,0004
            537E 0004
            5380 C1C8    MOV  R8,R7
            5382 C202    MOV  R2,R8
```

```
           5384 06A0    BL   @LOADAD1
           5386 4144 AD
           5388 04C6    CLR  R6
           538A D81D    MOVB *R13,@838D
           538C 838D
           538E 0582    INC  R2
           5390 C207    MOV  R7,R8
           5392 0460  ` B    @526C
           5394 526C Rl

           5396 04C6    CLR  R6
           5398 D832  2 MOVB *R2+,@838D
           539A 838D
           539C 0460  ` B    @526C
           539E 526C Rl

; Assign string
ASTR       53A0 830E    .WORD LDUBB
           53A2 C07A  z MOV  *SP+,R1
           53A4 C0BA    MOV  *SP+,R2
           53A6 04C5    CLR  R5
           53A8 C104    MOV  R4,R4
           53AA 1314    JEQ  53D4
           53AC A060  ` A    @SEGBASE,R1
           53AE 2AE2 *
           53B0 C3CF    MOV  R15,R15
           53B2 1310    JEQ  53D4                    ; CPU RAM
           53B4 C108    MOV  R8,R4
           53B6 C201    MOV  R1,R8
           53B8 06A0    BL   @LOADAD1
           53BA 4144 AD
           53BC C204    MOV  R4,R8
           53BE D81D    MOVB *R13,@838B
           53C0 838B
           53C2 80C5    C    R5,R3
           53C4 1510    JGT  ERRSTR
           53C6 DCA0    MOVB @838B,*R2+
           53C8 838B
LOOP25     53CA DC9D    MOVB *R13,*R2+
           53CC 0605    DEC  R5
           53CE 15FD    JGT  LOOP25
           53D0 0460  ` B    @LOADADDR
           53D2 4142 AB
           53D4 D811    MOVB *R1,@838B
           53D6 838B
           53D8 80C5    C    R5,R3
           53DA 1505    JGT  ERRSTR
           53DC 0585    INC  R5
LOOP26     53DE CCB1    MOV  *R1+,*R2+
           53E0 0645  E DECT R5
           53E2 15FD    JGT  LOOP26
           53E4 045C  \ B    *R12         ; PME fetch
ERRSTR     53E6 06A0    BL   @FAULGEN
           53E8 484A HJ
           53EA 000D    .WORD 13          ; String overflow

; Check string index
CSTR       53EC 53EE    .WORD CSTR+2
```

```
           53EE C09A    MOV  *SP,R2
           53F0 C06A  j MOV  @2(SP),R1
           53F2 0002
           53F4 04C3    CLR  R3
           53F6 D811    MOVB *R1,@8387    ; R3 LSBy
           53F8 8387
           53FA 0282    CI   R2,0001
           53FC 0001
           53FE 1103    JLT  ERRCSTR
           5400 80C2    C    R2,R3
           5402 1501    JGT  ERRCSTR
           5404 045C  \ B    *R12         ; PME fetch
ERRCSTR    5406 06A0    BL   @FAULGEN
           5408 484A HJ
           540A 0001    .WORD 1 ; Value range error, invalid index

; Adjust set
ADJ        540C 8316    .WORD LDUB
           540E C05A  Z MOV  *SP,R1
           5410 80C1    C    R1,R3
           5412 1103    JLT  541A
           5414 151D    JGT  5450
           5416 05CA    INCT SP
           5418 045C  \ B    *R12         ; PME fetch

           541A 60C1 `  S    R1,R3
           541C C003    MOV  R3,R0
           541E 0A13    SLA  R3,1
           5420 C808    MOV  R8,@IPCSAVE
           5422 2ACC *
           5424 0660  ` DECT @IPCSAVE
           5426 2ACC *
           5428 C820    MOV  @ERECP,@CUREREC
           542A 2AB4 *
           542C 2AE0 *
           542E C803    MOV  R3,@STKMISS
           5430 2AF0 *
           5432 06A0    BL   @CHKSTK
           5434 4E98 N
           5436 05CA    INCT SP
           5438 C08A    MOV  SP,R2
           543A 6283 b  S    R3,SP
           543C C0CA    MOV  SP,R3
           543E C041  A MOV  R1,R1
           5440 1303    JEQ  5448
LOOP16     5442 CCF2    MOV  *R2+,*R3+
           5444 0601    DEC  R1
           5446 15FD    JGT  LOOP16
LOOP17     5448 04F3    CLR  *R3+
           544A 0600    DEC  R0
           544C 15FD    JGT  LOOP17
           544E 045C  \ B    *R12         ; PME fetch

           5450 05CA    INCT SP
           5452 C083    MOV  R3,R2
           5454 0A13    SLA  R3,1
           5456 A0CA    A    SP,R3
           5458 0643  C DECT R3
```

```
           545A 0A11    SLA  R1,1
           545C A04A  J A    SP,R1
           545E 0641  A DECT R1
LOOP18     5460 C453  S MOV  *R3,*R1
           5462 0643  C DECT R3
           5464 0641  A DECT R1
           5466 0602    DEC  R2
           5468 15FB    JGT  LOOP18
           546A C281    MOV  R1,SP
           546C 05CA    INCT SP
           546E 045C  \ B    *R12         ; PME fetch

; Subrange set
SRS        5470 5472    .WORD SRS+2
           5472 C05A  Z MOV  *SP,R1
           5474 C0EA    MOV  @2(SP),R3
           5476 0002
           5478 1167  g JLT  5548
           547A 8043  C C    R3,R1
           547C 1565  e JGT  5548
           547E 04C2    CLR  R2
           5480 3CA0 <  DIV  @CON016,R2
           5482 40BA @
           5484 C802    MOV  R2,@2AF4
           5486 2AF4 *
           5488 0A13    SLA  R3,1
           548A 0204    LI   R4,FFFF
           548C FFFF
           548E 2923 )# XOR  @MASKTAB(R3),R4
           5490 40BE @
           5492 C804    MOV  R4,@2AC4
           5494 2AC4 *
           5496 C820    MOV  @CONFALS,@2AD8
           5498 40A8 @
           549A 2AD8 *
           549C 0582    INC  R2
           549E 0A42  B SLA  R2,4
           54A0 8081    C    R1,R2
           54A2 1506    JGT  54B0
           54A4 1305    JEQ  54B0
           54A6 C820    MOV  @CONTRUE,@2AD8
           54A8 40B6 @
           54AA 2AD8 *
           54AC 0222  " AI   R2,FFF0
           54AE FFF0
           54B0 6042 `B S    R2,R1
           54B2 04C0    CLR  R0
           54B4 3C20 <  DIV  @CON016,R0
           54B6 40BA @
           54B8 C800    MOV  R0,@2AD4
           54BA 2AD4 *
           54BC 0581    INC  R1
           54BE 0A11    SLA  R1,1
           54C0 04C3    CLR  R3
           54C2 28E1 (  XOR  @MASKTAB(R1),R3
           54C4 40BE @
           54C6 C803    MOV  R3,@2AD0
           54C8 2AD0 *
```

```
54CA C0A0    MOV  @2AD4,R2
54CC 2AD4 *
54CE A0A0    A    @2AF4,R2
54D0 2AF4 *
54D2 8820    C    @2AD8,@CONTRUE
54D4 2AD8 *
54D6 40B6 @
54D8 1301    JEQ  54DC
54DA 0582    INC  R2
54DC 0582    INC  R2
54DE C142  B MOV  R2,R5
54E0 0A12    SLA  R2,1
54E2 C802    MOV  R2,@STKMISS
54E4 2AF0 *
54E6 C808    MOV  R8,@IPCSAVE
54E8 2ACC *
54EA 0620    DEC  @IPCSAVE
54EC 2ACC *
54EE C820    MOV  @ERECP,@CUREREC
54F0 2AB4 *
54F2 2AE0 *
54F4 06A0    BL   @CHKSTK
54F6 4E98 N
54F8 022A  * AI   SP,0004
54FA 0004
54FC 8820    C    @2AD8,@CONFALS
54FE 2AD8 *
5500 40A8 @
5502 1308    JEQ  5514
5504 C0A0    MOV  @2AC4,R2
5506 2AC4 *
5508 28A0 (  XOR  @2AD0,R2
550A 2AD0 *
550C 0542  B INV  R2
550E 064A  J DECT SP
5510 C682    MOV  R2,*SP
5512 100F    JMP  5532
5514 064A  J DECT SP
5516 C6A0    MOV  @2AD0,*SP
5518 2AD0 *
551A C060  ` MOV  @2AD4,R1
551C 2AD4 *
551E 1306    JEQ  552C
5520 0203    LI   R3,FFFF
5522 FFFF
5524 064A  J DECT SP
5526 C683    MOV  R3,*SP
5528 0601    DEC  R1
552A 15FC    JGT  5524
552C 064A  J DECT SP
552E C6A0    MOV  @2AC4,*SP
5530 2AC4 *
5532 04C2    CLR  R2
5534 C060  ` MOV  @2AF4,R1
5536 2AF4 *
5538 1304    JEQ  5542
553A 064A  J DECT SP
553C C682    MOV  R2,*SP
```

```
           553E 0601    DEC  R1
           5540 15FC    JGT  553A
           5542 064A  J DECT SP
           5544 C685    MOV  R5,*SP
           5546 045C  \ B    *R12
           5548 05CA    INCT SP
           554A 04DA    CLR  *SP
           554C 045C  \ B    *R12         ; PME fetch

; Set membership
INN        554E 5550    .WORD INN+2
           5550 C03A  : MOV  *SP+,R0
           5552 C04A  J MOV  SP,R1
           5554 A040  @ A    R0,R1
           5556 A040  @ A    R0,R1
           5558 C0D1    MOV  *R1,R3
           555A 111B    JLT  5592
           555C 04C2    CLR  R2
           555E 3CA0 <  DIV  @CON016,R2
           5560 40BA @
           5562 8080    C    R0,R2
           5564 1216    JLE  5592
           5566 0A12    SLA  R2,1
           5568 A08A    A    SP,R2
           556A C812    MOV  *R2,@SAVPACK
           556C 2AF2 *
           556E C0D1    MOV  *R1,R3
           5570 04C2    CLR  R2
           5572 3CA0 <  DIV  @CON016,R2
           5574 40BA @
           5576 0A13    SLA  R3,1
           5578 C123  # MOV  @2D92(R3),R4
           557A 2D92 -
           557C 0544  D INV  R4
           557E C0A0    MOV  @SAVPACK,R2
           5580 2AF2 *
           5582 4084 @  SZC  R4,R2
           5584 1306    JEQ  5592
           5586 C281    MOV  R1,SP
           5588 05CA    INCT SP
           558A 064A  J DECT SP
           558C C6A0    MOV  @CONTRUE,*SP
           558E 40B6 @
           5590 045C  \ B    *R12         ; PME fetch

5592 C281    MOV  R1,R10
5594 05CA    INCT R10
5596 064A  J DECT R10
5598 C6A0    MOV  @CONFALS,*R10
559A 40A8 @
559C 045C  \ B    *R12         ; PME fetch

559E C07A  z MOV  *SP+,R1
55A0 C08A    MOV  SP,R2
55A2 A081    A    R1,R2
55A4 A081    A    R1,R2
55A6 8072  r C    *R2+,R1
55A8 1320    JEQ  55EA
```

```
55AA 151F    JGT  55EA
55AC 064A  J DECT SP
55AE C05A  Z MOV  *SP,R1
55B0 0642  B DECT R2
55B2 C812    MOV  *R2,@SAVPACK
55B4 2AF2 *
55B6 C481    MOV  R1,*R2
55B8 C081    MOV  R1,R2
55BA 60A0 `  S    @SAVPACK,R2
55BC 2AF2 *
55BE C802    MOV  R2,@2AF4
55C0 2AF4 *
55C2 A060  ` A    @SAVPACK,R1
55C4 2AF2 *
55C6 05C1    INCT R1
55C8 C80A    MOV  SP,@SAVPACK
55CA 2AF2 *
55CC 0A12    SLA  R2,1
55CE 6282 b  S    R2,SP
55D0 C08A    MOV  SP,R2
55D2 C0E0    MOV  @SAVPACK,R3
55D4 2AF2 *
LOOP13     55D6 CCB3    MOV  *R3+,*R2+
           55D8 0601    DEC  R1
           55DA 15FD    JGT  LOOP13
           55DC C060  ` MOV  @2AF4,R1
           55DE 2AF4 *
LOOP14     55E0 04F2    CLR  *R2+
           55E2 0601    DEC  R1
           55E4 15FD    JGT  LOOP14
           55E6 0460  ` B    @559E
           55E8 559E U

           55EA C041  A MOV  R1,R1
           55EC 045B  [ B    *R11

UNI        55EE 55F0    .WORD UNI+2
           55F0 06A0    BL   @559E
           55F2 559E U
           55F4 1303    JEQ  EXUNI
LOOP20     55F6 ECBA    SOC  *SP+,*R2+
           55F8 0601    DEC  R1
           55FA 15FD    JGT  LOOP20
EXUNI      55FC 045C  \ B    *R12         ; PME fetch

; Set intersection
INT        55FE 5600    .WORD INT+2
           5600 06A0    BL   @559E
           5602 559E U
           5604 C801    MOV  R1,@SAVPACK
           5606 2AF2 *
           5608 1304    JEQ  5612
LOOP22     560A 055A  Z INV  *SP
           560C 4CBA L  SZC  *SP+,*R2+
           560E 0601    DEC  R1
           5610 15FC    JGT  LOOP22
           5612 C05A  Z MOV  *SP,R1
           5614 6060 `` S    @SAVPACK,R1
```

```
          5616 2AF2 *
          5618 1303    JEQ  EXINT
LOOP21    561A 04F2    CLR  *R2+
          561C 0601    DEC  R1
          561E 15FD    JGT  LOOP21
EXINT     5620 045C  \ B    *R12         ; PME fetch

DIF       5622 5624    .WORD DIF+2
          5624 06A0    BL   @559E
          5626 559E U
          5628 1303    JEQ  EXDIF
LOOP23    562A 4CBA L  SZC  *SP+,*R2+
          562C 0601    DEC  R1
          562E 15FD    JGT  LOOP23
EXDIF     5630 045C  \ B    *R12         ; PME fetch

SETTEST   5632 565A VZ SZCB *SP,*R9
          5634 568E V  SZCB R14,*SP
          5636 56A0 V  SZCB @563A,*SP

; Equal set, less than or equal set and greater than or equal set
EQPWR     5638 563A    .WORD EQPWR+2
          563A C141  A MOV  R1,R5                  ; Opcode is B6 for
EQPWR. B6*2=364D
          563C 06A0    BL   @559E
          563E 559E U
          5640 0642  B DECT R2
          5642 C132  2 MOV  *R2+,R4
          5644 0A14    SLA  R4,1
          5646 A102    A    R2,R4
          5648 C804    MOV  R4,@2AD2
          564A 2AD2 *
          564C 0225  % AI   R5,-364       ; Subtract opcode*2
          564E FE94
          5650 C125  % MOV  @SETTEST(R5),R4
          5652 5632 V2
          5654 0642  B DECT R2
          5656 C0F2    MOV  *R2+,R3
          5658 0454  T B    *R4

EQSET     565A C041  A MOV  R1,R1
          565C 1305    JEQ  5668
          565E 8CBA    C    *SP+,*R2+
          5660 160A    JNE  5676
          5662 0603    DEC  R3
          5664 0601    DEC  R1
          5666 15FB    JGT  565E
          5668 C0C3    MOV  R3,R3
          566A 130B    JEQ  5682
LOOP15    566C C072  r MOV  *R2+,R1
          566E 1603    JNE  5676
          5670 0603    DEC  R3
          5672 15FC    JGT  LOOP15
          5674 1006    JMP  5682

          5676 C2A0    MOV  @2AD2,SP
          5678 2AD2 *
          567A 064A  J DECT SP
```

```
           567C C6A0    MOV  @CONFALS,*SP
           567E 40A8 @
           5680 045C  \ B    *R12         ; PME fetch

           5682 C2A0    MOV  @2AD2,SP
           5684 2AD2 *
           5686 064A  J DECT SP
           5688 C6A0    MOV  @CONTRUE,*SP
           568A 40B6 @
           568C 045C  \ B    *R12         ; PME fetch

LESET      568E C041  A MOV  R1,R1
           5690 1305    JEQ  569C
LOOP24     5692 4CBA L  SZC  *SP+,*R2+
           5694 16F0    JNE  5676
           5696 0603    DEC  R3
           5698 0601    DEC  R1
           569A 15FB    JGT  LOOP24
           569C 0460  ` B    @5668
           569E 5668 Vh

GESET      56A0 C041  A MOV  R1,R1
           56A2 13EF    JEQ  5682
           56A4 4EB2 N  SZC  *R2+,*SP+
           56A6 16E7    JNE  5676
           56A8 0601    DEC  R1
           56AA 15FC    JGT  56A4
           56AC 10EA    JMP  5682
           56AE 022A  * AI   SP,4
           56B0 0004
           56B2 045C  \ B    *R12         ; PME fetch

; Signal semaphore
SIGNAL     56B4 56B6    .WORD SIGNAL+2
           56B6 C07A  z MOV  *SP+,R1
           56B8 8821  ! C    @0002(R1),@40AA
           56BA 0002
           56BC 40AA @
           56BE 1603    JNE  56C6
           56C0 05A1    INC  @0(R1)
           56C2 0000
           56C4 102B  + JMP  SIGJUMP

           56C6 C121  ! MOV  @0002(R1),R4
           56C8 0002
           56CA 06A0    BL   @578C
           56CC 578C W
           56CE C844  D MOV  R4,@0002(R1)
           56D0 0002
           56D2 C960  ` MOV  @40AA,@0014(R5)
           56D4 40AA @
           56D6 0014
           56D8 C120    MOV  @READYQ,R4
           56DA 2AB6 *
           56DC 06A0    BL   @575C
           56DE 575C W\
           56E0 C804    MOV  R4,@READYQ
           56E2 2AB6 *
```

```
            56E4 C060  ` MOV  @CURTSK,R1
            56E6 2ABA *
            56E8 9865  e CB   @0003(R5),@0003(R1)
            56EA 0003
            56EC 0003
            56EE 1A16    JL   SIGJUMP
            56F0 C060  ` MOV  @CURSIB,R1
            56F2 2AB2 *
            56F4 A860  ` A    @40B4,@0004(R1)
            56F6 40B4 @
            56F8 0004
            56FA 1902    JNO  5700
            56FC 06A0    BL   @5A60
            56FE 5A60 Z`
            5700 06A0    BL   @UPDTIB
            5702 579A W
            5704 C120    MOV  @READYQ,R4
            5706 2AB6 *
            5708 06A0    BL   @575C
            570A 575C W\
            570C 06A0    BL   @578C
            570E 578C W
            5710 C804    MOV  R4,@READYQ
            5712 2AB6 *
            5714 C805    MOV  R5,@CURTSK
            5716 2ABA *
            5718 06A0    BL   @57BE
            571A 57BE W
SIGJUMP     571C 0460  ` B    @LOADADDR
            571E 4142 AB

; Wait for semaphore
WAIT        5720 5722    .WORD WAIT+2
            5722 C07A  z
            5724 C0A1    MOV  @0000(R1),R2
            5726 0000
            5728 1303    JEQ  5730
            572A 0621  ! DEC  @0000(R1)
            572C 0000
            572E 045C  \ B    *R12         ; PME fetch

            5730 06A0    BL   @UPDTIB
            5732 579A W
            5734 C121  ! MOV  @0002(R1),R4
            5736 0002
            5738 06A0    BL   @575C
            573A 575C W\
            573C C844  D MOV  R4,@0002(R1)
            573E 0002
            5740 C941  A MOV  R1,@0014(R5)
            5742 0014
            5744 C120    MOV  @READYQ,R4
            5746 2AB6 *
            5748 06A0    BL   @578C
            574A 578C W
            574C C804    MOV  R4,@READYQ
            574E 2AB6 *
            5750 C805    MOV  R5,@CURTSK
```

```
           5752 2ABA *
           5754 06A0    BL   @57BE
           5756 57BE W
           5758 0460  ` B    @LOADADDR
           575A 4142 AB

           575C C184    MOV  R4,R6
           575E C1E0    MOV  @40AA,R7
           5760 40AA @
           5762 8806    C    R6,@40AA
           5764 40AA @
           5766 1308    JEQ  5778
           5768 9966  f CB   @0003(R6),@0003(R5)
           576A 0003
           576C 0003
           576E 1A04    JL   5778
           5770 C1C6    MOV  R6,R7
           5772 C1A6    MOV  @0000(R6),R6
           5774 0000
           5776 10F5    JMP  5762
           5778 C946  F MOV  R6,@0000(R5)
           577A 0000
           577C 8807    C    R7,@40AA
           577E 40AA @
           5780 1303    JEQ  5788
           5782 C9C5    MOV  R5,@0000(R7)
           5784 0000
           5786 045B  [ B    *R11
           5788 C105    MOV  R5,R4
           578A 045B  [ B    *R11
           578C C144  D MOV  R4,R5
           578E C125  % MOV  @0000(R5),R4
           5790 0000
           5792 C960  ` MOV  @40AA,@0000(R5)
           5794 40AA @
           5796 0000
           5798 045B  [ B    *R11

;Update TIB with current register data
UPDTIB     579A C160  ` MOV  @CURTSK,R5
           579C 2ABA *
           579E C949  I MOV  R9,@000A(R5) ;Store frame pointer in TIB
           57A0 000A
           57A2 C948  H MOV  R8,@000E(R5) ;Store IPC in TIB
           57A4 000E
           57A6 6960 i` S    @SEGBASE,@000E(R5)      ; Segment relative
IPC
           57A8 2AE2 *
           57AA 000E
           57AC C960  ` MOV  @ERECP,@0010(R5)        ; Environment record
pointer
           57AE 2AB4 *
           57B0 0010
           57B2 C94A  J MOV  SP,@0008(R5) ; Stack pointer
           57B4 0008
           57B6 D960  ` MOVB @CURPRNO,@0013(R5)      ; Current procedure
number
           57B8 2AC3 *
```

```
          57BA 0013
          57BC 045B  [ B    *R11

; Update current registers from TIB
FROMTIB   57BE C80B    MOV  R11,@RETSAV2
          57C0 2ADC *
          57C2 C160  ` MOV  @CURTSK,R5   ; Current task
          57C4 2ABA *
          57C6 C265  e MOV  @AH(R5),R9   ; Local frame pointer
          57C8 000A
          57CA C225  % MOV  @EH(R5),R8   ; IPC
          57CC 000E
          57CE C2A5    MOV  @8(R5),SP    ; Stack pointer
          57D0 0008
          57D2 04E0    CLR  @CURPRNO-1
          57D4 2AC2 *
          57D6 D825  % MOVB @13H(R5),@CURPRNO
          57D8 0013
          57DA 2AC3 *
          57DC C165  e MOV  @10H(R5),R5
          57DE 0010
          57E0 C805    MOV  R5,@ERECP
          57E2 2AB4 *
          57E4 C825  % MOV  @2(R5),@EVECP
          57E6 0002
          57E8 2AB8 *
          57EA C825  % MOV  @4(R5),@CURSIB
          57EC 0004
          57EE 2AB2 *
          57F0 C3A5    MOV  @0(R5),R14   ; Global frame pointer
          57F2 0000
          57F4 04E0    CLR  @SEGBASE
          57F6 2AE2 *
          57F8 C808    MOV  R8,@IPCSAVE
          57FA 2ACC *
          57FC 06A0    BL   @4ECC
          57FE 4ECC N
          5800 C804    MOV  R4,@SEGBASE
          5802 2AE2 *
          5804 A204    A    R4,R8
          5806 C2E0    MOV  @RETSAV2,R11
          5808 2ADC *
          580A 0460  ` B    @4FCA
          580C 4FCA O
          580E C060  ` MOV  @2B6E,R1
          5810 2B6E +n
          5812 1601    JNE  5816
          5814 045C  \ B    *R12                  ;PME fetch

5816 06A0    BL   @FAULGEN
5818 484A HJ
581A 000A    .WORD 10          ; User IO error
581C C17A  z MOV  *SP+,R5
581E C808    MOV  R8,@IPCSAVE
5820 2ACC *
5822 05A0    INC  @2AD6
5824 2AD6 *
5826 06A0    BL   @4ECC
```

```
5828 4ECC N
582A 0460  ` B    @LOADADDR
582C 4142 AB
582E 04C4    CLR  R4
5830 C0A0    MOV  @2B88,R2
5832 2B88 +
5834 1308    JEQ  5846
5836 C0E2    MOV  @0008(R2),R3
5838 0008
583A 1302    JEQ  5840
583C A122  " A    @0012(R2),R4
583E 0012
5840 C0A2    MOV  @001C(R2),R2
5842 001C
5844 16F8    JNE  5836
5846 C0A0    MOV  @2782,R2
5848 2782 '
584A 60A0 `  S    @2780,R2
584C 2780 '
584E 0912    SRL  R2,1
5850 6084 `  S    R4,R2
5852 C682    MOV  R2,*SP
5854 045C  \ B    *R12                  ;PME fetch
5856 C0FA    MOV  *SP+,R3
5858 1112    JLT  587E
585A 1311    JEQ  587E
585C C07A  z MOV  *SP+,R1
585E A07A  z A    *SP+,R1
5860 C0BA    MOV  *SP+,R2
5862 A0BA    A    *SP+,R2
5864 C101    MOV  R1,R4
5866 E102    SOC  R2,R4
5868 E103    SOC  R3,R4
586A 0914    SRL  R4,1
586C 1804    JOC  5876
586E CC72  r MOV  *R2+,*R1+
5870 0643  C DECT R3
5872 15FD    JGT  586E
5874 045C  \ B    *R12                  ;PME fetch
5876 DC72  r MOVB *R2+,*R1+
5878 0603    DEC  R3
587A 15FD    JGT  5876
587C 045C  \ B    *R12
587E 022A  * AI   SP,0008
5880 0008
5882 045C  \ B    *R12
5884 C0FA    MOV  *SP+,R3
5886 11FB    JLT  587E
5888 13FA    JEQ  587E
588A C07A  z MOV  *SP+,R1
588C A07A  z A    *SP+,R1
588E C0BA    MOV  *SP+,R2
5890 A0BA    A    *SP+,R2
5892 C103    MOV  R3,R4
5894 0604    DEC  R4
5896 A044  D A    R4,R1
5898 A084    A    R4,R2
589A D452  R MOVB *R2,*R1
```

```
589C 0602    DEC  R2
589E 0601    DEC  R1
58A0 0603    DEC  R3
58A2 15FB    JGT  589A
58A4 045C  \ B    *R12
58A6 C0FA    MOV  *SP+,R3
58A8 06C3    SWPB R3
58AA C13A  : MOV  *SP+,R4
58AC 1111    JLT  58D0
58AE 1310    JEQ  58D0
58B0 C07A  z MOV  *SP+,R1
58B2 A07A  z A    *SP+,R1
58B4 C001    MOV  R1,R0
58B6 E004    SOC  R4,R0
58B8 0910    SRL  R0,1
58BA 1806    JOC  58C8
58BC D803    MOVB R3,@8387
58BE 8387
58C0 CC43  C MOV  R3,*R1+
58C2 0644  D DECT R4
58C4 15FD    JGT  58C0
58C6 045C  \ B    *R12
58C8 DC43  C MOVB R3,*R1+
58CA 0604    DEC  R4
58CC 15FD    JGT  58C8
58CE 045C  \ B    *R12
58D0 022A  * AI   SP,0004
58D2 0004
58D4 045C  \ B    *R12         ; PME fetch

; Kernel procedure TIME
58D6 C07A  z MOV  *SP+,R1
58D8 C460  ` MOV  @TIMELOW,*R1
58DA 2BA4 +
58DC C07A  z MOV  *SP+,R1
58DE C460  ` MOV  @TIMEHI,*R1
58E0 2BA6 +
58E2 045C  \ B    *R12         ; PME fetch

58E4 04C0    CLR  R0
58E6 05CA    INCT SP
58E8 C07A  z MOV  *SP+,R1
58EA A07A  z A    *SP+,R1
58EC C0BA    MOV  *SP+,R2
58EE 06C2    SWPB R2
58F0 C0FA    MOV  *SP+,R3
58F2 C13A  : MOV  *SP+,R4
58F4 1318    JEQ  5926
58F6 110B    JLT  590E
58F8 90B1    CB   *R1+,R2
58FA 1306    JEQ  5908
58FC C0C3    MOV  R3,R3
58FE 1613    JNE  5926
5900 0580    INC  R0
5902 0604    DEC  R4
5904 15F9    JGT  58F8
5906 100F    JMP  5926
5908 C0C3    MOV  R3,R3
```

```
590A 130D    JEQ  5926
590C 10F9    JMP  5900
590E 04C0    CLR  R0
5910 9091    CB   *R1,R2
5912 1307    JEQ  5922
5914 C0C3    MOV  R3,R3
5916 1607    JNE  5926
5918 0600    DEC  R0
591A 0601    DEC  R1
591C 0584    INC  R4
591E 11F8    JLT  5910
5920 1002    JMP  5926
5922 C0C3    MOV  R3,R3
5924 16F9    JNE  5918
5926 C680    MOV  R0,*SP
5928 045C  \ B    *R1
592A C6A0    MOV  @2B6E,*SP
592C 2B6E +n
592E 045C  \ B    *R12         ; PME fetch
5930 06A0    BL   @FAULGEN
5932 484A HJ
5934 000E    .WORD 14          ; Halt, breakpoint

5936 06A0    BL   @FAULGEN
5938 484A HJ
593A 000B    .WORD BH          ; Floating point error

593C C07A  z MOV  *SP+,R1
593E C801    MOV  R1,@2AEE
5940 2AEE *
5942 C0BA    MOV  *SP+,R2
5944 A052  R A    *R2,R1
5946 C802    MOV  R2,@2ADE
5948 2ADE *
594A C102    MOV  R2,R4
594C 05C2    INCT R2
594E 0224  $ AI   R4,0006
5950 0006
5952 0203    LI   R3,0004
5954 0004
5956 C144  D MOV  R4,R5
5958 CD20    MOV  @40A2,*R4+
595A 40A2 @
595C 0603    DEC  R3
595E 15FC    JGT  5958
5960 C105    MOV  R5,R4
5962 0203    LI   R3,0008
5964 0008
5966 D1B1    MOVB *R1+,R6
5968 9806    CB   R6,@409A
596A 409A @
596C 13FC    JEQ  5966
596E 9806    CB   R6,@4097
5970 4097 @
5972 120C    JLE  598C
5974 0246  F ANDI R6,DF00
5976 DF00
5978 9806    CB   R6,@4094
```

```
597A 4094 @
597C 1A0D    JL   5998
597E 9806    CB   R6,@4095
5980 4095 @
5982 1B0A    JH   5998
5984 0603    DEC  R3
5986 11EF    JLT  5966
5988 DD06    MOVB R6,*R4+
598A 10ED    JMP  5966
598C 9806    CB   R6,@4096
598E 4096 @
5990 14F9    JHE  5984
5992 9806    CB   R6,@409B
5994 409B @
5996 13F6    JEQ  5984
5998 0641  A DECT R1
599A 6060 `` S    @2AEE,R1
599C 2AEE *
599E C801    MOV  R1,@2AC8
59A0 2AC8 *
59A2 C105    MOV  R5,R4
59A4 04C1    CLR  R1
59A6 D054  T MOVB *R4,R1
59A8 06C1    SWPB R1
59AA 0221  ! AI   R1,FFBF
59AC FFBF
59AE 0A21  ! SLA  R1,2
59B0 C0E1    MOV  @3026(R1),R3
59B2 3026 0&
59B4 C061  a MOV  @3028(R1),R1
59B6 3028 0(
59B8 0460  ` B    @59BC
59BA 59BC Y
59BC 0200    LI   R0,0004
59BE 0004
59C0 C803    MOV  R3,@SAVPACK
59C2 2AF2 *
59C4 8CF4    C    *R4+,*R3+
59C6 1615    JNE  59F2
59C8 0600    DEC  R0
59CA 15FC    JGT  59C4
59CC C060  ` MOV  @2ADE,R1
59CE 2ADE *
59D0 C460  ` MOV  @2AC8,*R1
59D2 2AC8 *
59D4 04D2    CLR  *R2
59D6 D8B3    MOVB *R3+,@0001(R2)
59D8 0001
59DA 05C2    INCT R2
59DC 04D2    CLR  *R2
59DE D8B3    MOVB *R3+,@0001(R2)
59E0 0001
59E2 05C2    INCT R2
59E4 0204    LI   R4,0004
59E6 0004
59E8 CCA0    MOV  @40A2,*R2+
59EA 40A2 @
59EC 0604    DEC  R4
```

```
59EE 15FC    JGT  59E8
59F0 045C  \ B    *R12
59F2 0601    DEC  R1
59F4 1306    JEQ  5A02
59F6 C0E0    MOV  @SAVPACK,R3
59F8 2AF2 *
59FA A0E0    A    @40A4,R3
59FC 40A4 @
59FE C105    MOV  R5,R4
5A00 10DD    JMP  59BC
5A02 C0A0    MOV  @2ADE,R2
5A04 2ADE *
5A06 C4A0    MOV  @2AC8,*R2
5A08 2AC8 *
5A0A 05C2    INCT R2
5A0C 04F2    CLR  *R2+
5A0E 0201    LI   R1,000F
5A10 000F
5A12 C481    MOV  R1,*R2
5A14 045C  \ B    *R12
5A16 C07A  z MOV  *SP+,R1
5A18 C0AA    MOV  @0002(SP),R2
5A1A 0002
5A1C 8491    C    *R1,*R2
5A1E 160E    JNE  5A3C
5A20 88A1    C    @0002(R1),@0002(R2)
5A22 0002
5A24 0002
5A26 160A    JNE  5A3C
5A28 88A1    C    @0004(R1),@0004(R2)
5A2A 0004
5A2C 0004
5A2E 1606    JNE  5A3C
5A30 88A1    C    @0006(R1),@0006(R2)
5A32 0006
5A34 0006
5A36 1602    JNE  5A3C
5A38 04C1    CLR  R1
5A3A 100C    JMP  5A54
5A3C 1B06    JH   5A4A
5A3E C122  " MOV  @000A(R2),R4
5A40 000A
5A42 160C    JNE  5A5C
5A44 0201    LI   R1,FFFF
5A46 FFFF
5A48 1005    JMP  5A54
5A4A C122  " MOV  @0008(R2),R4
5A4C 0008
5A4E 1606    JNE  5A5C
5A50 0201    LI   R1,0001
5A52 0001
5A54 C13A  : MOV  *SP+,R4
5A56 C502    MOV  R2,*R4
5A58 C681    MOV  R1,*SP
5A5A 045C  \ B    *R12
5A5C C084    MOV  R4,R2
5A5E 10DE    JMP  5A1C
5A60 C801    MOV  R1,@2A26
```

```
5A62 2A26 *&
5A64 C800    MOV  R0,@2A24
5A66 2A24 *$
5A68 C060  ` MOV  @2B88,R1
5A6A 2B88 +
5A6C 1308    JEQ  5A7E
5A6E C021  ! MOV  @0004(R1),R0
5A70 0004
5A72 0910    SRL  R0,1
5A74 C840  @ MOV  R0,@0004(R1)
5A76 0004
5A78 C061  a MOV  @001C(R1),R1
5A7A 001C
5A7C 16F8    JNE  5A6E
5A7E C060  ` MOV  @2A26,R1
5A80 2A26 *&
5A82 C020    MOV  @2A24,R0
5A84 2A24 *$
5A86 045B  [ B    *R11
5A88 4201 B  SZC  R1,R8
5A8A 064A  J DECT SP
5A8C 04DA    CLR  *SP
5A8E 064A  J DECT SP
5A90 04DA    CLR  *SP
5A92 064A  J DECT SP
5A94 04DA    CLR  *SP
5A96 064A  J DECT SP
5A98 C6A0    MOV  @5A88,*SP
5A9A 5A88 Z
5A9C 045B  [ B    *R11
5A9E C80B    MOV  R11,@2A34
5AA0 2A34 *4
5AA2 020B    LI   R11,2A24
5AA4 2A24 *$
5AA6 CEC0    MOV  R0,*R11+
5AA8 CEC1    MOV  R1,*R11+
5AAA CEC2    MOV  R2,*R11+
5AAC CEC3    MOV  R3,*R11+
5AAE CEC4    MOV  R4,*R11+
5AB0 CEC5    MOV  R5,*R11+
5AB2 CEC6    MOV  R6,*R11+
5AB4 CEC7    MOV  R7,*R11+
5AB6 C121  ! MOV  @0012(R1),R4
5AB8 0012
5ABA C020    MOV  @2780,R0
5ABC 2780 '
5ABE 04C1    CLR  R1
5AC0 04C2    CLR  R2
5AC2 0203    LI   R3,40B4
5AC4 40B4 @
5AC6 C1A0    MOV  @2B88,R6
5AC8 2B88 +
5ACA 131C    JEQ  5B04
5ACC 88E6    C    @0004(R6),@0004(R3)
5ACE 0004
5AD0 0004
5AD2 1B07    JH   5AE2
5AD4 8806    C    R6,@CURSIB
```

```
5AD6 2AB2 *
5AD8 1304    JEQ  5AE2
5ADA C166  f MOV  @0008(R6),R5
5ADC 0008
5ADE 1601    JNE  5AE2
5AE0 C0C6    MOV  R6,R3
5AE2 C166  f MOV  @0002(R6),R5
5AE4 0002
5AE6 6140 a@ S    R0,R5
5AE8 0915    SRL  R5,1
5AEA 8105    C    R5,R4
5AEC 1560  ` JGT  5BAE
5AEE A085    A    R5,R2
5AF0 C046  F MOV  R6,R1
5AF2 C021  ! MOV  @0002(R1),R0
5AF4 0002
5AF6 A021  ! A    @0012(R1),R0
5AF8 0012
5AFA A021  ! A    @0012(R1),R0
5AFC 0012
5AFE C1A1    MOV  @001C(R1),R6
5B00 001C
5B02 16E4    JNE  5ACC
5B04 C160  ` MOV  @2782,R5
5B06 2782 '
5B08 6140 a@ S    R0,R5
5B0A 0915    SRL  R5,1
5B0C 8105    C    R5,R4
5B0E 154F  O JGT  5BAE
5B10 A085    A    R5,R2
5B12 8102    C    R2,R4
5B14 1511    JGT  5B38
5B16 04E3    CLR  @0002(R3)
5B18 0002
5B1A C0A3    MOV  @001C(R3),R2
5B1C 001C
5B1E C063  c MOV  @001A(R3),R1
5B20 001A
5B22 1303    JEQ  5B2A
5B24 C842  B MOV  R2,@001C(R1)
5B26 001C
5B28 1002    JMP  5B2E
5B2A C802    MOV  R2,@2B88
5B2C 2B88 +
5B2E C082    MOV  R2,R2
5B30 13C4    JEQ  5ABA
5B32 C881    MOV  R1,@001A(R2)
5B34 001A
5B36 10C1    JMP  5ABA
5B38 C1A0    MOV  @2B88,R6
5B3A 2B88 +
5B3C C020    MOV  @2780,R0
5B3E 2780 '
5B40 C166  f MOV  @0012(R6),R5
5B42 0012
5B44 0A15    SLA  R5,1
5B46 C0A6    MOV  @0002(R6),R2
5B48 0002
```

```
5B4A 8002    C    R2,R0
5B4C 1602    JNE  5B52
5B4E A005    A    R5,R0
5B50 102A  * JMP  5BA6
5B52 C980    MOV  R0,@0002(R6)
5B54 0002
5B56 0204    LI   R4,8800
5B58 8800
5B5A D920    MOVB @8385,@0402(R4)
5B5C 8385
5B5E 0402
5B60 0203    LI   R3,FAC
5B62 834A  J
5B64 D902    MOVB R2,@0402(R4)
5B66 0402
5B68 0207    LI   R7,0023
5B6A 0023  #
5B6C 81C5    C    R5,R7
5B6E 1A02    JL   5B74
5B70 6147 aG S    R7,R5
5B72 1002    JMP  5B78
5B74 C1C5    MOV  R5,R7
5B76 04C5    CLR  R5
5B78 C047  G MOV  R7,R1
5B7A DCD4    MOVB *R4,*R3+
5B7C 0607    DEC  R7
5B7E 15FD    JGT  5B7A
5B80 0204    LI   R4,8C02
5B82 8C02
5B84 C1C0    MOV  R0,R7
5B86 0267  g ORI  R7,4000
5B88 4000 @
5B8A D520    MOVB @838F,*R4
5B8C 838F
5B8E 0203    LI   R3,FAC
5B90 834A  J
5B92 D507    MOVB R7,*R4
5B94 C1C1    MOV  R1,R7
5B96 A087    A    R7,R2
5B98 A007    A    R7,R0
5B9A 0644  D DECT R4
5B9C D533  3 MOVB *R3+,*R4
5B9E 0607    DEC  R7
5BA0 15FD    JGT  5B9C
5BA2 C145  E MOV  R5,R5
5BA4 16D8    JNE  5B56
5BA6 C046  F MOV  R6,R1
5BA8 C1A1    MOV  @001C(R1),R6
5BAA 001C
5BAC 16C9    JNE  5B40
5BAE C0A0    MOV  @2A26,R2
5BB0 2A26 *&
5BB2 C041  A MOV  R1,R1
5BB4 130A    JEQ  5BCA
5BB6 C021  ! MOV  @0012(R1),R0
5BB8 0012
5BBA 0A10    SLA  R0,1
5BBC A021  ! A    @0002(R1),R0
```

```
5BBE 0002
5BC0 C0E1    MOV  @001C(R1),R3
5BC2 001C
5BC4 C842  B MOV  R2,@001C(R1)
5BC6 001C
5BC8 1006    JMP  5BD6
5BCA C020    MOV  @2780,R0
5BCC 2780 '
5BCE C0E0    MOV  @2B88,R3
5BD0 2B88 +
5BD2 C802    MOV  R2,@2B88
5BD4 2B88 +
5BD6 C880    MOV  R0,@0002(R2)
5BD8 0002
5BDA 04E2    CLR  @0004(R2)
5BDC 0004
5BDE C881    MOV  R1,@001A(R2)
5BE0 001A
5BE2 C883    MOV  R3,@001C(R2)
5BE4 001C
5BE6 1302    JEQ  5BEC
5BE8 C8C2    MOV  R2,@001A(R3)
5BEA 001A
5BEC C122  " MOV  @0016(R2),R4
5BEE 0016
5BF0 064A  J DECT SP
5BF2 C694    MOV  *R4,*SP
5BF4 064A  J DECT SP
5BF6 C680    MOV  R0,*SP
5BF8 1003    JMP  5C00
5BFA 0000    UNDEFINED

GROMRD    5BFC 0006    ;GROM read data
GROMRA    5BFE 6906 i  ;GROM read address

5C00 064A  J
5C02 C6A0    MOV  @40BC,*SP
5C04 40BC @
5C06 C122  " MOV  @0012(R2),R4
5C08 0012
5C0A 0A14    SLA  R4,1
5C0C 064A  J DECT SP
5C0E C684    MOV  R4,*SP
5C10 064A  J DECT SP
5C12 C6A2    MOV  @0014(R2),*SP
5C14 0014
5C16 0204    LI   R4,4000
5C18 4000 @
5C1A 064A  J DECT SP
5C1C C684    MOV  R4,*SP
5C1E 06A0    BL   @34AC
5C20 34AC 4
5C22 C80C    MOV  R12,@2A3E
5C24 2A3E *>
5C26 C80D    MOV  R13,@2A3A
5C28 2A3A *:
5C2A C80F    MOV  R15,@2A3C
5C2C 2A3C *<
```

```
5C2E C820    MOV  @PMEFLIP,@2A40
5C30 2AE8 *
5C32 2A40 *@
5C34 05A0    INC  @PMEFLIP
5C36 2AE8 *
5C38 020F    LI   R15,FFFF
5C3A FFFF
5C3C 020D    LI   R13,8800
5C3E 8800
5C40 C160   ` MOV  @2A26,R5
5C42 2A26 *&
5C44 C020    MOV  @2B6E,R0
5C46 2B6E +n
5C48 1612    JNE  5C6E
5C4A C0C5    MOV  R5,R3
5C4C C023   # MOV  @0002(R3),R0
5C4E 0002
5C50 C040   @ MOV  R0,R1
5C52 0221   ! AI   R1,0004
5C54 0004
5C56 0223   # AI   R3,000A
5C58 000A
5C5A 0204    LI   R4,0004
5C5C 0004
5C5E 06A0    BL   @GETDAT2
5C60 4178 Ax
5C62 8CC2    C    R2,*R3+
5C64 1604    JNE  5C6E
5C66 0604    DEC  R4
5C68 15FA    JGT  5C5E
5C6A C040   @ MOV  R0,R1
5C6C 1046   F JMP  5CFA
5C6E C085    MOV  R5,R2
5C70 0222   " AI   R2,000A
5C72 000A
5C74 0203    LI   R3,2B00
5C76 2B00 +
5C78 CCF2    MOV  *R2+,*R3+
5C7A CCF2    MOV  *R2+,*R3+
5C7C CCF2    MOV  *R2+,*R3+
5C7E C4D2    MOV  *R2,*R3
5C80 C065   e MOV  @0016(R5),R1
5C82 0016
5C84 C0F1    MOV  *R1+,R3
5C86 04C2    CLR  R2
5C88 3CA0 <  DIV  @2B6C,R2
5C8A 2B6C +l
5C8C 06C2    SWPB R2
5C8E D0C2    MOVB R2,R3
5C90 0223   # AI   R3,3030
5C92 3030 00
5C94 C803    MOV  R3,@2B30
5C96 2B30 +0
5C98 0204    LI   R4,2B1F
5C9A 2B1F +
5C9C 0203    LI   R3,0007
5C9E 0007
5CA0 D0B1    MOVB *R1+,R2
```

```
5CA2 1305    JEQ  5CAE
5CA4 0982    SRL  R2,8
5CA6 60C2 `  S    R2,R3
5CA8 DD31  1 MOVB *R1+,*R4+
5CAA 0602    DEC  R2
5CAC 15FD    JGT  5CA8
5CAE 0201    LI   R1,2000
5CB0 2000
5CB2 C0C3    MOV  R3,R3
5CB4 1303    JEQ  5CBC
5CB6 DD01    MOVB R1,*R4+
5CB8 0603    DEC  R3
5CBA 15FD    JGT  5CB6
5CBC 0201    LI   R1,2B50
5CBE 2B50 +P
5CC0 C0E0    MOV  @2B6A,R3
5CC2 2B6A +j
5CC4 6283 b  S    R3,SP
5CC6 C08A    MOV  SP,R2
5CC8 CCB1    MOV  *R1+,*R2+
5CCA 0643  C DECT R3
5CCC 15FD    JGT  5CC8
5CCE 04CF    CLR  R15
5CD0 06A0    BL   @34DC
5CD2 34DC 4
5CD4 06A0    BL   @34C4
5CD6 34C4 4
5CD8 06A0    BL   @34AC
5CDA 34AC 4
5CDC C320    MOV  @2A3E,R12
5CDE 2A3E *>
5CE0 C360  ` MOV  @2A3A,R13
5CE2 2A3A *:
5CE4 C3E0    MOV  @2A3C,R15
5CE6 2A3C *<
5CE8 C820    MOV  @2A40,@PMEFLIP
5CEA 2A40 *@
5CEC 2AE8 *
5CEE C0A0    MOV  @2A26,R2
5CF0 2A26 *&
5CF2 C022  " MOV  @0002(R2),R0
5CF4 0002
5CF6 0460  ` B    @5BEC
5CF8 5BEC [
5CFA C1C1    MOV  R1,R7
5CFC 0221  ! AI   R1,000C
5CFE 000C
5D00 06A0    BL   @GETDAT2
5D02 4178 Ax
5D04 04E0    CLR  @PMEFLIP
5D06 2AE8 *
5D08 0282    CI   R2,0001
5D0A 0001
5D0C 1602    JNE  5D12
5D0E 05A0    INC  @PMEFLIP
5D10 2AE8 *
5D12 06A0    BL   @GETDAT2
5D14 4178 Ax
```

```
5D16 C082    MOV  R2,R2
5D18 136F  o JEQ  5DF8
5D1A 0A12    SLA  R2,1
5D1C A087    A    R7,R2
5D1E C042  B MOV  R2,R1
5D20 06A0    BL   @GETDAT2
5D22 4178 Ax
5D24 C082    MOV  R2,R2
5D26 1368  h JEQ  5DF8
5D28 0641  A DECT R1
5D2A 0A12    SLA  R2,1
5D2C A042  B A    R2,R1
5D2E 06A0    BL   @GETDAT2
5D30 4178 Ax
5D32 C182    MOV  R2,R6
5D34 1361  a JEQ  5DF8
5D36 C1C1    MOV  R1,R7
5D38 C047  G MOV  R7,R1
5D3A C801    MOV  R1,@2A44
5D3C 2A44 *D
5D3E 06A0    BL   @GETDAT2
5D40 4178 Ax
5D42 C802    MOV  R2,@2A38
5D44 2A38 *8
5D46 06A0    BL   @GETDAT2
5D48 4178 Ax
5D4A C1C1    MOV  R1,R7
5D4C 064A  J DECT SP
5D4E C682    MOV  R2,*SP
5D50 C820    MOV  @2D96,@2A42
5D52 2D96 -
5D54 2A42 *B
5D56 020C    LI   R12,5D5E
5D58 5D5E ]^
5D5A 0460  ` B    @5F26
5D5C 5F26 _&
5D5E C047  G MOV  R7,R1
5D60 06A0    BL   @GETDAT2
5D62 4178 Ax
5D64 C1C1    MOV  R1,R7
5D66 C082    MOV  R2,R2
5D68 111A    JLT  5D9E
5D6A C0DA    MOV  *SP,R3
5D6C 1502    JGT  5D72
5D6E 1301    JEQ  5D72
5D70 0502    NEG  R2
5D72 C802    MOV  R2,@2A36
5D74 2A36 *6
5D76 06A0    BL   @5A8A
5D78 5A8A Z
5D7A 020C    LI   R12,5D82
5D7C 5D82 ]
5D7E 0460  ` B    @5ECA
5D80 5ECA ^
5D82 064A  J DECT SP
5D84 C6A0    MOV  @2A36,*SP
5D86 2A36 *6
5D88 020C    LI   R12,5D90
```

```
5D8A 5D90 ]
5D8C 0460  ` B    @5F26
5D8E 5F26 _&
5D90 020C    LI   R12,5D98
5D92 5D98 ]
5D94 0460  ` B    @5EB2
5D96 5EB2 ^
5D98 0620    DEC  @2A42
5D9A 2A42 *B
5D9C 15E0    JGT  5D5E
5D9E 022A  * AI   SP,-10
5DA0 FFF6
5DA2 C6A0    MOV  @2A38,*SP
5DA4 2A38 *8
5DA6 1109    JLT  5DBA
5DA8 020C    LI   R12,5DB0
5DAA 5DB0 ]
5DAC 0460  ` B    @5F5C
5DAE 5F5C _\
5DB0 020C    LI   R12,5DB8
5DB2 5DB8 ]
5DB4 0460  ` B    @5ECA
5DB6 5ECA ^
5DB8 1009    JMP  5DCC
5DBA 051A    NEG  *SP
5DBC 020C    LI   R12,5DC4
5DBE 5DC4 ]
5DC0 0460  ` B    @5F5C
5DC2 5F5C _\
5DC4 020C    LI   R12,5DCC
5DC6 5DCC ]
5DC8 0460  ` B    @5EC2
5DCA 5EC2 ^
5DCC 0203    LI   R3,8C02
5DCE 8C02
5DD0 C020    MOV  @2A44,R0
5DD2 2A44 *D
5DD4 0260  ` ORI  R0,4000
5DD6 4000 @
5DD8 D4E0    MOVB @8381,*R3
5DDA 8381
5DDC 0202    LI   R2,0008
5DDE 0008
5DE0 D4C0    MOVB R0,*R3
5DE2 0643  C DECT R3
5DE4 8820    C    @2D96,@2A42
5DE6 2D96 -
5DE8 2A42 *B
5DEA 1601    JNE  5DEE
5DEC 05C7    INCT R7
5DEE D4FA    MOVB *SP+,*R3
5DF0 0602    DEC  R2
5DF2 15FD    JGT  5DEE
5DF4 0606    DEC  R6
5DF6 15A0    JGT  5D38
5DF8 C360  ` MOV  @2A3A,R13
5DFA 2A3A *:
5DFC C3E0    MOV  @2A3C,R15
```

```
5DFE 2A3C *<
5E00 C320    MOV  @2A3E,R12
5E02 2A3E *>
5E04 C820    MOV  @2A40,@PMEFLIP
5E06 2A40 *@
5E08 2AE8 *
5E0A 020B    LI   R11,2A24
5E0C 2A24 *$
5E0E C03B  ; MOV  *R11+,R0
5E10 C07B  { MOV  *R11+,R1
5E12 C0BB    MOV  *R11+,R2
5E14 C0FB    MOV  *R11+,R3
5E16 C13B  ; MOV  *R11+,R4
5E18 C17B  { MOV  *R11+,R5
5E1A C1BB    MOV  *R11+,R6
5E1C C1FB    MOV  *R11+,R7
5E1E C2E0    MOV  @2A34,R11
5E20 2A34 *4
5E22 C121  ! MOV  @0002(R1),R4
5E24 0002
5E26 045B  [ B    *R11

5E28 0202    LI   R2,FARG
5E2A 835C  \
5E2C 1002    JMP  MOVFLT
5E2E 0202    LI   R2,FAC
5E30 834A  J
MOVFLT    5E32 CCBA    MOV  *SP+,*R2+
5E34 CCBA    MOV  *SP+,*R2+
5E36 CCBA    MOV  *SP+,*R2+
5E38 C4BA    MOV  *SP+,*R2
5E3A 045B  [ B    *R11
5E3C 0201    LI   R1,FAC
5E3E 834A  J
5E40 022A  * AI   SP,-8
5E42 FFF8
5E44 C08A    MOV  SP,R2
5E46 CCB1    MOV  *R1+,*R2+
5E48 CCB1    MOV  *R1+,*R2+
5E4A CCB1    MOV  *R1+,*R2+
5E4C C491    MOV  *R1,*R2
5E4E 045B  [ B    *R11

; Console floating point routine call
FLTCONS   5E50 04E0    CLR  @8354
          5E52 8354  T
          5E54 C81B    MOV  *R11,@83E4   ; GPLWS R2 given address to
console routine
          5E56 83E4
          5E58 06A0    BL   @5E2E
          5E5A 5E2E ^.
          5E5C 06A0    BL   @5E28
          5E5E 5E28 ^(
          5E60 02E0    LWPI GPLWS
          5E62 83E0
          5E64 0692    BL   *R2          ; Execute console real
arithmetic
          5E66 02E0    LWPI PASCALWS
```

```
           5E68 8380
           5E6A 06A0    BL   @5E3C
           5E6C 5E3C ^<
           5E6E D020    MOVB @8354,R0
           5E70 8354  T
           5E72 1601    JNE  5E76
           5E74 045C  \ B    *R12         ; PME fetch
           5E76 0280    CI   R0,0002
           5E78 0002
           5E7A 1603    JNE  5E82
           5E7C 06A0    BL   @FAULGEN
           5E7E 484A HJ
           5E80 0006    .WORD 6 ; Divide by zero
           5E82 06A0    BL   @FAULGEN
           5E84 484A HJ
           5E86 000C    .WORD 12          ; Floating point math error

UNUSED1    5E88 8300    .WORD FETCH

; Load real
LDRL       5E8A 5E8C ^  .WORD LDRL+2
           5E8C C07A  z MOV  *SP+,R1      ;Address
           5E8E 022A  * AI   SP,-8
           5E90 FFF8
           5E92 C08A    MOV  SP,R2
           5E94 CCB1    MOV  *R1+,*R2+
           5E96 CCB1    MOV  *R1+,*R2+
           5E98 CCB1    MOV  *R1+,*R2+
           5E9A C491    MOV  *R1,*R2
           5E9C 045C  \ B    *R12         ;PME fetch

; Store real
STRL       5E9E 5EA0 ^  .WORD STRL+2
           5EA0 C06A  j MOV  @8(SP),R1    ;Address
           5EA2 0008
           5EA4 CC7A  z MOV  *SP+,*R1+
           5EA6 CC7A  z MOV  *SP+,*R1+
           5EA8 CC7A  z MOV  *SP+,*R1+
           5EAA C47A  z MOV  *SP+,*R1
           5EAC 05CA    INCT SP           ;Pop address
           5EAE 045C  \ B    *R12         ;PME fetch

; Add real
ADR        5EB0 5EB2    .WORD ADR+2
           5EB2 06A0    BL   @FLTCONS
           5EB4 5E50 ^P
           5EB6 0D80    .WORD D80H        ; Console Floating ADD

; Subtract real
SBR        5EB8 5EBA    .WORD SDR+2
           5EBA 06A0    BL   @FLTCONS
           5EBC 5E50 ^P
           5EBE 0D7C    .WORD D7CH        ; Console Floating SUB

; Divide real
DVR        5EC0 5EC2    .WORD DVR+2
           5EC2 06A0    BL   @FLTCONS
           5EC4 5E50 ^P
```

```
          5EC6 0FF4    .WORD FF4H        ; Console Floating DIV

; Multiply real
MPR       5EC8 5ECA    .WORD MPR+2
          5ECA 06A0    BL   @FLTCONS
          5ECC 5E50 ^P
          5ECE 0E88    .WORD E88H        ; Console Floating MUL

; Absolute value real
ABR       5ED0 5ED2    .WORD ARB+2
          5ED2 075A  Z ABS  *SP
          5ED4 045C  \ B    *R12         ;PME fetch

; Negate real
NGR       5ED6 5ED8    .WORD NGR+2
          5ED8 051A    NEG  *SP
          5EDA 045C  \ B    *R12         ;PME fetch

; Common entry for equal, less than or equal and greater than or equal
real
EQREAL    5EDC 5EDE    .WORD EQREAL+2
          5EDE 0221  ! AI   R1,5146      ; Opcode is CDH (205) *2
          5EE0 5146 QF
          5EE2 04E0    CLR  @8354
          5EE4 8354  T
          5EE6 06A0    BL   @5E2E
          5EE8 5E2E ^.
          5EEA 06A0    BL   @5E28
          5EEC 5E28 ^(
          5EEE 064A  J DECT SP
          5EF0 02E0    LWPI GPLWS
          5EF2 83E0
          5EF4 0203    LI   R3,EQRCONT
          5EF6 5EFC ^
          5EF8 0460  ` B    @0D50
          5EFA 0D50  P
EQRCONT   5EFC 02E0    LWPI PASCALWS
          5EFE 8380
          5F00 0460  ` B    @TESTS
          5F02 52C4 R

; Round real
RND       5F04 5F06    .WORD RND+2
          5F06 04E0    CLR  @8354
          5F08 8354  T
          5F0A 06A0    BL   @5E2E
          5F0C 5E2E ^.
          5F0E 02E0    LWPI GPLWS
          5F10 83E0
          5F12 06A0    BL   @12B8                  ; Console floating
point
          5F14 12B8
          5F16 02E0    LWPI PASCALWS
          5F18 8380
          5F1A 064A  J DECT SP
          5F1C C6A0    MOV  @FAC,*SP
          5F1E 834A  J
          5F20 0460  ` B    @5E6E
```

```
          5F22 5E6E ^n

; Float
FLT       5F24 5F26    .WORD FLT+2
          5F26 C83A  : MOV  *SP+,@FAC
          5F28 834A  J
          5F2A 06A0    BL   @5F94
          5F2C 5F94  _
          5F2E 06A0    BL   @5E3C
          5F30 5E3C ^<
          5F32 045C  \ B    *R12         ; PME fetch

; Truncate real
TNC       5F34 5F36    .WORD TNC+2
          5F36 C01A    MOV  *SP,R0
          5F38 075A  Z ABS  *SP
          5F3A 04C1    CLR  R1
          5F3C D81A    MOVB *SP,@8383
          5F3E 8383
          5F40 0221  ! AI   R1,FFC1
          5F42 FFC1
          5F44 1107    JLT  5F54
          5F46 0281    CI   R1,0006
          5F48 0006
          5F4A 1504    JGT  5F54
          5F4C C08A    MOV  SP,R2
          5F4E A081    A    R1,R2
          5F50 0582    INC  R2
          5F52 D481    MOVB R1,*R2
          5F54 0540  @ INV  R0
          5F56 11D7    JLT  RND+2
          5F58 051A    NEG  *SP
          5F5A 10D5    JMP  5F06
          5F5C C03A  : MOV  *SP+,R0
          5F5E 0280    CI   R0,FF81
          5F60 FF81
          5F62 1114    JLT  5F8C
          5F64 0280    CI   R0,007F
          5F66 007F  
          5F68 1511    JGT  5F8C
          5F6A 0910    SRL  R0,1
          5F6C 1803    JOC  5F74
          5F6E 0201    LI   R1,0001
          5F70 0001
          5F72 1002    JMP  5F78

          5F74 0201    LI   R1,000A
          5F76 000A
          5F78 0220    AI   R0,0040
          5F7A 0040  @
          5F7C D060  ` MOVB @8381,R1
          5F7E 8381
          5F80 C08A    MOV  SP,R2
          5F82 CC81    MOV  R1,*R2+
          5F84 04F2    CLR  *R2+
          5F86 04F2    CLR  *R2+
          5F88 04D2    CLR  *R2
          5F8A 045C  \ B    *R12         ; PME fetch
```

```
          5F8C 06A0    BL   @FAULGEN
          5F8E 484A HJ
          5F90 000C    .WORD 12          ; Floating point math error

5F92 0064  d UNDEFINED
5F94 0204    LI   R4,FAC
5F96 834A  J
5F98 C014    MOV  *R4,R0
5F9A C084    MOV  R4,R2
5F9C 04F2    CLR  *R2+
5F9E 04F2    CLR  *R2+
5FA0 04F2    CLR  *R2+
5FA2 04D2    CLR  *R2
5FA4 C140  @ MOV  R0,R5
5FA6 1321  ! JEQ  5FEA
5FA8 0740  @ ABS  R0
5FAA 0203    LI   R3,0040
5FAC 0040  @
5FAE 0280    CI   R0,0064
5FB0 0064  d
5FB2 1A13    JL   5FDA
5FB4 0280    CI   R0,2710
5FB6 2710 '
5FB8 1A08    JL   5FCA
5FBA 0583    INC  R3
5FBC C040  @ MOV  R0,R1
5FBE 04C0    CLR  R0
5FC0 3C20 <  DIV  @5F92,R0
5FC2 5F92 _
5FC4 D920    MOVB @8383,@0003(R4)
5FC6 8383
5FC8 0003
5FCA 0583    INC  R3
5FCC C040  @ MOV  R0,R1
5FCE 04C0    CLR  R0
5FD0 3C20 <  DIV  @5F92,R0
5FD2 5F92 _
5FD4 D920    MOVB @8383,@0002(R4)
5FD6 8383
5FD8 0002
5FDA D920    MOVB @8381,@0001(R4)
5FDC 8381
5FDE 0001
5FE0 D520    MOVB @8387,*R4
5FE2 8387
5FE4 0545  E INV  R5
5FE6 1101    JLT  5FEA
5FE8 0514    NEG  *R4
5FEA 045B  [ B    *R11
5FEC 0000    UNDEFINED
5FEE 0000    UNDEFINED
5FF0 0000    UNDEFINED
5FF2 0000    UNDEFINED
5FF4 0000    UNDEFINED
5FF6 0000    UNDEFINED
5FF8 0000    UNDEFINED
5FFA 0000    UNDEFINED
```

```
GROMWD     5FFC 0606    ;GROM write data (never used)
GROMWA     5FFE 0606    ;GROM write address

; Standard page ends here. Secondary page not in this disassembly

Scratch pad RAM under the p-system

; First the inner PME, which also handles immediate data
; Version loaded below for code in CPU RAM
FETCH      8300 D078  x MOVB *R8+,R1      ; Instruction fetch
           8302 0971  q SRL  R1,7                    ; Make word
           8304 C0A1    MOV  @OPCODE(R1),R2 ;2DB2 Fetch address to
interpreter
           8306 2DB2 -
           8308 C032  2 MOV  *R2+,R0      ; Fetch execution pointer
           830A 0450  P B    *R0                     ; Branch to
instruction interpreter/data fetch
           830C 1000    NOP
LDUBB      830E 04C4    CLR  R4
           8310 D838  8 MOVB *R8+,@8389   ; Lsby R4 Load unsigned byte
when there are two
           8312 8389
           8314 1000    NOP
LDUB       8316 04C3    CLR  R3
           8318 D838  8 MOVB *R8+,@8387   ; Lsby R3 Load unsigned byte
           831A 8387
           831C 0452  R B    *R2                     ; Interpret
instruction
           831E 1000    NOP
LDUBBG     8320 04C5    CLR  R5
           8322 D838  8 MOVB *R8+,@838B   ; Lsby R5 Load unsigned byte
when there are three
           8324 838B
           8326 1000    NOP
LDUBIG     8328 04C4    CLR  R4
           832A D838  8 MOVB *R8+,@8389   ; Lsby R4 Load unsigned byte
           832C 8389
           832E 1000    NOP
LDBIG      8330 04C3    CLR  R3
           8332 D0F8    MOVB *R8+,R3
           8334 1102    JLT  833A                   ; "Big" value? (15
bits)
           8336 06C3    SWPB R3           ; Unsigned byte value 0..127
           8338 0452  R B    *R2                     ; Interpret
instruction
           833A 0243  C ANDI R3,7F00      ; Remove sign bit
           833C 7F00 
           833E D838  8 MOVB *R8+,@8387   ; Lsby R3 low byte Make 15 bit
value
           8340 8387
           8342 0452  R B    *R2                     ; Interpret
instruction
           8344 1D02    SBO  2
8346 D820    MOVB @8802,@0300
8348 8802
; Floating point accumulator (FAC)
FAC        834A 0300
```

```
          834C 3F00 ?  DIV  R0,R12
          834E 0101    UNDEFINED
          8350 0000    UNDEFINED

8352 0019    UNDEFINED
8354 0007    UNDEFINED
8356 3EA9 >  DIV  @0000(R9),R10; First period in device name
8358 0000
835A 0000    UNDEFINED
; Floating point argument (ARG)
FARG      835C 0000    UNDEFINED
          835E 0000    UNDEFINED
          8360 0000    UNDEFINED
          8362 0000    UNDEFINED

8364 346C 4l STCR @41F4(R12),1
8366 41F4 A
8368 453E E> SZC  *R14+,*R4
836A 4580 E  SZC  R0,*R6
836C 4320 C  SZC  @0000,R12
836E 0000
8370 3EDF >  DIV  *R15,R11     ; VDP RAM top pointer (GPL)
8372 FF7C  | SOCB *R12+,*R13+  ; Sub stack pointer (GPL)
8374 00FF    UNDEFINED                    ; Initialized to 04FF
8376 0000    UNDEFINED
8378 0001    UNDEFINED
837A 00C2    UNDEFINED
GPLST     837C 0000    UNDEFINED                  ; GPL Status
837E 6002 `  S    R2,R0

; Pascal main workspace
PASCALWS  8380 .WORD    ; R0
8382 BCE2    ; R1
8384 000C    ; R2
8386 0002    ; R3
8388 D930  0 ; R4
838A AD20    ; R5
838C AD96    ; R6
838E FA4A  J ; R7
8390 AF79  y ; R8 IPC
8392 FB5A  Z ; R9 Frame pointer for activation record on stack.
8394 FA3E  > ; R10 SP
8396 4E28 N( ; R11 Return link.
8398 .WORD FETCH ; R12 PME fetch instruction address.
839A 0000    ; R13 Read data address for currently executing code. VDPRD,
PGRMRD or 0 (for CPU RAM).
839C A91C    ; R14 Global data frame pointer for current segment.
839E 0000    ; R15 Code flag. =0: CPU RAM, <0: VDP RAM, >0: GROM.

;Workspace
83A0 0000    UNDEFINED
83A2 2970 )p XOR  *R0+,R5
83A4 297C )| XOR  *R12+,R5
83A6 0000    UNDEFINED
83A8 2780 '  CZC  R0,R14
83AA 0000    UNDEFINED
83AC 0000    UNDEFINED
83AE 0000    UNDEFINED
```

```
83B0 0000    UNDEFINED
83B2 3E98 >  DIV  *R8,R10
83B4 8C02    C    R2,*R0+
83B6 4784 G  SZC  R4,*R14
83B8 0000    UNDEFINED
83BA 2964 )d XOR  @348E(R4),R5
83BC 348E 4
83BE C400    MOV  R0,*R0

;Workspace
83C0 3E9D >  DIV  *R13,R10
83C2 0A00    SLA  R0,0
83C4 5AD8 Z  SZCB *R8,@0000(R11)
83C6 0000
83C8 0000    UNDEFINED
83CA 0BB9    SRC  R9,11
83CC E1E6    SOC  @0000(R6),R7
83CE 0000
83D0 8C02    C    R2,*R0+
83D2 0001    UNDEFINED
83D4 29A8 )  XOR  @5B02(R8),R6
83D6 5B02 [
83D8 0000    UNDEFINED
83DA 8380    C    R0,R14
83DC 507A Pz SZCB *R10+,R1
83DE C000    MOV  R0,R0

;GPL workspace
GPLWS     83E0 .WORD
83E2 7E9D ~  SB   *R13,*R10+
83E4 4540 E@ SZC  R0,*R5
83E6 0000    UNDEFINED
83E8 83E0    C    @42FA,R15
83EA 42FA B
83EC 0A4E  N SLA  R14,4
83EE 0000    UNDEFINED
83F0 4076 @v SZC  *R6+,R1
83F2 83DA    C    *R10,R15
83F4 0002    UNDEFINED
83F6 346C 4l STCR @0006(R12),1
83F8 1300
83FA 9800    .WORD GRMRD
83FC 0108
83FE 8C02    .WORD VDPWA
```